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RIGID FOUNDATIONS FOR VIBRATORY EQUIPMENT

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RIGID FOUNDATIONS FOR VIBRATORY EQUIPMENT

1.0 PURPOSE

The purpose of this guideline is to document a uniform computer-assisted approach for the design of rigid-block type foundations for rotating and reciprocating vibratory machines (compressors, pumps, inline pumps, critical service pumps).

2.0 APPLICATION

This guideline applies to:

- Rigid-block type foundations supported by soil or piles. Mat or pile cap rigidity is judged relative to the ground's stiffness. The design of flexible structures and foundations for vibrating machinery is described in Guideline 000.215.1236.
- Machines subjected to periodic harmonic loading which are considered significant enough to merit a dynamic analysis. This generally applies to reciprocating machines greater than 200 horsepower (150 kW) and centrifugal machines greater than 500 horsepower (375 kW). The design of foundations for vibrating machinery below the above limits is described in Guideline 000.215.1227.

Supplemental information can be found in PIP STC01015 (section 4.3.8), ACI 351-3R, API RP 686 (chapter 4) and ASCE 136.

3.0 INTERFACING DISCIPLINES

- Mechanical Engineering
- Electrical Engineering
- Piping

4.0 DEFINITIONS

Not Used

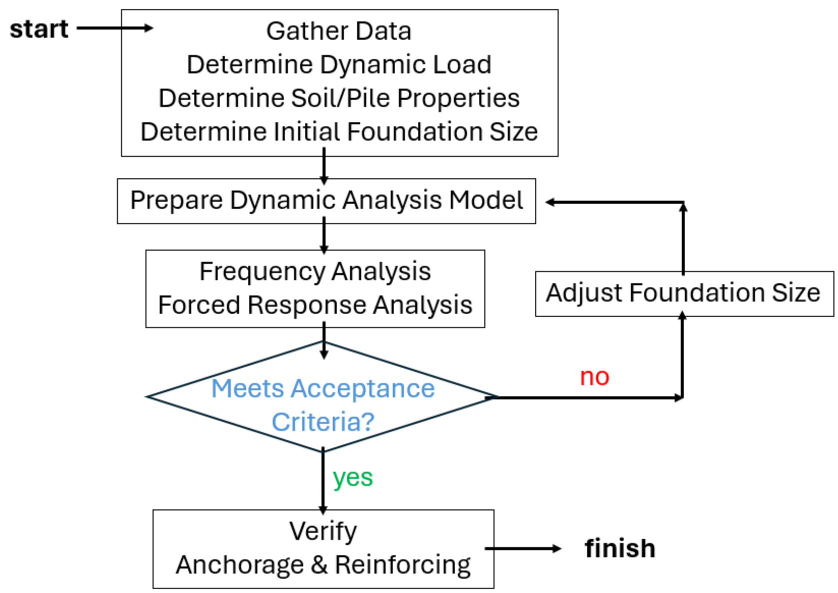
5.0 REQUIREMENTS

Not Used

6.0 EXECUTION OF GUIDELINES

6.1 Design Sequence

Following is a graphical description of how a vibration analysis fits into the design of a foundation mat.



6.2 Information Necessary for Design

Request machine data from the machine supplier. Refer to Specification 000.215.00920.

Obtain soil information from the geotechnical consultant.

Client and project data are in the form of project specifications, client specifications, or meeting notes.

6.2.1 Basic Machine Data

<u>Operating Speed:</u> This is the frequency or range of frequencies of the dynamic machine forces. For rotating machines, provide individual rotor speeds if different from overall machine speed. For reciprocating machines, provide primary and secondary speeds.

<u>Outline Drawing:</u> A layout of the machine provides a basic arrangement, dimensions to the cylinders, shaft, and other components.

<u>Anchor Bolts and Layout:</u> Provide bolt materials and sizes to verify bolt forces and pier dimensions. Whenever possible, place anchor bolts for large compressors 300 mm minimum from bolt centerline to face of concrete.

<u>Pier Layout:</u> This layout provides all dimensions and elevations for machine support. If not provided, pier dimensions may be determined using a minimum of 4 inches (100 mm) from the machine base to the face of the pier.

<u>Jacking Post Locations and Details:</u> This provides installation methods to properly align machine components.

6.2.2 Machine Forces

<u>Rotating Machines:</u> This data can consist of dynamic force magnitudes, or rotor weights and eccentricities. Refer to Attachment 01. Sometimes a shaft machine tolerance is indicated. If shaft

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machine tolerance or offset eccentricities are not provided, inform the supplier of the criterion that is being used for dynamic response checks of the support. Note that some machines, such as centrifuges, have unbalanced forces that are not based solely on the rotor weight and eccentricity. Such machines have unique acceptable limits on vibration of the support.

Reciprocating Machines: Primary and secondary machine forces are provided for reciprocating machines. Refer to Attachment 02.

Location of Dynamic Forces: This indicates the location at which reciprocating forces or rotor eccentricities are applied. Normally, this is a location along the shaft. Sometimes force values are provided at the anchor bolt locations, which can then be resolved at the center of shaft location.

Static Operating Forces: These include rotor torque, thermal expansion, and gas pressure.

Machine Upset Forces: These can include short circuit and blade loss.

6.2.3 Machine Component Data

Component Weights: The motor and compressor/pump are often broken out as separate parts.

Component Center of Gravity: Indicate a center of gravity for each machine component, usually on the machine outline drawing. Sometimes, one center of gravity is provided for the entire machine.

Component Mass Moments of Inertia: If provided, mass moments about each axis are used in the vibration analysis. If not provided, mass moments can be estimated based on the size and dimensions of the machine, or it could be judged as minor enough to be neglected.

6.2.4 Additional Machine Criteria

Plot Location: This determines how much area the foundation mat can cover without interfering with other structures and equipment. Reciprocating machines with an odd number of cylinders require larger mats because the combined cylinder unbalanced forces are larger. Consider separation of compressor foundations from vibration sensitive equipment or structures. Work with Piping to resolve any problems as early as possible.

Grout Requirements: The machine supplier is the most important source for grout requirements. Carefully examine supplier criteria versus client or construction preferences.

Bolt Tensioning Criteria: Insufficient bolt tensioning can easily cause excessive movement between the machine and base. Set anchor bolt tensioning in accordance with the machine supplier's instructions and note on our construction drawings.

Allowable Amplitudes: If provided, use supplier criteria instead of the criteria provided in this guideline. If not provided, inform the supplier of the criterion that is being used for dynamic checks of the foundation.

Measurement Point Locations: Amplitude measurement locations are designated to verify either of two objectives.

- Machine Reliability: Locations are typically at the bearings or bearing supports, or at the machine-foundation interface.
- Personnel Comfort: Design measurement point locations to be near the machine where operators would spend significant time (i.e., hours) while the machine is running. ACI 351.3R

indicates that many engineers do not consider human perception unless the machine is in the proximity of office areas.

6.2.5 Geotechnical Data

Soil Properties: Provide Unit Weight, Shear Modulus, and Poisson's Ratio values for each layer. Separate values may be needed for backfill if embedment is to be included. Representative values, for comparison purposes only, are as follows:

Table 1: Comparison Soil Properties

Description:	Unit Weight: k/ft^3 (kN/m^3)	Shear Modulus: k/ft^2 (kN/m^2)	Poisson's Ratio:
Granite	0.15 to 0.16 (23.6 to 25.1)	5.76e5 to 8.65e5 (2.76e7 to 4.14e7)	0.15 to 0.2
Limestone	0.145 to 0.155 (22.8 to 24.3)	2.88e5 to 7.21e5 (1.38e7 to 3.45e7)	0.16 to 0.22
Sandstone	0.145 to 0.155 (22.8 to 24.3)	1.44e5 to 5.76e5 (6.89e6 to 2.76e7)	0.17 to 0.24
Dense Sand	0.115 to 0.14 (18.1 to 22.0)	1,440 to 2,740 (68.900 to 131,000)	0.28 to 0.34
Medium Sand	0.11 to 0.13 (17.3 to 20.4)	1,150 to 2,150 (55,100 to 103,000)	0.30 to 0.36
Loose Sand	0.95 to 0.125 (14.9 to 19.6)	710 to 1,590 (34,000 to 76,100)	0.32 to 0.38
Hard Clay	0.125 to 0.145 (19.6 to 22.8)	1,590 to 2,150 (76,100 to 103,000)	0.38 to 0.41
Medium Clay	0.115 to 0.135 (18.1 to 21.2)	1,000 to 1,590 (47,900 to 76,100)	0.41 to 0.44
Soft Clay	0.1 to 0.125 (15.7 to 19.6)	418 to 1,000 (20,000 to 46,700)	0.44 to 0.47

Some vibratory analysis software programs require the shear wave velocity instead of the shear modulus. In addition, a coefficient of variation (COV) in the shear modulus value is used to account for possible variations in soil properties from one location to another. ASCE 136 indicates a value of 0.5 can be used for COV if the shear wave velocity has not been measured, and 0.3 if it has been measured. Applicable formulas are:

$$G_S = (V_S)^2 (U_S) / g \qquad \text{(Equation 1a)}$$

$$G_{LB} = G_S / (1 + COV) \qquad \text{(Equation 1b)}$$

$$G_{UB} = G_S * (1 + COV) \qquad \text{(Equation 1c)}$$

$$V_S = \text{sqrt} [G_S\ g / U_s] \qquad \text{(Equation 1d)}$$

$$V_{LB} = \text{sqrt} [G_{LB}\ g / U_s] \qquad \text{(Equation 1e)}$$

$$V_{UB} = \text{sqrt} [G_{UB}\ g / U_s] \qquad \text{(Equation 1f)}$$

where,

G_S = nominal soil shear modulus, k/ft^2 (kN/m^2)

G_{LB} ,G_{UB} = lower and upper bound shear modulus, k/ft^2 (kN/m^2)

V_S = nominal soil shear wave velocity, ft/s (m/s)

V_{LB} ,V_{UB} = lower and upper bound shear wave velocity, ft/s (m/s)

U_S = soil unit weight, k/ft^3 (kN/m^3)

g = gravitational acceleration, ft/s^2 (m/s^2)

Soil Material Damping: Also called internal or hysteric damping, this is the energy loss within the soil due to interparticle friction. The damping, expressed as a ratio, depends on vibration magnitude, with a higher value for large seismic movements and a lower value for machine vibrations. Though damping can be measured in a laboratory test, a damping ratio of 2% to 3% (ASCE 136) is typically assumed for machine design.

Allowable Soil Bearing Pressure / Pile Capacity: This is a standard component of every geotechnical investigation.

Construction Recommendations: Also a standard component; recommendations concerning excavation, backfill compaction, and vibration isolation from other foundations may be provided.

Embedment Recommendations: Provide embedment recommendations to confirm the characteristics of backfill materials.

Soil Layer Evaluation: Some geotechnical consultants are able to compute soil stiffness and damping values. This might be considered if highly unusual soil conditions are encountered. Geotechnical consultants can also supply an analysis reference system file (i.e., Dyna 6) containing recommended geotechnical property input. This file can then be edited for each specific machine situation.

6.2.6 Material Properties

Pile Material: This is important in determining stiffness properties for soil-pile interaction. Typical material properties follow:

Table 2: Typical Pile Material Properties

Description	Unit Weight k/ft^3 (kN/m^3)	Modulus of Elasticity k/ft^2 (kN/m^2)	Shear Modulus k/ft^2 (kN/m^2)
Wood (Douglas Fir)	0.036 (5.66)	215,000 (10,300,000)	13,500 (648,000)
Wood (Southern Pine)	0.040 (6.28)	215,000 (10,300,000)	13,500 (648,000)
Wood (Red Oak)	0.044 (6.91)	180,000 (8,620,000)	11,200 (538,000)
Wood (Red Pine)	0.040 (6.28)	184,000 (8,830,000)	11,500 (552,000)
Concrete (3 ksi)	0.015 (23.6)	449,000 (21,500,000)	282,000 (13,500,000)

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Description	Unit Weight k/ft^3 (kN/m^3)	Modulus of Elasticity k/ft^2 (kN/m^2)	Shear Modulus k/ft^2 (kN/m^2)
Concrete (4 ksi)	0.015 (23.6)	249,000 (24,900,000)	324,000 (15,500,000)
Steel (A36)	0.490 (77.0)	4,170,000 (200,000,000)	2,980,000 (14,300,000)

Pile Material Damping: In lieu of measurements, a damping ratio between 0% and 2.5% (Campbell 2018) may be used.

Concrete Strength and Reinforcing Grade: Standard material strengths for foundations are normally used, grade 60 (grade 420) reinforcing and f'C of 3 to 4 ksi (20 to 28 MPa) concrete. A higher strength of concrete may be advisable because tensile strength is important in preventing cracking where oil leaks are possible.

6.3 Initial Trial Foundation Size

General rules of thumb are used for an initial trial foundation size. The analysis determines acceptability.

6.3.1 Soil Supported Mat

Design the mat width perpendicular to the shaft to be 1.5 times the height from the shaft to the bottom of the mat (0.75 times for centrifugal machines). Second, design the mat length parallel to the shaft to be 2 ft (0.6 m) longer than the length of the pier. The location of the mat relative to the pier may have to be adjusted in order to reduce foundation eccentricities. Refer to basic criteria for alignment offsets.

6.3.2 Pile Layout and Pile Cap Dimensions

For the first trial, this is largely an educated guess based on past experience. As the trial layout and analysis cycle is repeated, a basis for better layout dimensions becomes apparent from analysis results.

The pile cap is sized to accommodate the pile pattern, with normal minimum center of pier to edge of pile cap distances. The location of the piles and pile cap relative to the pier may have to be adjusted to reduce foundation eccentricities. Refer Section 6.4.5 for criteria.

6.3.3 Mass Ratio

Mass ratios are used periodically as a traditional gage of how much foundation mass is being provided relative to the machine. The following guideline (PIP STC01015) may be used as an indicator of foundation performance:

for reciprocating machines, $MR \geq 5$ (Equation 2)

for centrifugal machines, $MR \geq 3$

where,

MR = Weight of foundation divided by weight of machine

Foundations with a low mass ratio do not necessarily need to be resized. Instead, carefully examine the dynamic analysis input and results for use.

6.4 Analytical Model

6.4.1 Mat or Pile Cap Rigidity

Check the mat thickness for rigidity because the stiffness and damping determination and closed-form simplified analysis procedures implement a common assumption that the foundation and machine are rigid relative to the soil. If the mat and/or machine are not rigid, then Guideline 000.215.1236 applies. Note that there is a tradeoff between using a thicker mat to meet rigidity requirements versus a greater effort to analyze a thinner flexible mat. Verify mat rigidity manually because the computer software does not confirm this assumption.

Confirm the mat thickness is not less than 0.6 m. In addition, apply one of the following alternative equations:

1. The following formula is derived from beam on elastic foundation theory using a flexural deflection pattern that approximates essentially rigid mat behavior:

$$L_Z \geq 0.02\left[\frac{G_S\, L^4}{b}\right]^{1/3} \qquad \text{(Equation 3, imperial)}$$

$$L_Z \geq 0.01\left[\frac{G_S\, L^4}{b}\right]^{1/3} \qquad \text{(Equation 3, metric)}$$

   where,

   L_z = thickness of foundation mat, ft (m)

   G_S = soil shear modulus, lb/in^2 (kN/m^2)

   L = mat cantilever dimension in rocking direction, ft (m)

   b = width of mat perpendicular to rocking direction, ft (m)

2. The following equation is commonly specified in many client specifications:

$$L_Z \geq 0.6 + \frac{L}{30} \qquad \text{(Equation 4, metric)}$$

   where,

   L = longest mat dimension, ft (m)

3. The following formula from Gazetas is based on a published study of the subject:

$$\left(\frac{E_c}{E_s}\right)\left(\frac{L_Z}{d}\right)^3 \geq 1.0 \qquad \text{(Equation 5)}$$

   where,

   E_c = Modulus of elasticity for concrete, k/ft^2 (kN/m^2)

   E_s = Modulus of elasticity for soil, k/ft^2 (kN/m^2)

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d = Mat cantilever beyond face of pier, in either direction, ft (m)

6.4.2 Machine Rigidity

From a global analysis perspective, machines supported on rigid mats are considered as rigid (or at least held together by the rigid mat). Where machine rigidity is considered as globally significant, guideline 000.215.1236 applies. Section 6.8.2 provides further information for pier design for machines that rely on the mat rigidity.

6.4.3 Soil Pressure

Keep static soil pressure low to maintain linear soil behaviour. The following from PIP STC01015 is generally considered acceptable:

$SB_{net} \leq 0.5\ (SB_{allow})$ (Equation 6)

where,

SB_{allow} = net static allowable soil pressure, k/ft^2 (kN/m^2)

SB_{net} = maximum net soil contact pressure, k/ft^2 (kN/m^2)

If this criterion is not met, try a larger mat.

6.4.4 Pile Load

Keep pile loads low to maintain elastic soil behavior. The following from PIP STC01015 is generally considered acceptable:

$PL \leq 0.5$ (allowable pile load) (Equation 7)

where,

PL = pile load, kip (kN)

If this criterion is not met, try additional piles or a larger pattern.

6.4.5 Center of Gravity Offsets

Keep offset values from PIP STC01015 to a minimum to reduce differential settlement.

$OS_x \leq 5$ percent (Equation 8)

$OS_y \leq 5$ percent

where,

OS_x, OS_y = distance between overall CG and center of mat divided by mat horizontal dimension in the direction of the offset. For pile groups, the group centroid and distance to outermost pile is used.

If these criteria are not met, relocate the mat or piles, with respect to the machine as appropriate.

6.4.6 Mass

Mass and mass moments of inertia are used to compute the translational and rotational dynamic inertia forces. The calculation is usually performed by a computer program.

6.4.7 Stiffness & Damping

Stiffness and damping result from the elastic nature of the supporting soil and piles. The magnitudes typically vary with frequency, though simpler constant formulas can be used under certain circumstances. The use of any method of calculating stiffness and damping may be used provided that the method is recognized, documented, and verified with respect to more comprehensive procedures or in-place measurements (refer to Attachment 03).

6.4.8 Embedment

Increased stiffness and damping results from soil along the vertical sides of a mat or pile cap embedded into the ground. Design embedment carefully; do not just select embedment as an option in a computer program. Use compacted cohesionless soils (sand)to provide full embedment. Cohesive soils (clay) can easily shrink away from the sides of a foundation mat. If isolation details are not provided, analyze foundations in cohesive soils for the extremes of full embedment and no embedment.

Use a conservative value of the embedded depth in the analysis. Normally, 2/3 of the total embedded height is used.

6.4.9 Water Table

Unless otherwise provided, evaluate the effect of soil properties below a water table as follows,

Shear Modulus – Use the shear modulus, as measured.

Unit Weight – Use the dry unit weight, as measured.

Poisson's Ratio – Use a value at or near 0.5.

6.5 Frequency Analysis

A frequency response analysis uses the machine's forcing function, machine-foundation inertia, and stiffness to calculate natural frequencies. For the rigid-block type foundation assumed in this guideline, there are 6 degrees of freedom and 6 natural frequencies.

This calculation is usually performed by a computer program. If a frequency independent impedance approach is used, then the natural frequencies can be calculated using explicit equations. If a frequency-dependent impedance approach is used, then natural frequencies can be determined by the response plot over a range of frequencies. Using reduced damping helps natural frequencies appear as sharp peaks.

6.6 Forced Response Analysis

A forced response analysis uses the machine's forcing function, machine-foundation inertia, damping, and stiffness to calculate a dynamic response. Responses are usually determined in terms of displacements or velocities.

6.6.1 Coastdown Effects

Centrifugal machines with high operating speeds are usually started and shut down slowly enough to be subject to temporary periods of resonance. If any of the 6 vibration modes have natural frequencies less than the machine speed, investigate coastdown amplitudes. Natural frequencies are used as coastdown machine speeds and reduced dynamic forces are calculated using the following formula:

$$F_C = (F_O)(CS / \Omega)2 \quad \text{(Equation 9)}$$

where,

F_O = peak force at machine speed, kip (kN)

F_C = peak force at coastdown speed, kip (kN)

CS = coastdown speed, rad/s

Ω = machine circular frequency, rad/s

The forced response analysis is repeated for each applicable coastdown frequency.

6.6.2 Multiple Machines on a Common Foundation

For multiple machines on a common mat, the vibration amplitude calculations are based on simultaneous operation of the maximum number of machines representing the design condition Spare and standby machines are assumed stopped. For those machines operating, all unbalance forces and moments are assumed to be acting in whichever reasonable combination produces the highest amplitudes.

For relatively small machines spaced close together, the mat may be thick enough to be considered rigid. This evaluation needs to be carefully studied and confirmed. For other cases, the mat is flexible, and this guideline does not apply. Refer instead to Guideline 000.215.1236.

6.6.3 Multiple Rotors

Multiple rotors can consist of a compressor and motor rotor. Large compressors can consist of multiple casings, each with their own rotor. Combining rotor dynamic forces becomes complicated because the orientation of rotor eccentricities cannot be predicted. ASCE 136 includes further discussion of this subject. There are four approaches to determining and combining resultant magnitudes.

- If the phase relationship between the rotors is somehow known, then both rotor forces can be combined into a single analysis. This is the most accurate approach.
- Use one or several assumed phase relationships. A separate analysis would be needed for each relationship. Traditional specification criteria often describe in-phase (aligned), out-of-phase (opposed) cases. The drawbacks are that peak dynamic forces are almost never aligned with peak deformations, and additional rotors adds complexity.

  Note that the common use of phase angles equal to zero corresponds to the in-phase (aligned) case.
- Combine resultant magnitudes from each rotor using a simple summation. A separate analysis is needed for each rotor. This is an upper-bound solution that could be unlikely to occur.

The choice of which option to implement is judgmental. Considerations can include how the analysis computes deformations at each resultant (vector addition or simple summation), software capability, and experience.

6.7 Dynamic Acceptance Criteria

6.7.1 Frequency

To further prevent the possibility of excessive amplitudes, check that machine-foundation natural frequencies (PIP STC01015 and API RP 686) are outside the resonant range as follows:

$FR \leq 0.8$ or $FR \geq 1.2$ (Equation 10)

where,

FR = machine speed divided by natural frequency

Note that client or manufacturers criteria can vary. Some specifications may use different range values (such as 0.7 and 1.4). Other specifications use a resonant frequency ratio (machine speed divided by resonant frequency). For low to moderate amounts of damping, the difference is typically negligible.

Sometimes, the frequency ratio criteria is difficult to achieve. For reciprocating machines, 6 primary and 6 secondary frequency ratios need to fall outside the resonant range.

If a reasonable foundation size does not result in permissible frequency ratios, then an analysis can be run using a further reduced level of damping (recommend half of what is being used otherwise). The reduced level of damping and the result are to be agreed upon by the client, the geotechnical engineer, and the machine supplier.

Trying to live within the resonant range carries with it a certain level of risk. On one hand, computed damping levels are usually high enough to preclude significant resonant amplitudes. On the other hand, machines have failed before due to resonance, especially under unusual soil conditions.

6.7.2 Forced Response

Vibration response limits, expressed in terms of amplitude, velocity, or acceleration, depend on considerations of machine performance and human comfort. Measurement locations are described in Section 6.2.4. If the permissible response is provided in terms of peak velocity or acceleration, then Equation 11 can be used.

$V_{PEAK} = (\Delta_{PEAK})(\Omega)$ (Equation 11a)

$A_{PEAK} = (\Delta_{PEAK})(\Omega)^2$ (Equation 11b)

If a root-mean-square value permissible response is provided, then Equation 12 can be used:

$\Delta_{RMS} = ((\Delta_{PEAK})(0.707)$ (Equation 12a)

$V_{RMS} = (V_{PEAK})(0.707)$ (Equation 12b)

$A_{RMS} = (A_{PEAK})(0.707)$ (Equation 12c)

where,

Δ_{PEAK} = peak amplitude computed by the analysis, ft (m)

Δ_{RMS} = root mean square deformation, ft (m)

A_{PEAK} = peak acceleration, ft/s^2 (m/s^2)

A_{RMS} = root mean square acceleration, ft/s^2 (m/s^2)

V_{PEAK} = peak velocity, ft/s (m/s)

V_{RMS} = root mean square velocity, ft/s (m/s)

Ω = machine circular frequency, rad/s

For machine performance, keep vibration amplitudes low to avoid unscheduled shutdowns for maintenance. If machine supplier or client requirements are not provided, the following peak velocity limits from PIP STC01015 can be used:

rotating machines	0.12 inch/s	(3.0 mm/s)
reciprocating machines	0.15 inch/s	(3.8 mm/s)

For personnel comfort, a limiting value depends on duration of exposure, configuration/orientation, and machine frequency. For continuous exposure, a velocity limit of 0.09 in/s (2.4 mm/s) may be used. This is equivalent to the boundary between “easily noticeable” and “troublesome to persons” in Figure 1.

If continuous exposure criterium becomes problematic, a limited exposure duration can be considered based on ISO 2631-1. Refer to Attachment 07.

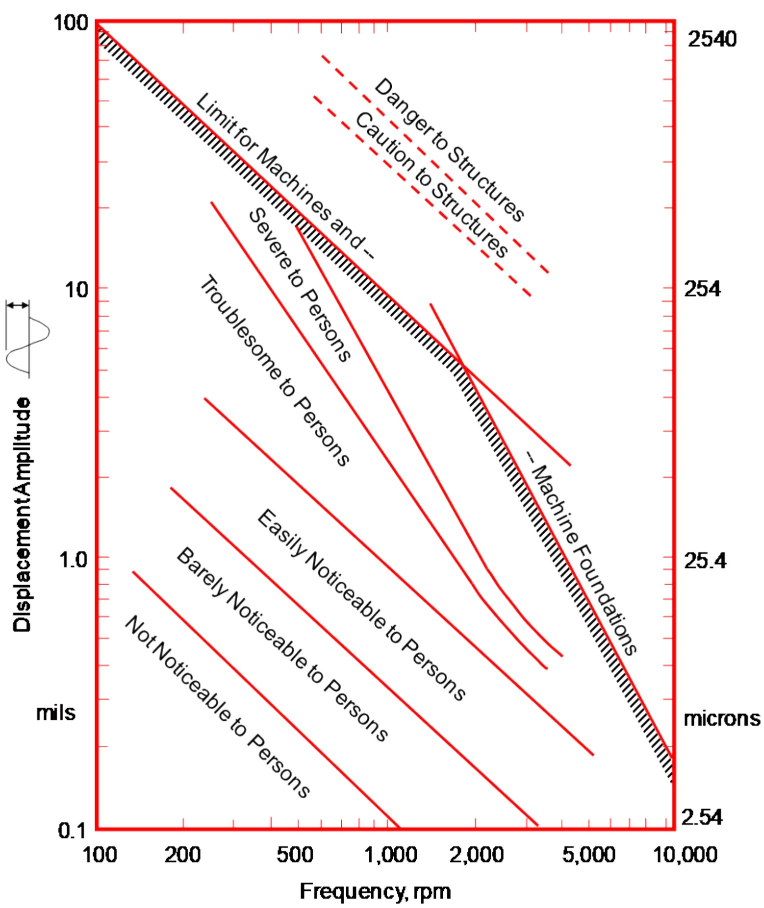


Figure 1: Reiher-Meister Chart (ACI 351-3R)

If response criteria are not met, revise the foundation size and embedment criteria. The vibration analysis is performed again until response criteria are met. In unusual cases, acceptable response values may not be achievable with a reasonable foundation size. Response criteria may then need to be evaluated again with the client and supplier until a mutually agreeable solution is obtained.

6.8 Additional Design Considerations

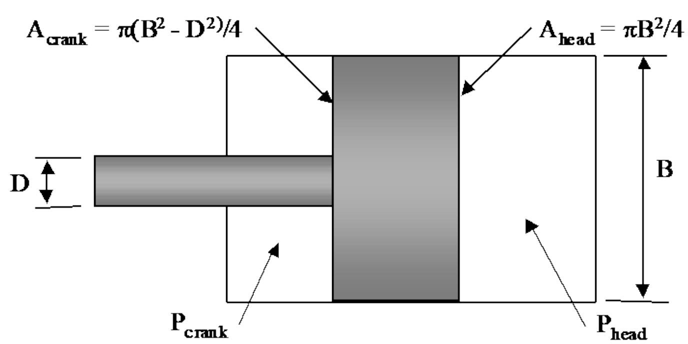
The vibration analysis is usually of primary concern in designing foundations for vibrating machines. However, several other considerations need to be addressed in completing the design.

6.8.1 Reinforcing Design

Reinforcing is used to eliminate or reduce concrete cracking. As a minimum, provide mats with temperature reinforcing (top and bottom bars > 0.0018 times gross area), and provide piers with a cage of #5 at 300 mm each way. For foundation mats thicker than 4 ft (1.2 m), consider mass concrete provisions from ACI 351.3R, section 7.3.2,.

6.8.2 Piers for Reciprocating Machines

Consider gas pressure forces in the design of reciprocating machine piers. The magnitudes of these forces depend on the rigidity of the machine frame; the more rigid the frame, the lower the force transmitted to the pier. For large machines, frames are rarely rigid enough to cancel these forces. In lieu of a detailed finite element analysis of the machine frame, the following equation from ACI 351-3R may be used to compute pier forces:



$$F_{fdn} = [(P_{head})(A_{head}) - (P_{crank})(A_{crank})]\ F_{cr} / F_{red} \qquad \text{(Equation 13)}$$

where,

A_{head} = Area of piston head, in^2 (mm^2)

A_{crank}= Area of piston head on crank side, in^2 (mm^2)

B = Cylinder bore diameter, in (mm)

D = Rod diameter, in (mm)

P_{head} = Instantaneous head pressure, k/in^2 (kN/mm^2)

P_{crank} = Instantaneous crank pressure, k/in^2 (kN/mm^2)

F_{cr} = Correction factor (1.15 to 1.2)

F_{fdn} = Lateral force on foundation, tributary to cylinder, kip (kN)

F_{red} = Reduction factor, use 2.0 unless better data is available

The instantaneous head and crank pressures can be taken from the maximum and minimum suction and discharge pressures obtained from the mechanical engineer.

The anchor bolts tributary to the cylinder are used to resist F_{fdn}.

6.8.3 Grouting Methods

Grout requirements are determined in accordance with the project grout specification, the compressor supplier's requirements, and the grout supplier's criteria. Conflicts with supplier, client, construction, or design requirements need to be resolved and a mutual agreement obtained. Normally, epoxy grout is used for compressor foundations. Steel chocks or epoxy chocks may also be used to thermally isolate the machine from the supporting foundation. Anchor bolt sleeves are normally not be filled with grout; consult machine supplier drawings on this matter.

6.8.4 Anchor Bolt Design

Bolt requirements are provided by the machine supplier. This always includes the number and diameter of anchor bolts. Bolt material and post-tensioning requirements may or may not be included. If not, then these requirements are selected based on the applied forces and coordinated with the supplier. In addition, because slippage cannot be tolerated, provide lateral resistance through sufficient clamping force between machine and foundation.

$$T_{min} = F / \mu - W_r \qquad \text{(Equation 14)}$$

where,

T_{min} = minimum tensile clamping force per bolt, kip (N)

F = applied lateral force per bolt, kip (N)

μ = friction coefficient (use 0.15 based on oily steel on cast iron)

W_r = applicable machine weight (could conservatively use zero), kip (N)

For centrifugal machines, the applied force per bolt can be determined from the overall unbalanced dynamic force. For reciprocating machines, the applied force per bolt is determined from the gas pressure forces on each cylinder.

If post tensioning is required, provide bolt tightening instructions to the field with due consideration for tension measurement procedures and for retensioning at a later time to minimize relaxation.

Provide concrete embedment in accordance with PIP STE05121. Because the level of actual post tensioning in the field is difficult to control, and because cracking the pier due to overtightened bolts is to be avoided, provide bolts with a ductile embedment. Follow the ductile criteria for embedment for tension and lateral bursting embedment without the use of additional reinforcing. Avoid additional embedment reinforcing because it requires cracking to become effective. Lateral forces are transmitted between the machine and the foundation by shear keys or by post-tensioning bolts and friction between the machine base and the foundation, not by shear through the anchor bolts. If not dictated by the machine supplier, place anchor bolts for large compressors 12 in (300 mm) minimum from bolt centerline to face of concrete.

6.8.5 Elevated Pipe Anchors

Dynamic forces from connected piping are rarely significant enough to be used for the mat design.

Truly rigid anchor points to resist mechanical or pulsation loads are nearly impossible to provide by tall, slender piers without the addition of congested, unsightly bracing. Thus, it is inadvisable to provide elevated pipe anchors. However, if required, design such pipe supports attached to the foundation pier or mat so that their natural frequencies of horizontal vibration are either less than 0.5 times the compressor primary frequency or greater than 1.5 times the compressor secondary frequency.

7.0 RESOURCES

Not Used

8.0 REFERENCES

Guidelines

Guideline 000.215.1227	Pump Foundation
Guideline 000.215.1236	Non-Rigid Support for Vibratory Equipment

Specifications

Specification 000.215.00920	Structural Data For Mechanical Equipment

Non-Fluor Documents

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9.0 ATTACHMENTS / ADDENDA

Attachments

Attachment 01	Rotating Machine Forces
Attachment 02	Reciprocating Machine Forces
Attachment 03	Impedance from the Ground
Attachment 04	Notes on Using Dyna 6
Attachment 05	Sample Design: Reciprocating Machine Foundation
Attachment 06	Sample Design: Centrifugal Machine Foundation
Attachment 07	Spreadsheet Utilities (not attached, included in zip file)

Addenda

Not Used

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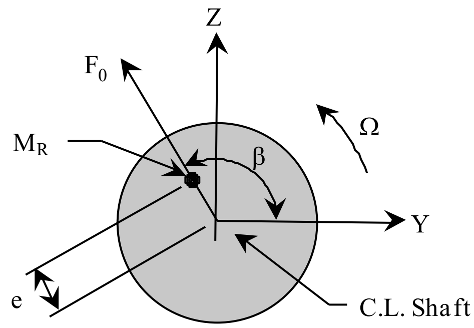
RIGID FOUNDATIONS FOR VIBRATORY EQUIPMENT

Rotating Machine Forces

Inertia forces are caused by the imperfect balancing of a rotor due to manufacture or due to usage. Forces from this imbalance are calculated as follows:



$F_0 = (M_R)(e)(\Omega)^2 / 12$ (A1-1 imperial)

$F_0 = (M_R)(e)(\Omega)^2 / 1000$ (A1-1 metric)

$F_X(t) = 0$ (A1-2a)
$F_Y(t) = F_0 \cos [(\Omega)(t) + \beta]$ (A1-2b)
$F_Z(t) = F_0 \sin [(\Omega)(t) + \beta]$ (A1-2c)

where,

F_0 = inertia force of unbalanced rotor, lb (N)
M_R = rotor mass, slug (kg)
e = eccentricity of rotor, in (mm)
Ω = machine frequency (rad/s)
t = time (s)
F_X, F_Y, F_Z = forces in each coordinate direction, lb (N)
β = crank angle (rad)

The preferred choice is to obtain the unbalanced force (F_0) directly from the machine supplier. This is nearly always supplied for the largest machines, but rarely supplied for small machines. If not, then Equations A1-1 and A1-2 may be applied.

The machine weight and frequency (Ω) are always supplied. If the rotor mass (M_R) is not supplied, then it may need to be estimated as a percentage of machine weight. The crank angle (β) becomes relevant only when adding the effects of more than one machine, such as a pump and motor. Refer to guideline section 6.5.3.

The eccentricity (e) is never directly provided, but it can be determined from the designated quality grade of the rotor. The quality grade is associated with a value (Q_G) representing eccentricity multiplied by frequency. Values of the quality grade and associated Q_G are shown in Table A1. This, along with the rotor mass enables the unbalanced force to be expressed as

$F_0 = (M_R)(Q_G)(\Omega)\ S_F / 12$ (A1-3 imperial)

$F_0 = (M_R)(Q_G)(\Omega)\ S_F / 1000$ (A1-3 metric)

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Rotating Machine Forces

The service factor (S_F) is included as an allowance for long-term degradation. ACI 351.3R states that this value is usually 2 or more. Alternatively, ASCE 136 mentions the use of one quality grade higher.

If the quality grade is not available, Mechanical can assist with a determination, or Table A1 descriptions can be applied. Alternatively, the initial test eccentricity of the rotor can be used along with Equation A1-1. Vibration sensors may also be examined to give an indication of an upper limit of shaft misalignment. If such sensors are used, then an eccentricity may be computed based on the shaft vibration level that causes the machine to be shut down. Apply judgment on an individual case basis with the assistance of the Mechanical Engineer.

The next choice is to use a common rule-of-thumb formula derived from Equation A1-3 and based on a Q_G value of 0.62 in/s (16 mm/s).

$$\text{force} = \frac{(\text{rotor weight})(\text{rotor speed, rpm})}{6{,}000} \tag{A1-4}$$

Whichever formula is selected in lieu of supplier information, review the design criteria with the machine supplier for acceptance.

Note that the quality grade for the machine being purchased may not correspond to the generic descriptions in this table. It is possible that pump rotors are G 2.5 or G1. Also, the motor rotor quality may be better than the pump rotor quality. This can make a significant difference to the resulting vibration amplitudes.

Table A1: Rotor Quality Grade and Q_G (from ISO 21940-11)

Generic Description:	Quality Grade:	Q_G (mm/s)	Q_G (in/s)
Crushing Machines	G 16	16	0.63
General Machinery, fans, gears, pumps	G 6.3	6.3	0.25
Compressors, gas/steam turbines	G 2.5	2.5	0.01
Grinding machinery drives	G 1	1.0	0.04

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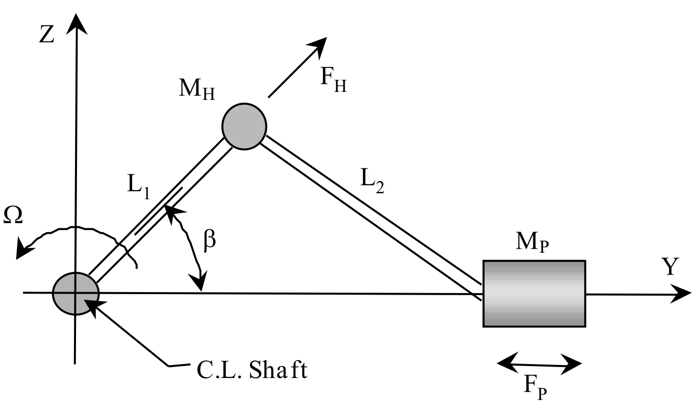
Reciprocating Machine Forces

Two types of forces act within a reciprocating compressor: inertial forces and gas forces.

For the dynamic analysis of a mat, gas forces tend to cancel each other within the machine frame and within the foundation pier. Thus, only inertia forces need be considered. However, for machine anchorage and pier design purposes, the magnitude of gas forces transmitted into the pier is very important and is dependent on machine rigidity. To obtain reasonably accurate anchorage design forces, the rigidity of the machine must be realistically considered.

Inertia forces result from varying accelerations of rotating and reciprocating machine parts and act at the crankshaft bearings without any opposing reaction on the frame. The magnitude of inertia forces depends on speed, rotating inertia, and on reciprocating inertias of crosshead, piston rod, and piston as follows:



For one cylinder:

$$F_X(t) = 0 \quad \text{(A2-1a)}$$
$$F_Y(t) = (M_H + M_P)(L_1)(\Omega)^2 \cos[(\Omega)(t) + \beta] + (M_P / L_2)(L_1\,\Omega)^2 \cos 2\,[(\Omega)(t) + \beta] \quad \text{(A2-1b)}$$
$$F_Z(t) = (M_H)(L_1)(\Omega)^2 \sin[(\Omega)(t) + \beta] \quad \text{(A2-1c)}$$

Where:

M_H = hinge mass, slug (kg)
M_P = piston mass, slug (kg)
L_1 = length from shaft to hinge, ft (m)
L_2 = length from hinge to piston, ft (m)

Ω = machine circular frequency (rad/s)
t = time (s)
F_X, F_Y, F_Z = forces in each coordinate direction, lb (N)

F_P = inertia force from piston, lb (N)

F_H = inertia force from hinge, lb (N)
β = crank angle (rad)

The last term in Equation A2-1b results in a harmonic force operating at twice the machine frequency. This is the secondary force that is a part of reciprocating machine design.

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Reciprocating Machine Forces

For multiple cylinders, the individual piston forces are combined using the appropriate distances between cylinders and the appropriate crank angles. These combined forces are usually provided midway between the cylinders by the machine supplier. If phase angles are not provided by the supplier along with peak forces, then request them.

Gas forces result from the action of piston motion and valve action which generate time varying head and crank end pressures. The magnitude of gas forces depends on the differential between suction and discharge pressure, the area of the cylinder bore, rod area, pulsations, and external resistance. The near uniform pressure within each compressed volume makes the force on the cylinder a direct reaction to the force on the piston. Thus, gas forces act on the crankshaft with an equal and opposite reaction on the cylinder. Refer to ACI 351.3R for a more detailed description and equations.

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RIGID FOUNDATIONS FOR VIBRATORY EQUIPMENT

Impedance from the Ground

This attachment provides conceptual information on the calculation of impedance (stiffness and damping). The simplifying assumption of a rigid machine & foundation means that soil impedance is of fundamental importance because it is the sole source of stiffness and damping.

The academic topic of soil dynamic impedance ranges from low strain conditions (vibrating machinery) to high strain conditions (seismic). This attachment focuses on low strain conditions, harmonic loading, and Dyna version 6.1. References are listed for further information.

Soil impedance software is often perceived as a "black box" where input provided by a user is then mysteriously determined by the software. A basic knowledge and inherent feel for the relationship between input and results is what this attachment attempts to address.

CONTENTS

Impedance from the Ground

5) Nomenclature

6) References

Impedance from the Ground

1) Basic Concepts

1A) Equation of Motion – The basic dynamic equation of motion is,

$$M\,\ddot{y} + C\,\dot{y} + K\,y = F(t) \tag{1}$$

Where, M, C, and K are mass, damping, and stiffness respectively. Displacement, velocity, and acceleration are designated as y, $\dot{y}$, and $\ddot{y}$. F(t) is the applied dynamic force.

Mass is understood to be the mass of the foundation and machine. A "fictitious", separately computed, soil mass may be described in a few simplified methods to achieve an improved alignment with comprehensive theoretical results.

Although a time-history analysis could be used to implement Equation 1, a steady-state analysis is more efficiently applied for harmonic loading, especially for rigid block foundations. This involves a couple of substitutions into Equation 1.

$$F(t) = F_0 \cos(\Omega t + \theta_F) \tag{2}$$

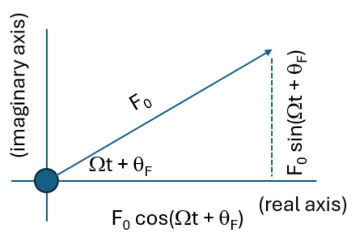

Figure 1 (Complex Dynamic Force)

In these equations, Ω is the forcing function frequency, F_0 is the peak value of harmonic force, θ_F is the force phase angle, and t is time. The following derivation uses the following mathematical equivalents,

$$e^{im} = \cos m + i \sin m \tag{3a}$$

$$\sin(m + n) = \sin(m)\cos(n) + \cos(m)\sin(n) \tag{3b}$$

$$\cos(m + n) = \cos(m)\cos(n) - \sin(m)\sin(n) \tag{3c}$$

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Impedance from the Ground

Equation 2 is expanded into its complex equivalent (refer to Figure 1). Though complex numbers, where i = $\sqrt{-1}$, are not encountered in a software's user interface, they are efficiently used in theoretical derivations and calculations.

$$F(t) = F_0 \{ \cos(\Omega t + \theta_F) + i \sin(\Omega t + \theta_F) \}$$

Observation: *If ever needed, Microsoft Excel includes built-in cell equations for handling complex numbers.*

Substituting Equations 3b and 3c,

$$F(t) = F_0 \{ [\cos(\Omega t) \cos(\theta_F) - \sin(\Omega t) \sin(\theta_F)] + i [\sin(\Omega t) \cos(\theta_F) + \cos(\Omega t) \sin(\theta_F)] \}$$

Re-arranging terms inside the brackets,

$$F(t) = F_0 \{ \cos(\theta_F) \cos(\Omega t) + i \cos(\theta_F) \sin(\Omega t) + i \sin(\theta_F) \cos(\Omega t) - \sin(\theta_F) \sin(\Omega t) \}$$

This is equal to,

$$F(t) = F_0 \{ \cos(\theta_F) + i \sin(\theta_F)\} \{ \cos(\Omega t) + i \sin(\Omega t) \}$$

Substituting Equation 3a,

$$F(t) = F_0 \{ \cos(\theta_F) + i \sin(\theta_F)\} \{ e^{i\Omega t} \}$$

Defining $J = F_0 \{ \cos(\theta_F) + i \sin(\theta_F) \}$

$$F(t) = J e^{i\Omega t} \qquad (4)$$

Note that if $\theta_F = 0$, then $J = F_0$

The deformation (y) can undergo a similar substitution.

$$y = Q e^{i\Omega t} \qquad (5a)$$
$$\dot{y} = Q i \Omega e^{i\Omega t} \qquad (5b)$$
$$\ddot{y} = -Q \Omega^2 e^{i\Omega t} \qquad (5c)$$

In these equations, Q is a complex deformation equal to Qr + i Qi.

Substitution of Equation 4 and Equation 5 into Equation 1, and elimination of common terms results in,

$$Q \{-M \Omega^2 + i C \Omega + K\} = J \qquad (6)$$

RIGID FOUNDATIONS FOR VIBRATORY EQUIPMENT

Impedance from the Ground

To design a foundation, the mass, damping, stiffness, and force are calculated. Equation 6 can be used to solve for Q.

$$Q = J / \{-M\,\Omega^2 + i\,C\,\Omega + K\} \qquad (7)$$

The deformation (y) can then be calculated using Equations 5a and 3,

$$y = Q\,e^{i\Omega t} = \Delta_0\,(\cos\theta_D + i\sin\theta_D)\,(\cos\Omega t + i\sin\Omega t) = \Delta_0\,\{\cos(\Omega t + \theta_D) + i\sin(\Omega t + \theta_D)\}$$

Where Δ_0 is the peak amplitude, equal to the modulus of Q, and θ_D is the deformation phase angle, equal to the argument of Q. The complex deformation is then determined. The real portion of the above equation is,

$$y = \Delta_0 \cos(\Omega t + \theta_D) \qquad (8)$$

where,

$$\Delta_0 = \sqrt{Qr^2 + Qi^2} \qquad (9a)$$

$$\theta_D = \arctan(Qi / Qr) \qquad (9b)$$

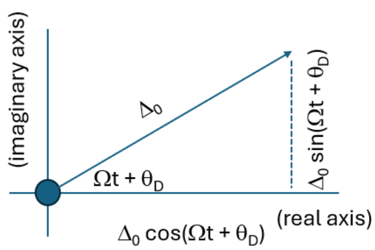


Figure 2 (Complex Dynamic Displacement)

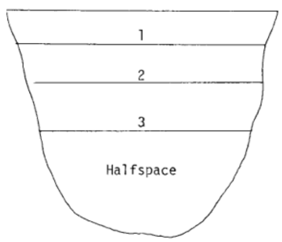
Alternatively, if the peak displacement and phase angle are measured for an actual soil, and with a known mass, applied force, and frequency, then Equation 6 can be used to determine stiffness and damping. This would be the process used to determine K and C from field test results. It can also be used to back check the mathematical accuracy of solved displacements.

<u>1B) Elastic Medium</u> – The use of a theoretical medium to represent the soil permits the development of mathematical models. Other than an expected scatter of measured data, there are differences between actual soil and an elastic medium. Such differences need to be considered (refer to Section 1I).

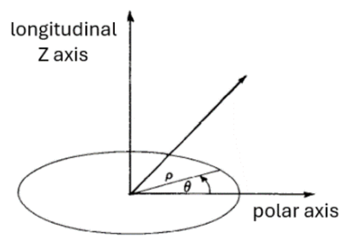
Impedance from the Ground

Observation: *The elastic medium concept has evolved along with advances in methodology and computer capabilities. Originally intended to represent an elastic half-space (uniform soil), soil layering was then considered. More recently, material nonlinearities evolved from a factored linear model factor to being directly considered.*

Elastic medium properties consist of one or more layers (Figure 3A), each with a shear wave velocity, shear modulus, unit weight, Poisson's ratio, and material damping.



A (Soil Layers)



B (Cylindrical Coordinates)

Figure 3 (Elastic Medium)

Observation: *Instead of cartesian coordinates (X, Y, Z), theoretical analytical derivations (simplified and comprehensive) often use cylindrical coordinates (ρ, θ, Z). As a result, circular footings are commonly described in impedance equations.*

For a rectangular foundation of dimensions L_X by L_Y, equivalent radius of a circular disk is based on equivalent cross-section properties,

Translation: $R_T = \sqrt{L_X\, L_Y/\pi}$ (10a)

Rocking about X axis: $R_X = \sqrt{L_X\, L_Y^2/3\,\pi}$ (10b)

Rocking about Y axis: $R_Y = \sqrt{L_X^2\, L_Y/3\,\pi}$ (10c)

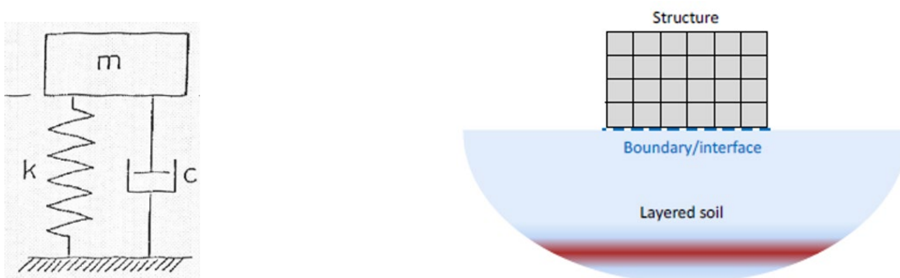
Torsion: $R_Z = \sqrt{L_X\, L_Y\left(L_X^2 + L_Y^2\right)/6\,\pi}$ (10d)

<u>1C) Lumped Parameters</u> – This is a simplification of the foundation and elastic medium intended to facilitate practical analysis methods. Basically, the soil impedance is computed as single

Impedance from the Ground

stiffness and damping values. The term was originally synonymous with frequency independent soil impedance.

The concept of lumped parameters has the effect of separating the soil impedance calculation from the foundation analysis. For soil structure interaction, the soil may be determined separately or combined with the supported structure.



A (Lumped Parameters) **B (Soil Structure Interaction)**

Figure 4 (Parameter Comparison)

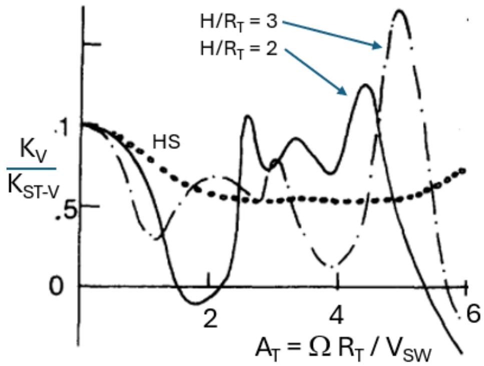
Lumped parameters (Figure 4A) are mainly used for rigid block foundations. Lumped parameters could also be used for tabletop foundations where the soil impedance is applied at several discrete mat locations (refer to Fluor Guideline 000.215.1236).

Soil Structure Interaction (Figure 4B) is used in comprehensive solutions involving non-rigid foundations. The interface between soil and structure consists of a series of nodes, each with direct and cross impedance values.

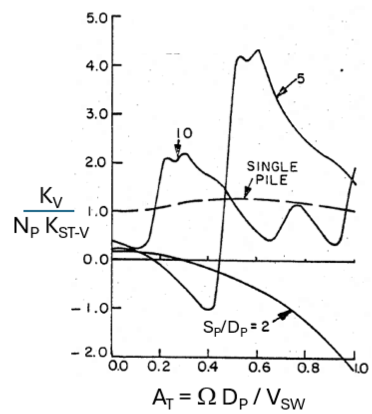
<u>1D) Frequency Dependence</u> – Soil and pile impedance is dependent on the frequency of the forcing function. This frequency variation is indicated in comprehensive calculations based on an elastic medium and in actual field measurements.

The extent of this frequency variation depends on several factors including: the mode of vibration, soil or pile support, pile group configuration, soil layering, and soil properties.

Impedance from the Ground



A (Soil Stiffness)



B (Piles Stiffness)

Figure 5 (Frequency Dependent Impedance)

For soil support, Figure 5A indicates the vertical stiffness for a half-space (labeled HS), and two stratum conditions (labeled H/R_T) where H is stratum height and R_T is mat equivalent radius. Stiffness frequency variations are due to the reflection of stress waves from the stratum-bedrock boundary.

For pile support, Figure 5B indicates the vertical stiffness for a single pile and for several 2-pile S_P/D_P spacing situations where S_P is pile spacing and D_P is pile diameter.

Figure 6 illustrates how pile group frequency dependence works. Indicated is a snapshot in time of the vertical deformation created by a source pile. If the deformation were put in timewise motion, the deformation curve would appear as a wave moving away from the source pile. The moving wave also diminishes with distance. A receiver pile at a fixed spacing from the source pile would be subjected to an applied soil deformation.

There is also a phasing effect because the receiver pile deformation peak may or may not align with the source pile's peak. The spacing of peak deformations would change with frequency. In Figure 6, the source pile and receiver pile B are in alignment and stiffness values would be additive. With receiver pile C at a downward position, stiffness values would counteract.

Impedance from the Ground

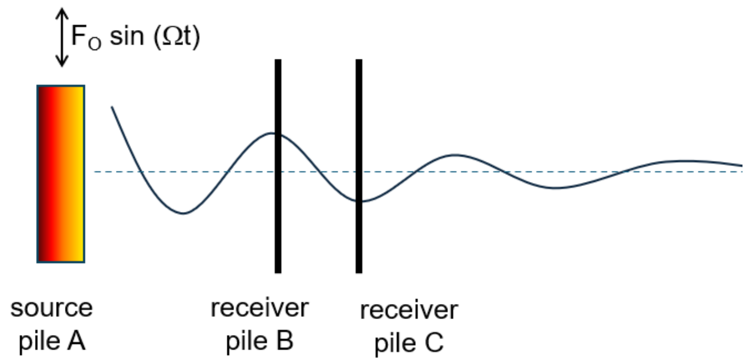
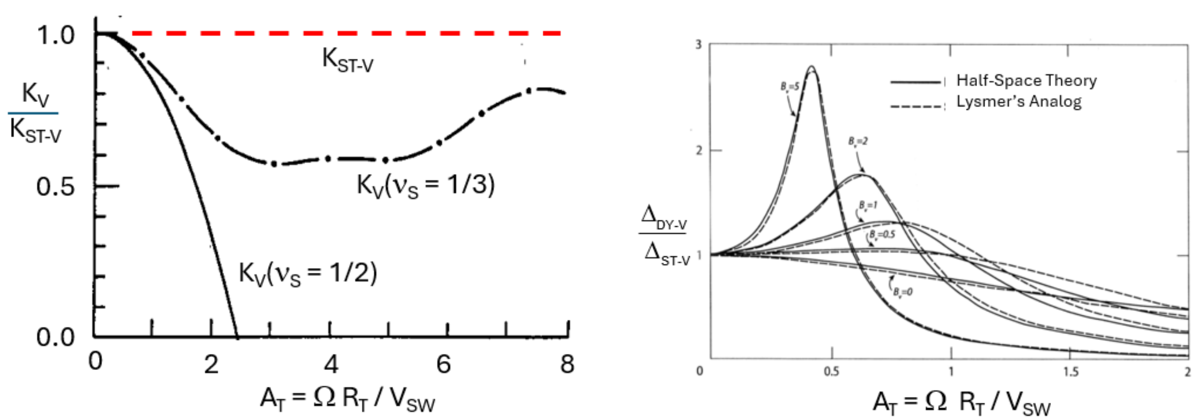


Figure 6 (Pile-to-Pile Interaction Effect)

1E) Lysmer's Analog – For uniform soil conditions, the calculated amplitude based on frequency dependent impedance is a close match to the calculated amplitude based on the soil's static stiffness in combination with a suitably determined level of damping.

Figure 7 illustrates the basic concept for vertical translation where K_V is the dynamic stiffness, $K_{ST\text{-}V}$ is the static stiffness, ν_S is Poisson's ratio, A_T is the nondimensional frequency, $\Delta_{DY\text{-}V}$ is the peak dynamic deflection, $\Delta_{ST\text{-}V}$ is the static deflection, and B_V is the mass ratio.



A (Stiffness Comparison) **B (Amplitude Comparison)**

Figure 7 (Vertical Soil Half-Space Comparison)

Observation: *This concept was developed in the 1960s, a time of manual calculations and slide rules. Prior to this development, theoretical solutions were determined using complicated continuum solutions – too tedious for common usage. Foundations for vibratory equipment had been determined largely from rules of thumb, mass ratios, and empirical methods. With this* *frequency independent approach, a "game changing" rational calculation method became feasible. This rational approach also led to a greater manual calculation effort.*

The frequency independent approach remains valid for uniform soil conditions, and is described in Fluor Guideline 000.215.1233

1F) Correspondence Principle

The correspondence principle is used to implement material damping ratios into an impedance calculation. A complex modifier is applied (typically to the shear modulus) and affects subsequent calculations. The exact substitution approach varies from one method to another. The modification can be derived from Equation 6 and the definition of a damping ratio.

damping ratio: $\beta_S = C / C_{CR} = C / \sqrt{K\,M}$

complex substitution: $G_S^* = G_S\,(1 + i\,2\,\beta_S)$ (11a)

When this substitution is inconvenient, or when analytical impedance expressions are not used, material damping may be applied directly into the computed complex stiffness.

stiffness adjustment: $(K + i\,C\,\Omega)(1 + i\,2\,\beta_S)$ (11b)

Software applies the correspondence principle in internal calculations so that it is not seen in the user interface. This is encountered when manually validating software results.

Observation: *The frequency independent approach, based on Lysmer's analog, applies soil material damping in a different manner (refer to Fluor Guideline 000.215.1233).*

1G) Negative Stiffness – From a static perspective, negative stiffness is illogical. From a dynamic perspective, negative stiffness values are logical and result when the impedance is more than 90 degrees out of phase with the applied dynamic force. This result can be demonstrated from comprehensive elastic medium calculations and from actual field measurements.

For soil supported foundations, negative stiffness can result from soil layering as illustrated in Figure 5A. This can occur at certain frequencies when reflected stress waves are out of phase, and counteract, with the applied dynamic load.

Negative stiffness values can also result in uniform soil for high values of Poisson's ratio as indicated in Figure 8. This situation occurs when a significant mass below the mat vibrates in phase with the foundation. An illustration of this begins with Equation 6,

Impedance from the Ground

$Q \{-M\ \Omega^2 + i\ C\ \Omega + K\} = F_0$

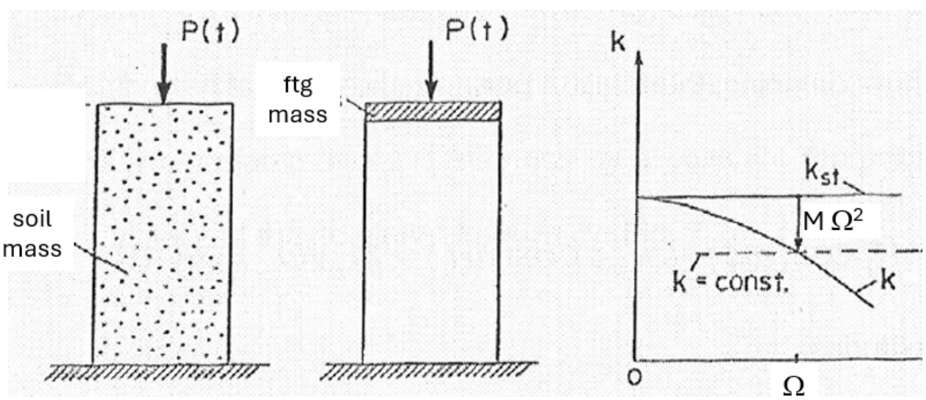
Now suppose that the in-phase mass is considered such that M is the original machine & foundation mass (Figure 8B), and M_A is the added in-phase soil mass (Figure 8A). Equation 6 becomes,

$$Q \{-(M + M_A)\ \Omega^2 + i\ C\ \Omega + K\} = F_0 \quad (12)$$

Re-arranging terms results in,

$$Q \{-M\ \Omega^2 + i\ C\ \Omega + (K - M_A\ \Omega^2)\} = F_0 \quad (13)$$

Substituting, $K \leftarrow K - M_A\ \Omega^2$, this would imply a reduction in stiffness with frequency (Figure 8C) if the mass was to remain as commonly defined.



A (Soil Half-Space) **B (Lumped Mass)** **C (Stiffness Effect)**

Figure 8 (Negative Stiffness from In-Phase Mass)

For pile foundations, negative stiffness can originate from pile group interactions. In Figure 6, the source pile is deformed upward while receiver pile C is deforming downward. If the opposing group downward effect is greater than the group upward effect, then a negative stiffness results.

<u>1H) Layered Soil</u>

Soil conditions commonly vary with depth, with layering properties being highly variable from one location to another. The number of potential situations is endless.

Impedance from the Ground

Comprehensive methods, both finite element and continuum, treat the analysis of soil layers in a general fashion. Novak's single pile method allows for any combination of layers within the pile length. Novak's embedment approach also considers layering.

In contrast, most practical soil support methods deal with a limited number of specific soil situations. Dyna 6 evaluated situations are,

- Half-Space (a uniform elastic layer of unlimited depth)
- Stratum (one elastic layer over rigid bedrock of unlimited depth)
- Composite (one elastic layer over an elastic layer of unlimited depth).

Wolf 2004 is the only widely known practical general solution for soil supported foundations. It is commonly called the Cone Method because cone shapes are used to approximate the spread, reflection, and refraction of stress waves though the soil (Figure 9). Though independent evaluations consider the Cone Method to be a reasonable approximation, acceptance remains limited.

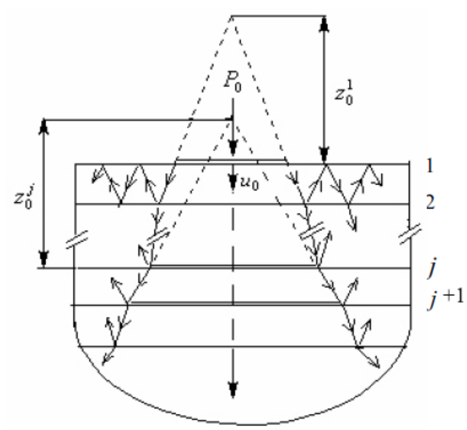


Figure 9 (Cone Method Diagram)

Observation: *Limited acceptance of the Cone Method appears to be partially based on a Wolf 2004 statement that impedance results are at least within 20% of theoretical values. Alternative situation-specific methods tend to provide graphical comparisons and are not quantitative about accuracy. From a user's perspective, comparisons must be made against available alternatives.*

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Impedance from the Ground

1l) Adjustments to Align with Measurements

Because of observed differences between results based on field measurements in comparison to theoretical calculations, some adjustments become necessary. Several of these adjustments are described in the following.

Resonant Amplitude: A damping "safety factor" is provided to adjust amplitudes near resonant peaks. This may be because field results indicate a difference or may be used as a safety measure if avoiding resonance is not practical.

Surface Effect: The field-measured shear modulus diminishes toward the surface due to lower confining soil pressures. This results in embedment impedance being overestimated. A reduced depth of embedment is commonly recommended.

Low Frequency Effect: Field measurements on pile and soil supported foundations indicate that radiation damping is greatly reduced for frequencies below the first natural frequency of the soil layer. Though the reduction in low frequency radiation damping is incorporated into the methodology, the response is overestimated unless material damping is included.

Slippage: Field measurements on piles and embedment is lower than elastic theory due to slippage between piles/mats and soil.

A weak zone can be applied to approximate this slippage. Dyna 6 uses an approach developed by Novak 1980 and Novak 1990 as illustrated in Figure 10 where R_P is the pile radius, T_W is the weak zone thickness, G_S is the soil shear modulus, and G_W is the weak zone soil shear modulus. Because the weak zone is massless, it may be necessary to partially or totally replace this missing mass by added mass at the pile/mat center.

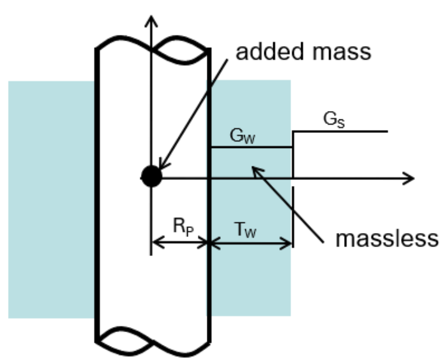

Figure 10 (Weak Zone Diagram)

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2) Empirical Methods (Synopsis)

Empirical methods are based on field test results. There is no mathematical model to generalize behavior. The approach is acceptable provided designs are within the limits of tested conditions.

Barkan 1962 is an empirical method for soil supported foundations with uniform soil properties. The method is based on the modulus of subgrade reaction and tables of factors for specific soil types. The method is referenced in British and Indian design specifications. An updated version of the method is used in the Russian and Chinese design codes.

This category of methods is not pursued beyond this section.

3) Practical Methods

Practical methods are those deemed widely usable for design purposes (as opposed to academic development or highly specialized situations). Though there are varying opinions on what fits this definition, the general considerations are ease of use and speed. Practicality originally considered methods that could be performed manually. The current concept of practicality encompasses methods that are software based. Nevertheless, the tradeoff for ease of use and speed is an inherent degree of approximation with respect to theoretical results.

Observation: *When introduced, the game-changing frequency independent approach had some drawbacks. Though rational, a large compressor foundation analysis was a lengthy sequence. It was time consuming to optimize dimensions or to correct early errors in a manual calculation. The rule-of-thumb initial foundation sizing criteria that remains today was used to minimize revisions.*

3A) Half-Space (Veletsos) – This method applies to soil supported foundations with uniform soil properties. Simple polynomial equations (Veletsos 1974a, Veletsos 1974b) are used to compute stiffness and damping based on an equivalent rigid circular footing. This method is used in Dyna 6 for soil supported foundations and for pile tip reactions.

- Approach: Veletsos polynomial equations were developed by curve fitting against stiffness and damping results calculated from a comprehensive (continuum) method.
- Calculation: Without material damping, the method can be computed manually. With material damping, the use of a spreadsheet is advised.

Impedance from the Ground

- Accuracy: Impedance from the source comprehensive method for three values of Poisson's ratio (0, 1/3, and 1/2) were used to compile three sets of polynomial coefficients. The method's accuracy is maximized at these exact values. For other values of Poisson's ratio, Dyna 6 uses, without interpolation, coefficients for 1/3 if Poisson's ratio if less than 0.4 and uses coefficients for 1/2 if greater.

With incorporation of material damping into the shear modulus using the correspondence principle, the calculation sequence is as follows.

Step 1: Determine material damping effect (G_S^*), shear wave velocity (V_{SW}) and dimensionless frequency (A), where # is a placeholder for mode.

$$G_S^* = G_S \,(1 + i\, 2\beta_S) \tag{14a}$$

$$V_{SW} = \sqrt{\frac{G_S^*\, g}{\gamma_S}} \tag{14b}$$

$$A_\# = \frac{\Omega\, R_\#}{V_{SW}} \tag{14c}$$

Step 2: Determine static stiffness (K_{ST}).

$$K_{ST\text{-}H} = \frac{8\, G_S^*\, R_T}{2 - \nu_S} \tag{15a}$$

$$K_{ST\text{-}V} = \frac{4\, G_S^*\, R_T}{1 - \nu_S} \tag{15b}$$

$$K_{ST\text{-}X} = \frac{8\, G_S^*\, (R_X)^3}{3(1 - \nu_S)} \tag{15c}$$

$$K_{ST\text{-}Y} = \frac{8\, G_S^*\, (R_Y)^3}{3(1 - \nu_S)} \tag{15d}$$

$$K_{ST\text{-}Z} = \frac{16\, G_S^*\, (R_Z)^3}{3} \tag{15e}$$

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Step 3: Determine polynomial coefficients (b_1 through b_4) for each mode of vibration.

Table 1 (Polynomial Coefficients)

Poisson's Ratio = 0	Vertical	Horizontal	Rocking	Torsion
b_1	0.250	0.000	0.525	0.425
b_2	1.000	0.000	0.800	0.687
b_3	0.000	0.000	0.000	0.000
b_4	0.850	0.775	0.000	0.000

Poisson's Ratio = 0.33	Vertical	Horizontal	Rocking	Torsion
b_1	0.350	0.000	0.500	0.425
b_2	0.800	0.000	0.800	0.687
b_3	0.000	0.000	0.000	0.000
b_4	0.750	0.650	0.000	0.000

Poisson's Ratio = 0.5	Vertical	Horizontal	Rocking	Torsion
b_1	0.000	0.000	0.400	0.425
b_2	0.000	0.000	0.800	0.687
b_3	0.170	0.000	0.027	0.000
b_4	0.850	0.600	0.000	0.000

Step 4: Determine stiffness coefficients (S) and damping coefficients (N).

$$S_{\#}=1 - \frac{b_1\,[b_2\,A_{\#}]^2}{1+[b_2\,A_{\#}]^2} - b_3\,(A_{\#})^2 \tag{16a}$$

$$N_{\#}=b_4+\frac{b_1\,b_2\,[b_2\,A_{\#}]^2}{1+[b_2\,A_{\#}]^2} \tag{16b}$$

Step 5: Determine dynamic stiffness (K_{DY}), stiffness (K) & damping (C),

$$K_{DY\text{-}\#} = K_{ST\text{-}\#}\,(S_{\#} + i\,N_{\#}\,A_{\#}) \tag{17}$$

$K_{\#}$ = real portion of $K_{DY\text{-}\#}$ (18a)

$C_{\#}$ = imaginary portion of $K_{DY\text{-}\#}$ / Ω (18b)

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3B) Stratum (Kausel) - This method applies to soil supported foundations with a uniform soil layer, or stratum, over bedrock. Simple equations (Kausel 1979) are used to compute stiffness and damping based on an equivalent rigid circular footing and depth to bedrock. This method is used in Dyna.

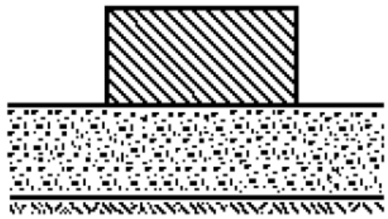

Figure 11 (Stratum Diagram)

- Approach: Kausel's equations were derived to match results from comprehensive (finite element) method. Despite the title of Kausel 1979, impedance formulas for vertical and torsional modes from Elsabee are provided in the conclusions. At low frequencies, below the stratum's first natural frequency, radiation damping is virtually eliminated. As the stratum height decreases, the stratum natural frequency increases – making it more likely that the low frequency adjustment applies.

- Calculation: The method can be computed manually; however, a spreadsheet is suggested.

- Accuracy: Impedance from the source comprehensive finite element method for specific ratios of foundation radius (R) and stratum height (H) were used to derive the simple equations. Limitations are $R/H \leq 0.5$, $E/R \leq 1.5$, and $E/H \leq 0.75$.

The calculation sequence is as follows.

Step 1: Calculate static stiffness, including stratum height (H) and embedment (E),

$$K_{ST\text{-}H} = \frac{8\ G_S\ R_T}{2\ -\ \nu_S}\left(1+\frac{R_T}{2\ H}\right)\left(1+\frac{2\ E}{3\ R_T}\right)\left(1+\frac{5\ E}{4\ H}\right) \tag{19a}$$

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Impedance from the Ground

$$K_{ST\text{-}V} = \frac{4\ G_S\ R_T}{1\ \text{-}\ \nu_S}\left(1+1.28\frac{R_T}{H}\right)\left(1+0.47\frac{E}{R_T}\right)\left(1+\left(0.85\text{-}0.28\frac{E}{R_T}\right)\frac{E/H}{1\text{-}E/H}\right) \quad (19b)$$

$$K_{ST\text{-}X} = \frac{8\ G_S\ R_X^3}{3(1\ \text{-}\ \nu_S)}\left(1+\frac{R_X}{6\ H}\right)\left(1+2\frac{E}{R_X}\right)\left(1+0.7\frac{E}{H}\right) \quad (19c)$$

$$K_{ST\text{-}Y} = \frac{8\ G_S\ R_Y^3}{3(1\ \text{-}\ \nu_S)}\left(1+\frac{R_Y}{6\ H}\right)\left(1+2\frac{E}{R_Y}\right)\left(1+0.7\frac{E}{H}\right) \quad (19d)$$

$$K_{ST\text{-}Z} = \frac{16\ G_S\ R_Z^3}{3}\left(1+2.67\frac{E}{R_Z}\right) \quad (19e)$$

$$K_{ST\text{-}CXZ} = (0.4\frac{E}{R_Y}\text{-}0.03)R_Y\ K_{ST\text{-}H} \quad (19f)$$

$$K_{ST\text{-}CYZ} = (0.4\frac{E}{R_X}\text{-}0.03)R_X\ K_{ST\text{-}H} \quad (19g)$$

Step 2: Determine the stiffness coefficient ($S_{\#}$) from Table 2.

Impedance from the Ground

Table 2 (Stiffness and Damping Coefficients)

Mode	Stiffness coefficient ($S_\#$)	Damping coefficient ($N_\#$)
Horizontal, Cross	1.0	0.6
Vertical	1.0	0.85
Rocking about X	$1 - 0.2\,A_X$ (for $A_X < 2.5$ or vs $\geq$ 0.45) 0.5 (for $A_X \geq 2.5$)	$\frac{0.3A_X}{1+A_X}$
Rocking about Y	$1 - 0.2\,A_Y$ (for $A_Y < 2.5$ or vs $\geq$ 0.45) 0.5 (for $A_Y \geq 2.5$)	$\frac{0.3A_Y}{1+A_Y}$
Torsion	$1 - 0.133\,A_Z$ (for $A_Z < 3$) 0.6 (for $A_Z \geq 3$)	$\frac{0.3A_Z}{1+A_Z}$

Where,
$A_X = \Omega\, R_X / V_{SW}$
$A_Y = \Omega\, R_Y / V_{SW}$
$A_Z = \Omega\, R_Z / V_{SW}$

Step 3: Calculate the stratum's first natural frequency (ω_{PW}, ω_{SW}) as,

$$\omega_{PW} = \frac{\pi\, V_{SW}}{2\, H}\sqrt{\frac{2(1-\nu_S)}{1-2\nu_S}} \tag{20a}$$

$$\omega_{SW} = \frac{\pi\, V_{SW}}{2\, H} \tag{20b}$$

Step 4: Determine the damping coefficient ($N_\#$)

Determine the frequency ratio (ξ) from Table 3,

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Table 3 (Low Frequency Parameters)

Mode:	Ψ	ξ
Vertical	0.67	Ω / ω_{PW}
Horizontal, Cross	0.65	Ω / ω_{SW}
Rocking about X	0.5	Ω / ω_{PW}
Rocking about Y	0.5	Ω / ω_{PW}
Torsion	0.15	Ω / ω_{SW}

If $\xi \geq 1$, then determine the damping coefficient from Table 2. Otherwise, determine the damping coefficient from,

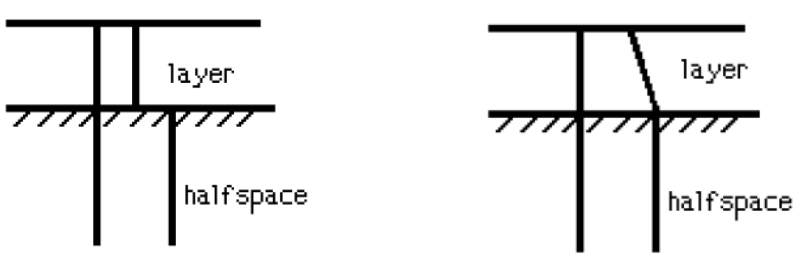
$$N_{\#} = \frac{\Psi \, \beta_S \, \xi}{1 - (1 - 2\,\beta_S)\,\xi^2} \tag{21}$$

Step 5: Calculate dynamic impedance ($K_{DY\text{-}\#}$) where # is a placeholder for mode,

$$K_{DY\text{-}\#} = K_{ST\text{-}\#}(S_{\#} + i\,A_{\#}\,N_{\#})(1 + i\,2\,\beta_S) \tag{22}$$

$K_{\#}$ = real portion of $K_{DY\text{-}\#}$ (23a)
$C_{\#}$ = imaginary portion of $K_{DY\text{-}\#} / \Omega$ (23b)

3C) Composite (Wong) - This method applies to soil supported foundations with a uniform or linear soil (composite) layer over a uniform half-space. This method is used in Dyna.



A (uniform layer) **B (linear layer)**

Figure 12 (Composite Diagram)

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- Approach: Wong 1985 provides normalized table coefficient values to compute stiffness and damping based on a rigid square footing, layer thickness, and relative strength of layer and half-space. Table values were determined from a comprehensive (analytical) method. Note that the upper layer is always weaker than the half-space, and composite layer & half-space only differ by shear wave velocity.

- Calculation: The method can be computed manually.

- Accuracy: The source reference provides 12 tables (6 uniform, 6 linear) for square footings with 5% material damping. Considered are 3 values of shear modulus ratio, 2 values of Poisson's ratio, 5 values of H/a, and 21 values of A_O. Dyna 6 documentation is not specific about how table data might have been interpolated and/or extrapolated.

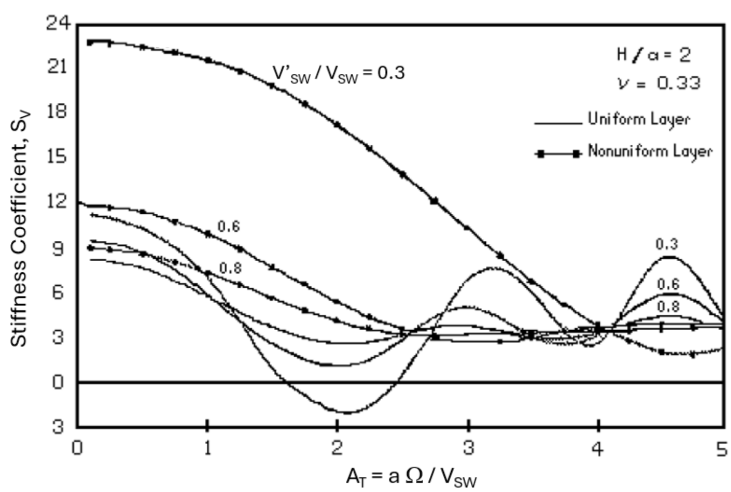

Figure 13 (Sample Plot of Vertical Stiffness Coefficient)

Impedance from the Ground

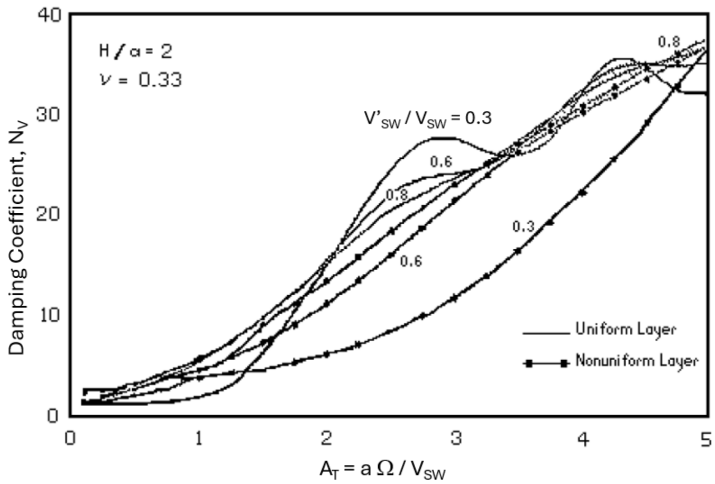

Figure 14 (Sample Plot of Vertical Damping Coefficient)

Where V_{SW} is the shear wave velocity within the half-space and V'_{SW} is the shear wave velocity at top of the composite layer.

The calculation sequence for vertical translation is,

Step 1: Determine stiffness coefficient (S_V) from Figure 13.

Step 2: Determine damping coefficient (N_V) from Figure 14.

Step 3: determine dynamic impedance ($K_{DY\text{-}V}$) from,

$$K_{DY\text{-}V} = G_S\, R_T\, (S_V + i\, N_V) \tag{24}$$

K_V = real portion of $K_{DY\text{-}V}$ (25a)

C_V = imaginary portion of $K_{DY\text{-}V} / \Omega$ (25b)

3D) Single Piles (Novak) – This method applies to single piles with one or more soil layers between pile head and tip, and a half-space layer below the tip. This method (Novak 1978a, Novak 1978b, Novak 1978c) is widely recognized and used in Dyna.

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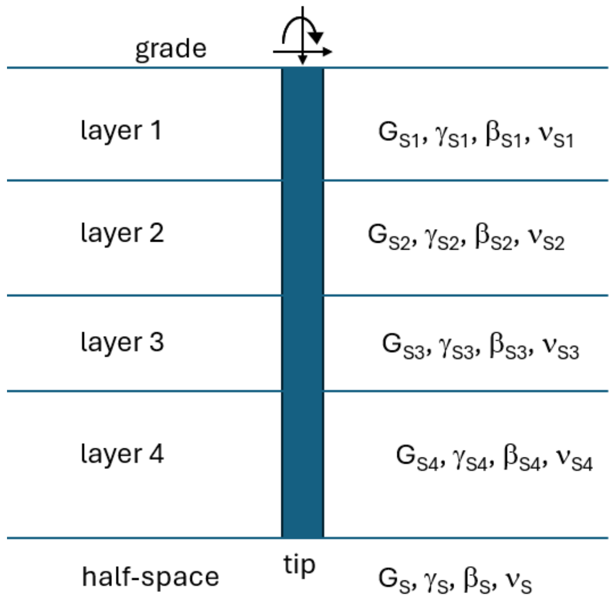

Figure 15 (Single Pile Diagram)

- Approach: For each layer, a dynamic impedance is determined using a continuum equation for strictly lateral spread of stress waves, and a vertical continuum equation for soil layer and pile. The resulting layer impedances are combined and solved using a coupled solution. Vertical and torsional modes are solved independently. Rocking and horizontal modes are coupled.

- Calculation: If performed manually, the method is lengthy and tedious. Use of a pre-prepared spreadsheet or compiled program is essential.

- Accuracy: The wave spread is lateral only. This means that the result is theoretically correct only if pile is infinitely long

To provide a feel for the methodology, the following calculation sequence is used for vertical translation for a floating pile without a weak zone.

Steps 1 thru 4 below are computed for each soil layer (U) and use the following equations for nondimensional frequency (A_O^*), modified Bessel functions (K_0 and K_1), and side impedance (K_{Q-V}),

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$$A_0^* = \frac{i\, A_0}{\sqrt{1 + i\, 2\, \beta_S}} \tag{26}$$

$$K_0(x) = -\left[\ln\frac{x}{2} + \gamma\right] + \sum_{s=1}^{\infty}\left(\frac{x}{2}\right)^{2x}\frac{1}{(s!)^2}\left[\sum_{r=1}^{s}\frac{1}{r} - \left(\ln\frac{x}{2} + \gamma\right)\right] \tag{27a}$$

$$K_1(x) = \frac{1}{x} + \sum_{s=1}^{\infty}\left(\frac{x}{2}\right)^{2s\text{-}1}\frac{1}{(s!)^2}\left[\frac{1}{2} + s\left(\ln\frac{x}{2} + \gamma - \sum_{r=1}^{s}\frac{1}{r}\right)\right] \tag{27b}$$

$$K_{U\text{-}V} = 2\,\pi\,(1+2\,\beta_S)\,A_0^*\,\frac{K_1(A_0^*)}{K_0(A_0^*)} \tag{28}$$

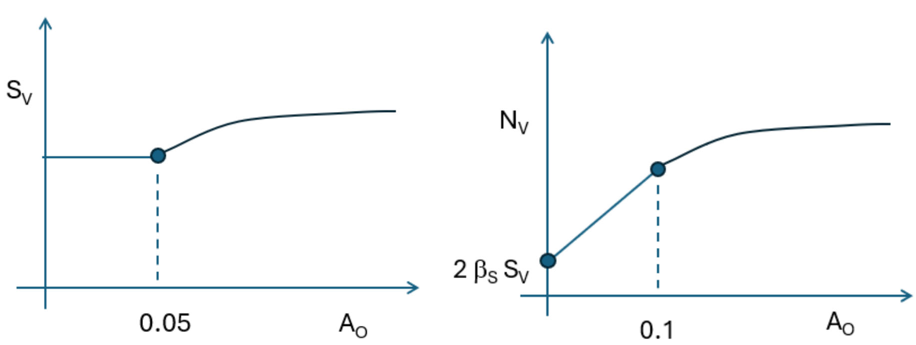
Where, x is a complex function argument, r and s are integer summation values, and γ is a constant equal to 0.5772.

Observation: *Incremental summation values of Equation 27 eventually gets smaller as s increases. The summation continues until incremental values become negligible, usually about 10 increments.*

Step 1: Calculate the layer non-dimensional frequency,

$$A_V = R_P\,\Omega\,/\,V_{SW} \tag{29}$$

Step 2: Calculate the layer side soil reaction. This is illustrated as,



A (stiffness) **B (damping)**

Figure 16 (Single Pile Vertical Impedance Coefficients)

(stiffness coefficient): If $A_V \geq 0.05$, then

solve for $K_{U\text{-}V}$ using $A_0 = A_V$.

otherwise

solve for $K_{U\text{-}V}$ using $A_O = 0.05$

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S_V = (real portion of K_{U-V})

(damping coefficient): If $A_V \geq 0.1$, then

solve for K_{U-V} using $A_0 = A_V$.

N_V = (imaginary portion of K_{U-V}).

otherwise,

solve for K_{U-V} using $A_O = 0.1$

$N_{0.1}$ = (imaginary portion of K_{U-V})

$N_{0.0} = 2\ \beta s\ S_V$

$N_V = N_{0.0} + (N_{0.1} - N_{0.0})\ (A_V / 0.1)$

Step 3: Compute the layer vertical frequency factor (Λ), where μ is the pile mass per unit length, and c is the coefficient of pile internal damping.

$$\Lambda = H \sqrt{\frac{1}{E_P\ A_P}\left[\mu\ \omega^2 - G_S\ (S_V + i\ N_V)\right]} \tag{30}$$

Step 4: Compute layer "U" stiffness matrix,

$$\left[K'_Q\right] = -\frac{E_P\ A_P}{H}\begin{bmatrix} \cot\Lambda & -\operatorname{cosec}\Lambda \\ -\operatorname{cosec}\Lambda & \cot\Lambda \end{bmatrix} \tag{31}$$

Step 5: Compute pile tip impedance using the half space method (section 3A) where A_T is based on the pile tip radius,

$$K_{TIP-V} = K_{ST-V}\ (S_V + i\ N_V\ A_T) \tag{32}$$

Step 6: Combine all layers and tip stiffness into a single matrix (K'_P). To illustrate how this works, a 3-layer example would be combined as,

$$\left[K'_P\right] = \begin{bmatrix} K'_1(1,1) & K'_1(1,2) & - & - \\ K'_1(2,1) & K'_1(2,2)+K'_2(1,1) & K'_2(1,2) & - \\ - & K'_2(2,1) & K'_2(2,2)+K'_3(1,1) & K'_3(1,2) \\ - & - & K'_3(2,1) & K'_3(2,2)+K_{TIP-V} \end{bmatrix} \tag{33}$$

With $\{P_P\}$ representing a column matrix for applied forces at each layer boundary and $\{\Delta_P\}$ representing a column matrix of deformations at each layer boundary,

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$\{P_P\} = [K'_P]\ \{\Delta_P\}$ (34)

Step 7: Invert the stiffness matrix, $[\varepsilon] = [K'_P]^{-1}$

Thus, $\{\Delta_P\} = [\varepsilon]\ \{P_P\}$ (35)

Step 8 Compute the single pile stiffness and damping from the pile butt dynamic deformation.

The only applied force is a unit load at the pile butt, or $P_P(1) = 1.0$, and the only deformation needed is also at the pile butt.

$K_{DY\text{-}V} = 1 / \varepsilon\ (1,1)$ (36)

K_V = (real portion of $K_{DY\text{-}V}$) (37a)
C_V= (imaginary portion of $K_{DY\text{-}V}$) / Ω (37b)

3E) Pile Groups (El-Marsafawi) – This is an approximate approach to compute stiffness and damping of a pile group. The procedure is from El-Marsafawi 1992 using static interaction charts from Poulos 1980 and dynamic interaction charts from Kaynia 1982. This is the procedure used by Dyna.

- Approach: El-Marsafawi determines the superimposed effect of every 2-pile situation of spacing and orientation in a pile group. Interaction factors apply between pile heads, using uniform pile properties, and uniform soil properties. Dyna 6 determines an equivalent uniform soil layer and/or equivalent pile properties as needed.

- Calculation: A compiled program or spreadsheet is recommended. Other than chart interpolations and extensions, matrix inversions are needed.

- Accuracy: For small pile groups, comparisons indicate a close correlation to comprehensive methods. For large pile groups, the correlation is less accurate because the effect of intervening piles is not considered. Static and dynamic chart limitations are described below. The exact interpolation/extrapolation scheme is not described in Dyna 6 documentation.

The approach is further described in the following.

Interaction factors (α) are defined as follows,

$\alpha_{SR} = \Delta_{SR} / \Delta_{SS}$ (38)

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Where,

Δ_{SR} = dynamic displacement at head of receiver pile "R" subjected to a unit harmonic load at head of source pile "S".

Δ_{SS} = dynamic displacement at head of source pile "S" subjected to a unit harmonic load at head of source pile "S".

Note 1: If the receiver and source pile are the same, then α = 1.0

Note 2: Group torsion is usually calculated based on pile group geometry. Torsional rotation of individual piles generates a minor resistance and does not affect other piles.

There are several methodologies for computing 2-pile interaction factors. Novak's method uses static and dynamic interaction factors. This is because the dynamic interaction factors are nominalized to a static load. Thus, interaction factor values are computed from,

α_{SR} = (static interaction factor) (frequency variation) (39)

The static interaction factor is obtained from interpolating between several charts from Poulos 1980. Refer to Figure 17 for one of these charts. Poulos 1980 charts are for settlement (i.e., vertical translation) for floating & end bearing piles based on calculations in a previous publication. The collection of charts covers length to diameter ratios (10, 25, 50, 100), Poisson's ratio (0, 0.15, 0.35, 0.5) and correction factors for stratum, bell pier, and linearly increasing soil elastic modulus.

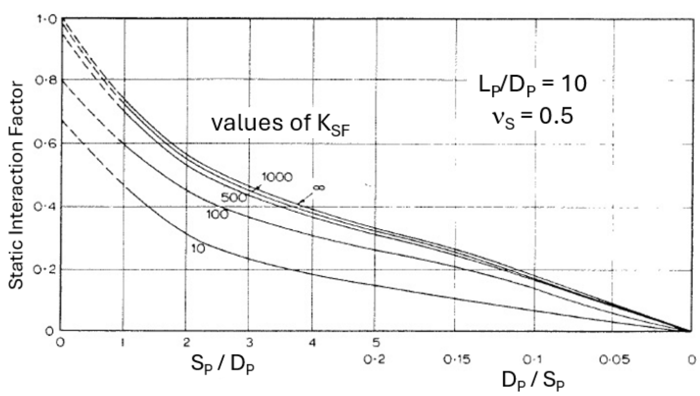

Figure 17 (Sample Poulos Vertical Static Interaction Chart)

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In Figure 17, K_{SF} is a pile stiffness factor equal to Ep Ra / Es. Ra is the pile cross-section area divided by pile shaft area. Thus, Ra is equal to 1.0 for a solid pile.

The frequency variation is obtained from curve fitting several charts from Kaynia 1982. Note that these factors are determined using Kaynia's comprehensive (analytical) methodology. Refer to Figure 18 for one of these charts. Two figures are provided based on L_P/D_P = 15, Es/Ep = 0.001, and γ_S/γ_P = 0.7. The maximum indicated non-dimensional frequency is 1.0.

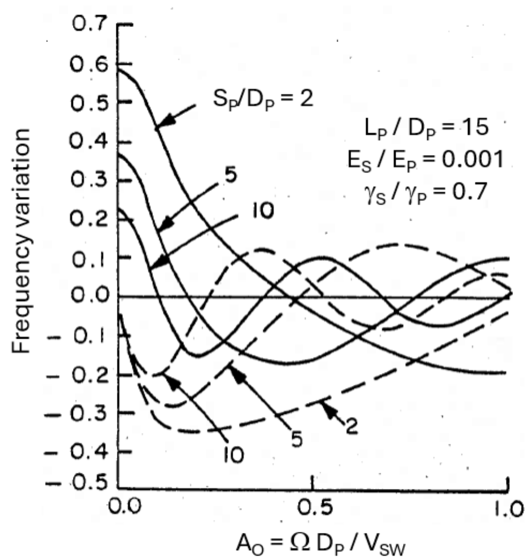

Figure 18 (Sample Kaynia Vertical Dynamic Interaction Chart)

To illustrate how interaction factors are applied, the following is a derivation for vertical loading.

The dynamic deformation at receiver pile "j" would be the summation of effects from any source pile in the group.

$$\Delta j = \Delta j1 + \Delta j2 + \Delta j3 + \ldots + \Delta ji + \ldots \tag{40}$$

Factor out Δjj (source = receiver) on the right side,

$$\Delta j = \{ (\Delta j1 / \Delta jj) + (\Delta j2 / \Delta jj) + (\Delta j3 / \Delta jj) + \ldots + (\Delta jj / \Delta jj) + \ldots \} \Delta jj \tag{41}$$

Use Equation 38 to substitute for the deformation ratios,

$$\Delta j = \{ \alpha j1 + \alpha j2 + \alpha j3 + \ldots + 1 + \ldots \} \Delta jj \tag{42}$$

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For the deformation at the source pile, substitute using $\Delta = K' / P$, where using single pile stiffness (K) and damping (C) results, $K' = K + i\,C\,\Omega$,

$$\Delta j = \{ (\alpha j1) + (\alpha j2) + (\alpha j3) + \ldots + (1) + \ldots \} P / K' \tag{43}$$

With all piles being the same, a NP x NP pile group can be expressed in matrix form as,

$$\begin{Bmatrix} \Delta 1 \\ \Delta 2 \\ \Delta 3 \\ \cdots \\ \Delta NP \end{Bmatrix} = \begin{bmatrix} 1 & \alpha 12 & \alpha 13 & \ldots & \alpha 1NP \\ \alpha 21 & 1 & \alpha 23 & \ldots & \alpha 2NP \\ \alpha 31 & \alpha 32 & 1 & \ldots & \alpha 3NP \\ \cdots & \cdots & \cdots & \cdots & \cdots \\ \alpha N1 & \alpha N2 & \alpha N3 & \ldots & 1 \end{bmatrix} \begin{Bmatrix} P1 \\ P2 \\ P3 \\ \cdots \\ PNP \end{Bmatrix} \frac{1}{K'} \tag{44}$$

With the group flexibility matrix, $f_G = [\alpha] / K'$, the above can be simplified to,

$$\{\Delta\} = [f_G] \{P\} \tag{45}$$

Re-arranging to solve for the applied force, and designating the group stiffness matrix as $[K_G] = [f_G]^{-1}$,

$$\{P\} = [K_G] \{\Delta\} \tag{46}$$

With a rigid block mat, the individual vertical deformations are the same ($\Delta 1 = \Delta 2 = \Delta 3 = \ldots \Delta NP = \Delta_G$) and Equation 46 can be solved for P1...PNP.

The vertical lumped parameter impedance ($K_{DY\text{-}V}$), stiffness (K_V), and damping (C_V) are,

$$K_{DY\text{-}V} = (P1 + P2 + P3 + \ldots + PNP) / \Delta_G \tag{47}$$

K_V = real portion of $K_{DY\text{-}V}$ (48a)
C_V = imaginary portion of $K_{DY\text{-}V} / \Omega$ (48b)

3F) Embedment (Novak) – Except for a stratum, the soil's dynamic reaction against the side of a pile cap or soil supported mat is the same as developed for the side stiffness of pile. Refer to Section 3D, steps 1 and 2. For stratum embedment, refer to Section 3B.

4) Comprehensive Methods (Synopsis)

Comprehensive methods to accurately evaluate an elastic medium use either numerical simulation (finite element) or analytical (continuum) models. Accuracy and attention to detail is required because not just any finite element or continuum implementation is considered

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comprehensive. Usage of a comprehensive approach often applies in certain situations determined by regulatory authorities (mainly nuclear & seismic applications). Comprehensive methods are also used to determine theoretical elastic medium impedance values. For most typical process plant situations, these methods are highly specialized and time consuming.

Observation: *The effort hour and schedule drawbacks of using comprehensive methods are significant. Implementation considerations include relevant experience and availability/familiarity with suitable software.*

This category of methods is not pursued beyond this section.

5) Nomenclature

Variables:
a = half width of square footing
$A_{\#}$ = nondimensional frequency
$A^*_{\#}$ = complex nondimensional frequency
A_P = pile area

b_1, b_2, b_3, b_4 = polynomial coefficients
B_V = mass ratio, refer to guideline 000.215.1233

c = coefficient of pile internal damping
$C_{\#}$ = viscous damping coefficient
C_{CR} = critical damping

D_P = pile diameter

E = embedment depth
E_P = pile modulus
E_S = soil modulus

f_G = pile group flexibility matrix
F(t) = time varying applied force
F_0 = peak value of harmonic force

g = gravitational acceleration
G_S = soil shear modulus

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G_W = weak zone shear modulus

H = layer thickness
HS = half-space

i = $\sqrt{-1}$

J = complex force

$K_\#$ = stiffness
K_{SF} = pile stiffness factor
$K_{ST\text{-}\#}$ = static stiffness
$K_{DY\text{-}\#}$ = dynamic impedance
K_P = pile impedance
K_Q = pile side impedance
K_0, K_1 = modified Bessel functions

L_X = mat or pile cap length in X direction
L_Y = mat or pile cap length in Y direction
L_P = pile length

m = symbolic usage
M = mass
M_A = added mass

n = symbolic usage
$N_\#$ = damping coefficient
NP = number of piles

P_P = pile load

Q = complex deformation
Qi = imaginary component of complex Q
Qr = real component of complex Q

r = integer summation value
$R_\#$ = equivalent circular disk radius
R_P = pile radius

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Ra = pile cross-section area divided by pile shaft area

s = integer summation value
$S_{\#}$ = stiffness coefficient
S_P = pile spacing

t = time
T_W = weak zone thickness

U = soil layer designation

V_{SW} = shear wave velocity
V'_{SW} = shear wave velocity at top of composite layer

x = complex function argument
X = horizontal cartesian axis

y = time-varying displacement
$\dot{y}$ = time-varying velocity
$\ddot{y}$ = time-varying acceleration
Y = horizontal cartesian axis

Z = vertical cartesian or cylindrical axis

α_{SR} = interaction factor between source pile “S” and receiver pile “R”

β_S = soil material damping ratio

γ = single pile constant equal to 0.5772
γ_S = soil unit weight
γ_P = pile unit weight

Δ = deformation
Δ_0 = peak deformation
Δ_P = pile dynamic deformation
$\Delta_{ST\text{-}\#}$ = static deformation
$\Delta_{DY\text{-}\#}$ = dynamic deformation
Δ_{SR} = dynamic deformation at receiver pile “R” due to load on source pile “S”

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Δ_{SS} = dynamic deformation at source pile "S" due to load on source pile "S"

ε = flexibility matrix equal to inverse of stiffness matrix

θ = horizontal cylindrical axis angle
θ_F = forcing phase angle
θ_D = deformation phase angle

Λ = complex frequency parameter

μ = pile mass per unit length

ν_S = soil Poisson's ratio

ξ = stratum frequency ratio

ρ= horizonal cylindrical axis

Ψ = stratum factor

Ω = forcing frequency

ω_{PW} = P-wave frequency
ω_{SW} = S-wave frequency

<u>Subscripts,</u>
C = cross
ST = static
CR = critical
DY = dynamic
H = horizontal translation
j = receiver pile number
P = pile
PW = primary (compression) wave
S = soil
SW = secondary (shear) wave

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T = translation
V = vertical translation
X = X horizontal axis
Y = Y horizontal axis
Z = Z vertical axis
0 = nominal
# = mode placeholder for T, V, H, X, Y, or Z as applicable

6) References

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000.215.1236, Non-Rigid Support for Vibratory Equipment

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(Barkan 1962) Dynamics of Bases and Foundations; D. Barkan; McGraw-Hill; 1962

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(Kausel 1979) *Vertical and Torsional Stiffness of Cylindrical Footings*; E. Kausel & R. Ushijima; Publication R79-6; MIT Department of Civil Engineering; 1979

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(Novak 1978c) *Dynamic Soil Reactions for Plane Strain Case*; M. Novak, T. Nogami, and F. Aboul-Ella; Journal of the Engineering Mechanics Division, ASCE, August 1978, pp 953-959

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(Novak 1980) Approximate Approach to Contact Effects of Piles; M. Novak and M. Sheta, ASCE Proceedings of Dynamic Response of Pile Foundations: Analytical Aspects; pp 53-79; October 1980

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(Veletsos 1974a) *Torsional Vibration of Viscoelastic Foundations*; A. Veletsos and D. Nair; Journal of the Geotechnical Engineering Division; ASCE, March 1974; pp 225-246

(Veletsos 1974b) *Basic Response Functions for Elastic Foundations*; A. Veletsos and B. Verbic; Journal of the Engineering Mechanics Division; ASCE; April 1974; pp 189-202

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Notes on Using Dyna Version 6.1

Dyna is a commercial program used for the vibratory analysis of rigid foundations. For non-rigid foundations, the program can also be used to calculate stiffness and damping values for input into SAP2000. This program performs the same basic function as DynaN, Dyna5, PVAP, and SVAP.

The following sections include notes, tips, and techniques focusing on the most commonly used options, and are intended as supplements to the program's help file documentation.

Contents

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Notes on Using Dyna Version 6.1

Part 1: General

1A) Program Notes

(a) Default Folder – A default data folder is created upon installation of the program (i.e., "Dyna 6.0 Projects") within the local Documents folder.

(b) Help File – Dyna program documentation is in the help file and can be accessed within the program using pulldown menu, "Help-Help…".

(c) Data Format – Data is stored by Dyna using a text file format. If need be, input values could be manipulated directly in the text file using Notepad, especially for the rapid input of large pile groups.

(d) Title Labels – The Project Title and Case Considered input fields (Figure 1) are used by Dyna to generate subdirectory names. These fields need to conform to directory naming rules and avoid the use of special characters.

(e) Dyna5 Files – Files created using Dyna version 5 can be loaded into Dyna. After loading a Dyna5 file, review and edit the Project Title and Case Considered data fields (Figure 1) for use as directory names by eliminating any special characters. If blank, complete the Case Considered field.

(f) Input Errors – A few checks are made and reported during data input. Examples include:

- Stratum ratios R/H, E/R, and E/H exceed maximum limit.

(g) Analysis Errors – Computational issues are not trapped and reported within the analysis routine. Such issues become apparent by the lack of results, incorrect results, or a rather cryptic error message. Examples include,

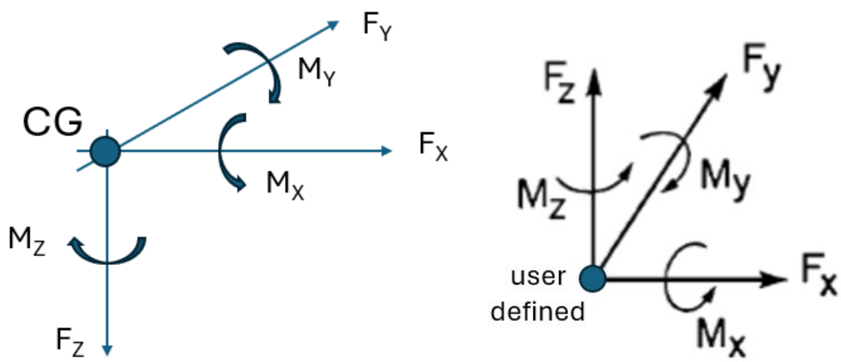
- Poisson's ratio is not between 0 and 0.5.
- Sum of layer depths are not equal to the pile length.
- Maximum and minimum frequencies values are reversed.
- Composite layer shear wave velocity is not less than that for the half-space below.

(h) Impedance Location – Calculated impedance values are reported at the center of gravity, not at the center of base.

Notes on Using Dyna Version 6.1

(i) Material Damping – Pile and soil material damping coefficients (D) are input as D = tan δ =2 β, where δ is a loss angle and β is the damping ratio. Thus, if the damping ratio is 2%, then 0.04 would be entered.

(j) Coordinate System – Locations are specified using cartesian coordinates. The X and Y axes are horizontal. Internally, the vertical Z axis is positive downward. Externally in several user input dialogs, the Z axis positive translations and rotations are reversed. Note the difference in origins.



A (Internal System) **B (External System)**

Figure 1 (Cartesian Coordinate Systems)

Item:	Coordinate System:
3D VIEW utility	External
Pile Location Input	External
Resultant Input	External
Pile Location Display	Internal
Force Input Dialog	Internal
Output Text File	Internal

(k) Units of Measure – Units are not always listed in Dyna 6 dialogs. Within this document, units are noted where needed. Note that force and moment units depend on the quadratic/non-quadratic option.

(l) Harmonic Loading – Dyna 6 analyzes simple harmonic forces and not configurations for centrifugal or reciprocating machines. Methods of dealing with this are described in Section 2H (c) and (d).

Notes on Using Dyna Version 6.1

1B) What's New in Version 6.1?

The current version is 6.1 and contains user interface enhancements. Analysis results are nearly identical.

General:

- The program name no longer includes version number (i.e., "Dyna5" to "Dyna")
- The program's copy protection scheme has been enhanced.
- The program's help file has been updated.

Interface:

- A directory & file naming sequence has been implemented.
- Program windows and dialog boxes have been redesigned.
- An optional pile cross-section properties calc has been added.

Analysis:

- With respect to Dyna5, there is a small difference in the half-space solution result.

1C) DynaN Differences

DynaN remains available for usage. There are several noteworthy technical differences.

- DynaN uses a different weak zone methodology.
- DynaN applies frequency-dependent stiffness and damping modifications. The modifications were derived for an improved match with field testing. However, modification values are not provided in program documentation.
- DynaN reports analysis errors when using high frequencies and a weak zone around a large pile cap.
- DynaN does not include a means for the user to reduce damping. This means that natural frequencies are more difficult to determine within the program.
- DynaN does not input or report phase angles. For example, a centrifugal machine has vertical and horizontal load components that are actually 90 degrees out of phase.

Notes on Using Dyna Version 6.1

- There is a significant impedance stiffness discrepancy with Dyna 5/6 for pile groups in layered soil. Both DynaN and Dyna 5/6 use group impedance factors which are based on uniform soil. It appears that different methodologies are used to determine equivalent uniform soil properties from layered soil.

Part 2: Input

The initial program window is shown in Figure 2 which indicates the main drop-down menu options (File, Project, Foundation…) and eight ribbon icons (New, Open, Save…).

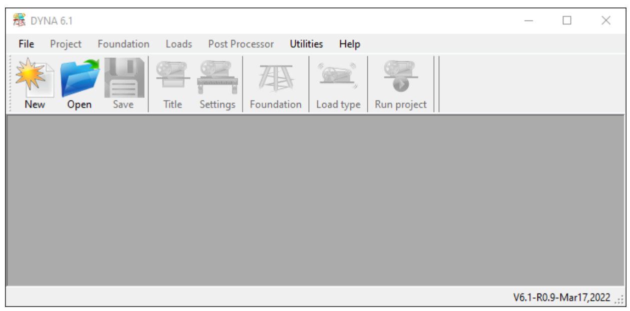


Figure 2 (Dyna Program Window)

<u>2A) Project Title</u>

The Title Dialog Box (Figure 3) is displayed upon initiating a new file. Otherwise, it can be accessed from ribbon icon “Title” or from drop down menu (Project – Title).

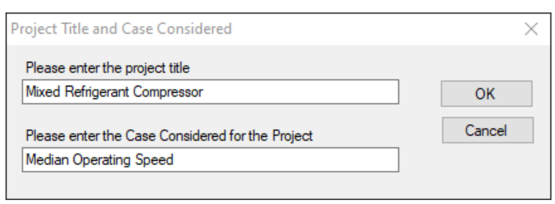


Figure 3 (Title Dialog Box)

Upon running an analysis, Dyna,

- Creates two subfolders if they do not already exist. The first uses the project title line (i.e., “Mixed Refrigerant Compressor”). The second uses the case considered (i.e., “Median Operating Speed”).
- Creates a copy of the input file within the case considered folder based on the two folder titles (i.e., “Mixed Refrigerant Compressor-Median Operating Speed.dy6”).

Notes on Using Dyna Version 6.1

- Creates a file for the text output within the case considered folder based on the two folder titles (i.e., “Mixed Refrigerant Compressor-Median Operating Speed_out.txt”).

- Creates additional analysis output files in the same sub-directory.

2B) Project Properties

The Project Properties dialog (Figure 4) is essentially a dashboard. It is displayed upon designating the title for a new file, or upon opening an existing file. Closing this window closes the file.

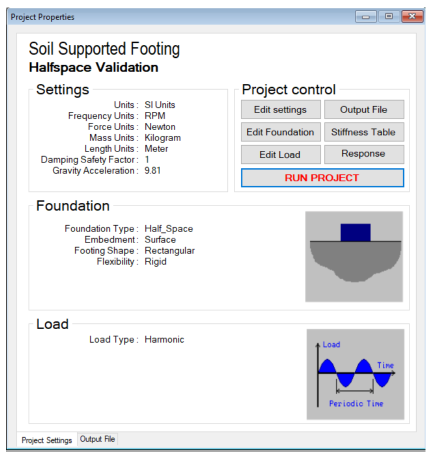


Figure 4 (Project Properties)

2C) Edit Settings

The Project Settings dialog (Figure 5) can be accessed using dashboard button “Edit Settings”, icon button “Settings”, or pulldown menu “Project-Settings”.

Other than frequency, consistent units of measure are used. The only unit-dependent value is the acceleration of gravity.

Notes on Using Dyna Version 6.1

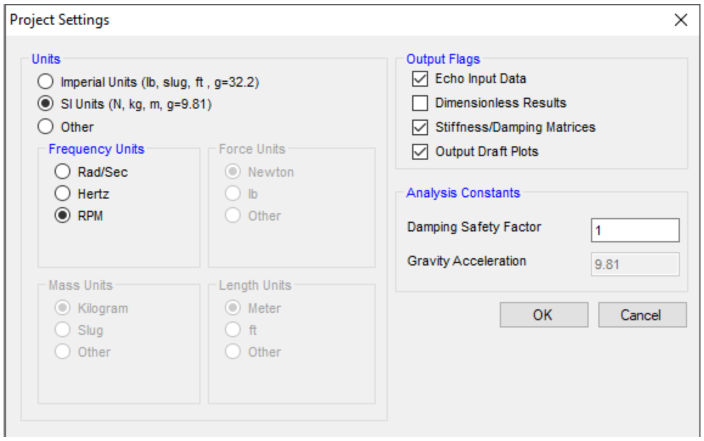


Figure 5 (Settings Dialog)

<u>Output Flags:</u> (these flags affect content of the output text file)

- Echo Input Data – if checked, lists all input data
- Dimensionless Results – if checked, listed foundation response values are dimensionless.
- Stiffness/Damping Matrices – if checked, lists stiffness and damping values at each frequency.
- Output Draft Plots – if checked, adds text-based graphic response plots

The damping safety factor is used to reduce computed damping values, with the greatest effect near resonance. Computed damping is divided by this value. Thus, a value of 1 has no effect and a value of 2 reports half the computed damping.

The gravity acceleration value must be consistent with the length units in use. This field is filled in automatically if standard imperial or SI metric units are selected.

<u>2D) Edit Foundation</u>

Pressing the Edit Foundation button displays the Foundation Type dialog (Figure 6). This dialog can also be accessed using pulldown menu "Foundation".

Notes on Using Dyna Version 6.1

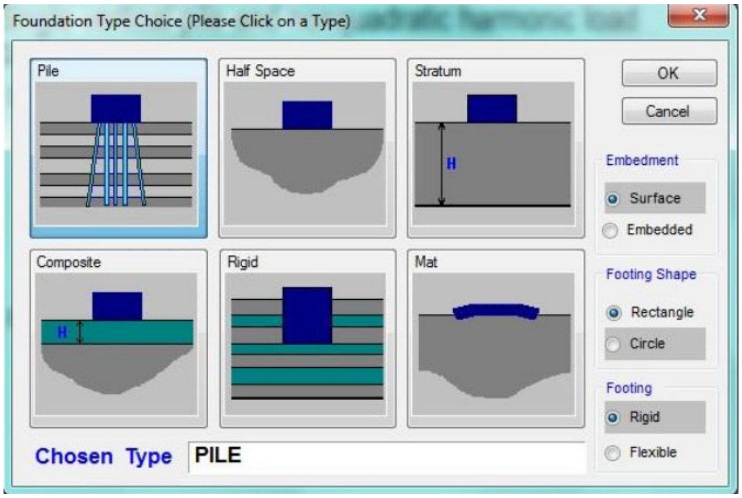


Figure 6 (Foundation Type Dialog)

Choices:

- Pile – a rigid or flexible pile cap supported by one or more piles
- Half-Space – a rigid mat supported by a uniform soil half-space
- Stratum – a rigid mat supported by a uniform soil layer over bedrock
- Composite – a rigid mat supported by a uniform soil layer over a stronger uniform half-space
- Rigid – a deep rigid body supported on layered soil
- Mat – a flexible mat supported on a uniform soil half-space

Options: (applicability is in accordance with Table 1)

- Surface/Embedded – designates if embedment impedance is to be included or excluded.
- Rectangle/Circle – designates if the mat or pile cap is circular or rectangular.
- Rigid/Flexible – designates if the mat or pile cap is rigid or flexible.

Table 1 (Availability of Options)

	Pile Cap	Half-Space	Stratum	Composite	Rigid	Mat
Embed	choice	choice	choice	choice	embed	surface
Shape	choice	choice	choice	choice	choice	rectangle
Rigidity	rigid	rigid	rigid	rigid	rigid	flexible

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Notes on Using Dyna Version 6.1

Note 1: Half-space and pile options are most commonly used and are explained below.

Note 2: Stratum & composite input are similar to half-space.

2E) Half-Space Foundation

Selecting “Half Space” in the Foundation Type dialog and then “OK” displays the Half-Space Foundation Properties dialog (Figure 7). This dialog may also be access using dropdown menu “Foundation – Half-Space Foundation”

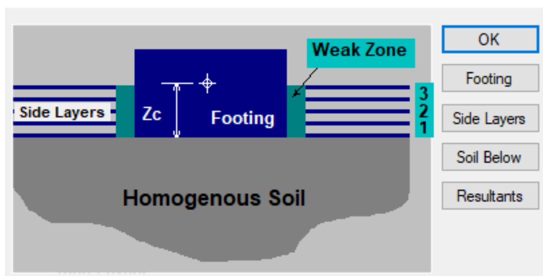


Figure 7 (Half-Space Foundation Properties)

The dialog buttons are described in the following subsections.

(a) “Footing” button – Pressing this button leads to the Footing Mass Properties dialog (Figure 8).

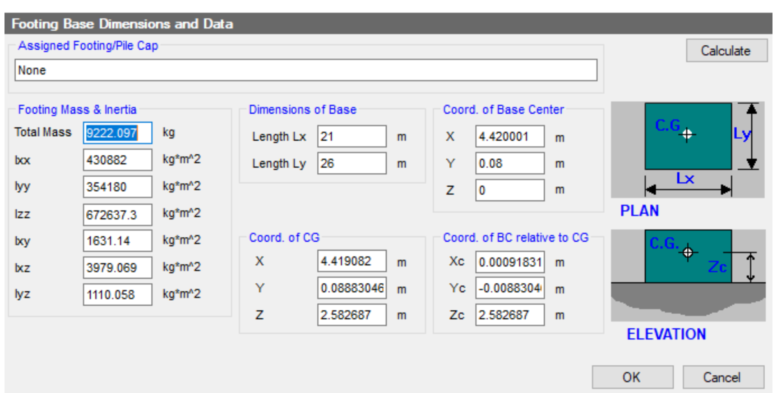


Figure 8 (Footing Mass Properties Dialog)

If known, footing properties may be entered directly into the data fields using the coordinate system indicated in Figure 1A. Alternatively, footing properties may be computed by pressing

Notes on Using Dyna Version 6.1

the "Calculate" button. This button leads to the Mass Calculation Utility as described in section 2G.

Footing Mass & Inertia – mass properties for foundation mat, pile cap, and machine. Includes total mass, mass moments (Ixx, Iyy, Izz), and cross mass moments (Ixy, Ixz, Iyz). All mass & inertia values are with respect to the CG.

- Dimensions of Base – used for base and embedment impedance calculation.
- Coordinates of CG (center of gravity) – location of the mat/pile cap/machine center of gravity.
- Coordinates of Base Center (BC) – location of the soil-mat interface centroid.
- Coordinates of BC relative to CG – not editable, calculated from BC and CG.

(b) "Side Layers" button – Pressing this button leads to the Embedment Layers dialog (Figure 9).

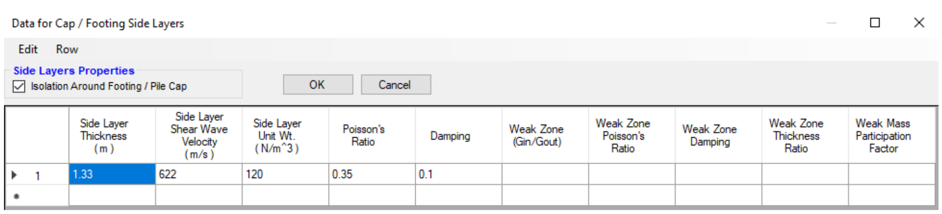


Figure 9 (Embedment Layers Dialog)

Checkbox "Isolation around footing/pile cap" shows or hides 5 columns for weak zone properties.

Side layer thickness – This is a vertical dimension. The summation of side layer thicknesses equals the embedment thickness.

Side layer shear wave velocity, unit weight, Poisson's ratio, and damping apply outside the weak zone.

Damping is soil material damping as described in Section 1A(i).

Weak zone (Gin/Gout) – This is a ratio of the inner weak zone shear modulus divided by the outer side layer shear modulus. The outer shear modulus is computed from the shear wave velocity and unit weight.

Notes on Using Dyna Version 6.1

Weak zone damping – Soil material damping inside the weak zone is as described in Section 1A(i).

Weak zone thickness ratio – This is a ratio of the weak zone horizontal width (between edge of weak zone and side layer) divided by foundation equivalent radius.

Weak zone mass participation factor – because weak zones do not include mass, this factor adds mass at a point at the center of foundation at mid layer thickness. For example, a factor of 0.5 adds half the weak zone mass. Typical values are 0.25 to 0.5 with a maximum of 0.75.

(c) “Soil Below” button – Pressing this button leads to Figure 10 dialog

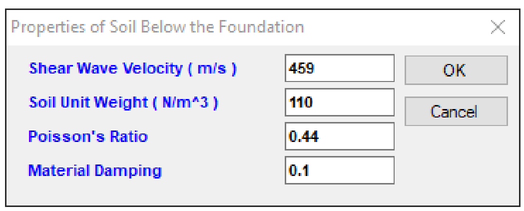


Figure 10 (Properties for Soil Below Foundation Dialog)

These are the properties of the half-space below the foundation. Material damping is as described in section 1A(i).

(d) “Resultants” button – Pressing this button leads to Figure 11 dialog

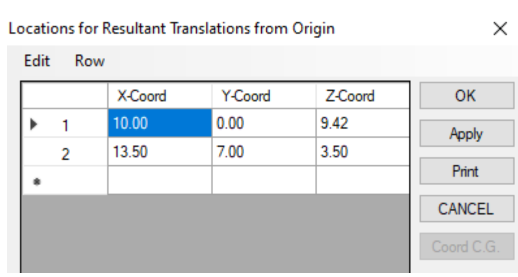


Figure 11 (Resultant Locations Dialog)

Used to define coordinate locations in m (ft) for the reporting of amplitudes, other than at the CG. The coordinate system of Figure 1A is used.

Notes on Using Dyna Version 6.1

2F) Pile Foundation

Selecting "Pile" in the Foundation Type dialog and then "OK" displays the Pile Foundation Properties dialog (Figure 12). This dialog may also be access using dropdown menu "Foundation – Pile Foundation"

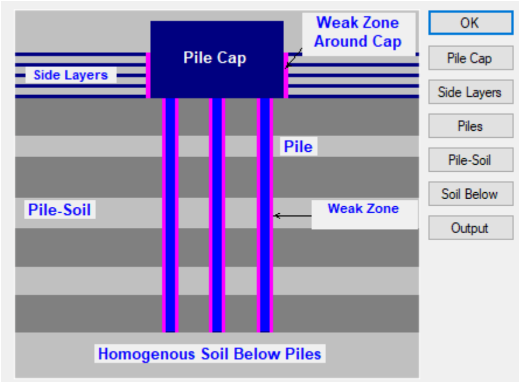


Figure 12 (Pile Foundation Properties)

(a) "Pile Cap" button – for input of mass and dimensional properties. Refer to Section 2E(a) and Figure 8.

(b) "Side Layers" button – for input of embedment properties. Refer to Section 2E(b) and Figure 9.

(c) "Piles" button – Pressing this button opens the Pile Properties Dialog (Figure 13). Pile locations shown in this dialog are with respect to the CG.

Notes on Using Dyna Version 6.1

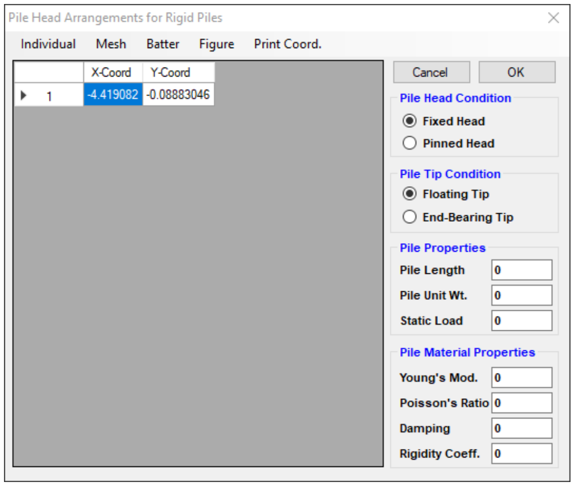


Figure 13 (Pile Properties Dialog)

Units of Measure – unlisted units are indicated in Table 2.

Table 2 (Figure 13 Units of Measure)

Item:	Imperial (metric):
Coordinates	ft (m)
Pile Length	ft (m)
Pile Unit Weight	lb/ft^3 (N/m^3)
Young's Modulus	lb/ft^2 (N/m^2)

Pile Coordinates – may be entered individually using the "Individual" menu option or as a group using the "Mesh" menu option. Within these options, pile locations are *not* with respect to the CG.

Pile Head Condition – this applies to the connection between top of pile and rigid pile cap. A pinned head results in zero rotational impedance.

Pile Tip Condition – this applies only to a low frequency modification affecting radiation damping.

Pile Damping – this is pile material damping as described in Section 1a(i)

Rigidity Coefficient – This is a Timoshenko shear coefficient and is basically a correction factor due to shear stresses not being uniformly distributed across a section. The traditional definition

Notes on Using Dyna Version 6.1

of this value is the average shear stress divided by the shear stress at the centroid. Some standard values are provided in Table 3.

Table 3 (Rigidity Coefficients)

Pile Shape	Rigidity Coefficient
Solid Circular	4/3
Solid Rectangle	3/2
Thin Tube	2
H or Box	(web area) / (cross-section area)

(d) “Pile-Soil” button – Pressing this button opens the Data for Pile-Soil System dialog (Figure 14A/B)

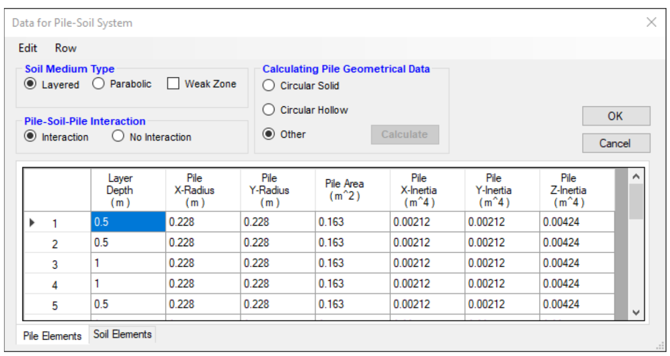


Figure 14A (Pile-Soil Dialog, Pile Elements Tab)

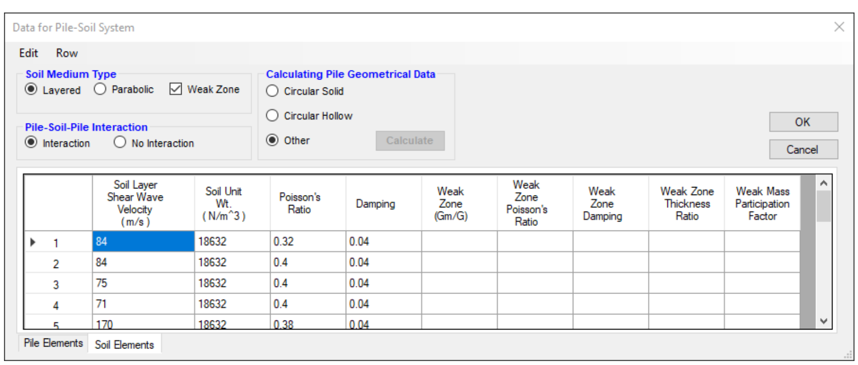


Figure 14B (Pile-Soil Dialog, Soil Elements Tab)

Layered/Parabolic – layered means layer data is provided by the user. Parabolic means layers is determined by Dyna based on a single set of maximum values.

Notes on Using Dyna Version 6.1

Weak Zone checkbox – if checked, shows additional weak zone data columns. These data fields are the same as used for embedment weak zones. Refer to Section2E(b) for details.

Interaction/No Interaction – includes or excludes pile interaction effects. Typically, interaction is included.

Calculating Pile Geometrical Data – this is an optional calculation for two cross-section types.

(e) “Soil Below” button – this is for properties of soil below pile tip and is similar to that for a half-space. Refer to Section 2E(c) and Figure 9.

(f) “Output” button – Pressing this button opens the Output Options Dialog (Figure 15)

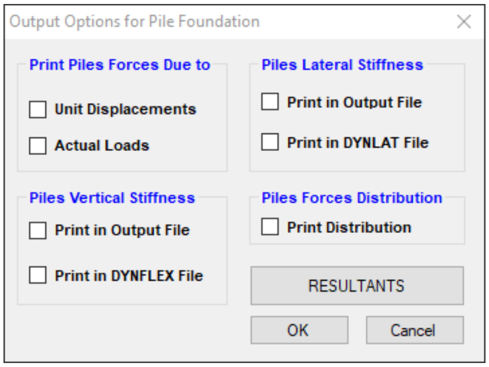


Figure 15 (Output Options Dialog)

<u>Checkboxes:</u> (provides data on individual piles)

- Unit Displacements – Leads to a dialog for frequency labels and pile numbers. The output text file includes absolute values at specified frequency numbers for specified pile numbers of load due to unit displacement of the rigid pile cap.
- Actual Loads – Leads to a dialog for frequency labels and pile numbers. The output text file includes absolute values at specified frequency numbers for specified pile numbers of load due to actual loads.
- Print in Output File (vertical stiffness) – at each frequency, lists the complex vertical stiffness matrix (upper triangle) in the output text file.
- Print in DYNFLEX File (vertical stiffness) – at each frequency, lists the complex vertical stiffness matrix (upper triangle) in a “DYNFLEX” text file.

Notes on Using Dyna Version 6.1

- Print in Output File (lateral stiffness) – at each frequency, lists the 2 (XZ, UZ) complex horizontal stiffness matrices (upper triangle) in the output text file.
- Print in DYNLAT file (lateral stiffness) – at each frequency, lists the 2 (XZ, UZ) complex horizontal stiffness matrices (upper triangle) in a "DYNLAT" text file.
- Print Distribution (pile forces) – at each frequency, lists absolute values of forces on individual piles due to unit displacements of rigid foundation.

"Resultants" button – same as described in Section 2E(d).

2G) Mass Calculation Utility

The Mass Calculation Utility (Figure 16) can be accessed from the Footing Mass Properties dialog (Figure 6) by pressing the "Calculate" button.

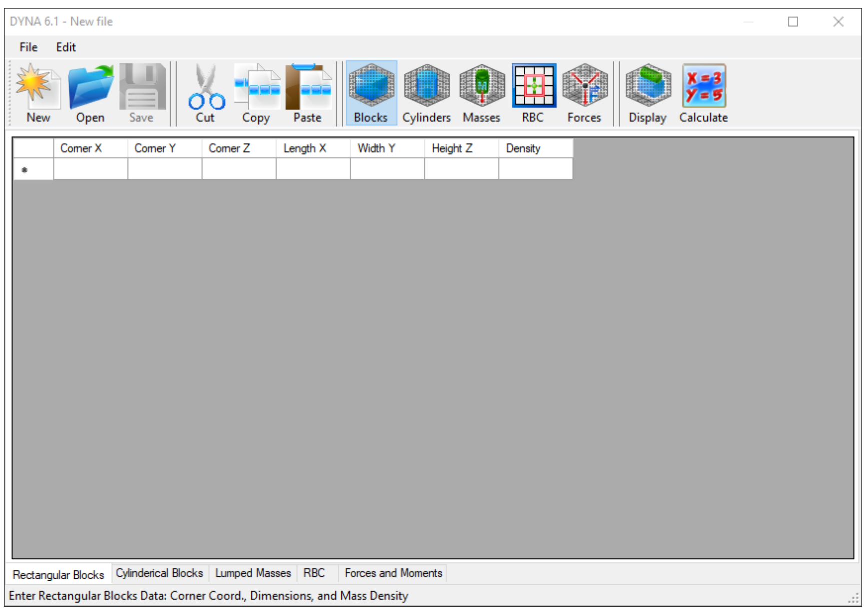


Figure 16 (Mass Calculation Utility)

This utility allows the input of foundation and machine components, and the calculation of associated properties and dimensions. Separate input tables are provided for,

Units of measure are not indicated within the utility but are as indicated in Table 4.

Notes on Using Dyna Version 6.1

Table 4 (Figure 16 Units of Measure)

Item:	**Imperial (metric):**
Coordinates, Dimensions	ft (m)
Mass	slug (kg)
Mass Moments	lb-ft^2 (kg-m^2)
Forces (quadratic)	slug-ft (kg-m)
Forces (non-quadratic)	lb (N)
Moments (quadratic)	slug-ft^2 (kg-m^2)
Moments (non-quadratic)	lb-ft (N-m)

- Rectangular Blocks – to represent foundation components. Locations are based on the corner closest to the cartesian axis origin.
- Cylindrical blocks – to represent circular foundation components (if any).
- Lumped masses – to represent machine components.
- RBC – to input footing/pile cap contact size and centroid.
- Forces/moments – to input magnitude of forces and moments, and applied location. This data partially duplicates loading input described in Section 2H. Note that input at this location is not provided for phase angles or for associated frequencies.

<u>Caveats:</u>

- Within this utility, the vertical Z axis is positive upward. Upon return to Dyna 6, all properties are transferred and converted accordingly.
- Mass density (i.e., slug/ft^3, kg/m^3) is used here instead of the unit weight (i.e., lb/ft^3, N/m^3) used for soil input.
- Moment loads are applied at the designated coordinates. The output text file reports input forces at the center of gravity.

<u>2H) Edit Load</u>

Pressing the Edit Load button from the Project Properties dialog displays the Loading Type dialog (Figure 17). This dialog can also be accessed using pulldown menu “Loads – Choose Load”.

Notes on Using Dyna Version 6.1

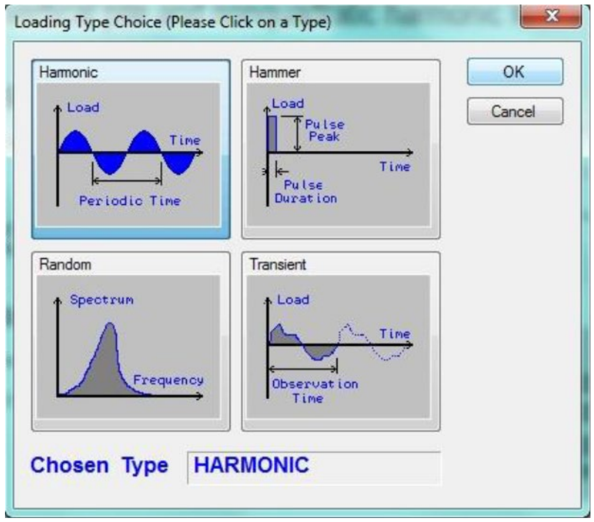
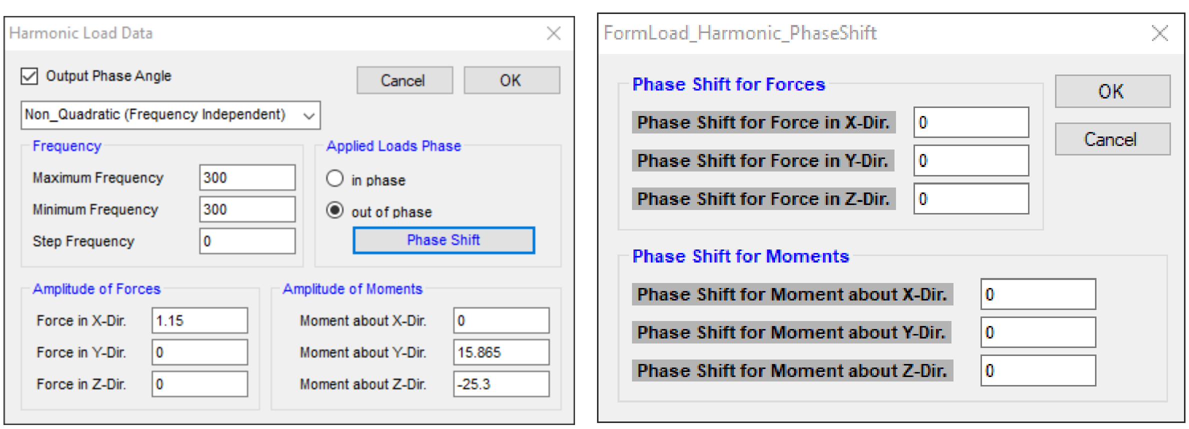


Figure 17 (Loading Type Dialog)

Options are provided for harmonic, hammer, random, and transient loading. Because compressors typically use the harmonic option, this will be described in further detail.

Note that there are 2 input screens for harmonic loading input. These screens are described in the following sub-sections.

(a) Harmonic Load dialog – Selecting the “Harmonic” option and then OK leads to the Harmonic Frequencies & Forces dialog (Figure 18A)



18A (Frequencies & Forces) **18B (phase angles)**

Figure 18 (Harmonic Loading Dialog)

Notes on Using Dyna Version 6.1

This dialog is for forces and moments at the center of gravity along with associated frequencies and phase angles.

- The quadratic option takes force input as a mass and dimension from the rotation axis (i.e., kilograms-meters) while the non-quadratic option takes force input as a force (i.e., newtons). For further information, refer to Attachment 1.
- Phase angles (Figure 17B) can be input by selecting "out of phase" and pressing the "Phase Shift" button. If "in phase", then all phase angles are set to zero.
- Units of measure are not indicated in Figure 18 but are as indicated in Table 5. Frequencies are determined by the user in the Settings Dialog (refer to Section 2C).

Table 5 (Figure 18 Units of Measure)

Item:	Imperial (metric):
Coordinates	ft (m)
Frequencies	user-determined
Forces (quadratic)	slug-ft (kg-m)
Forces (non-quadratic)	lb (N)
Moments (quadratic)	slug-ft^2 (kg-m^2)
Moments (non-quadratic)	lb-ft (N-m)
Phase Shifts	degrees

(b) Loads & Forces Dialog – Using the Mass Calculation Utility described in Section 2G, select the Forces & Moments tab (Figure 19).

Notes on Using Dyna Version 6.1

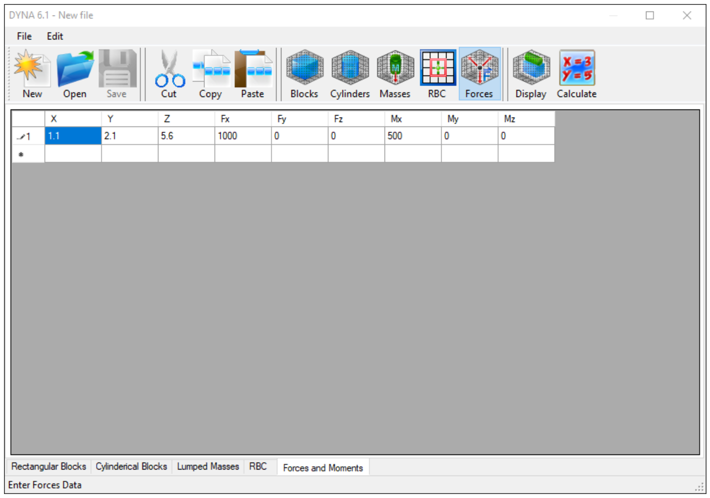


Figure 19 (Forces & Moments Dialog)

This dialog is used to input harmonic forces and moments based on specified coordinate locations.

- Cartesian coordinates are used as indicated in Figure 1B, with the vertical Z axis upward.
- Units of measure are indicated in Table 6.

Table 6 (Figure 19 Forces & Moments Tab, Units of Measure)

Item:	Imperial (metric):
Coordinates	ft (m)
Forces (quadratic)	slug-ft (kg-m)
Forces (non-quadratic)	lb (N)
Moments (quadratic)	slug-ft^2 (kg-m^2)
Moments (non-quadratic)	lb-ft (N-m)

- Associated frequencies and phase angles are not included in this dialog.

Upon exiting this utility, all data is transferred to Dyna's Harmonic Loading Dialog.

Locations, but not phase angles are provided in the 3D VIEW Forces & Moments Dialog (Figure 19), while forces and phase angles at the CG are editable in the Dyna 6 Harmonic Loading Dialog (Figure18). If needed, a utility (attachment 7) is provided to enter both locations and phase angles for direct input into the Dyna 6 Harmonic Loading Dialog.

Notes on Using Dyna Version 6.1

(c) Rotating Forces Input – Rotating dynamic forces are basically an offset mass rotating about a horizontal axis. This produces harmonic forces in the vertical and horizontal directions, with one being 90 degrees out of phase (refer to Attachment 1).

(d) Reciprocating Forces Input – Reciprocating forces consist of a set of forces at a primary frequency and another set of forces at a higher secondary frequency (refer to Attachment 2). Note that only one set of forces associated with a frequency can be handled in a single analysis run. In addition, the phase angles between individual load components are a combination from multiple cylinders and are often not automatically provided by the machine supplier.

Part 3: Analysis

3A) Run project – Pressing the “Run Project” button from the Project Properties dialog executes an analysis. An analysis can also be started by pressing ribbon icon “Run Project”.

The analysis includes a 6 dof coupled frequency domain solution of Attachment 3, Equation 3.

3B) Forced Response Analysis – Refer to main Guideline Section 6.5. Response results are computed as single amplitudes (zero to peak).

3C) Frequency Analysis – Refer to main Guideline Section 6.6. Natural frequencies are visually by using a large damping safety factor and viewing resulting amplitudes over a range of frequencies (Figure 20)

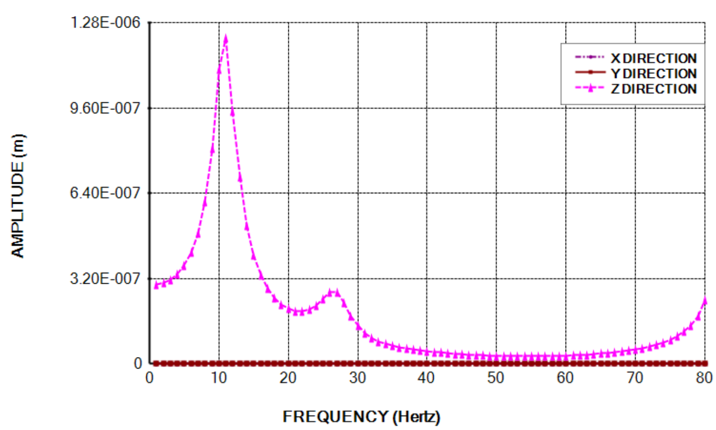


Figure 20 (Translation Response Chart)

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Notes on Using Dyna Version 6.1

Part 4: Results

4A) Text Results

After a successful analysis run, the Output Text file (Figure 21) is automatically displayed. Otherwise, it can be displayed using the “Output File button in the Project Properties dialog (Figure 4). Content of the Output Tex file is user-determined though the Settings Dialog (Figure 5) and Output Options Dialog (Figure 15).

```
*************************************************
*                                               *
*        D Y N A 6   S I M U L A T I O N        *
*                                               *
*          Copyright (c) Western - 2011         *
*             RUN DATE -  2024/ 8/19            *
*             TIME     -    10: 0:21            *
*             REVISION -  Jan.21/201            *
*                                               *
*************************************************

Guideline Example-First Case

                     DATA ECHO
                     *********

GRAVITATIONAL CONSTANT SET TO  32.20 ft/s**2
*****************************

FREQUENCY UNITS :  RPM
***************

LENGTH UNITS : ft
************

FORCE UNITS : lb
***********

MASS UNITS : slug
**********

STIFFNESS AND DAMPING CONSTANTS TO BE PRINTED
*********************************************


DAMPING SAFETY FACTOR -    2.00
*********************
```

Figure 21 (Sample of Output Text File)

4B) Graphical Results Utility

After a successful analysis run, the Graphical Results Utility (Figure 22) can be accessed using any of the listed items under drop down menu “Post Processing”.
The utility acts as an independent program and can store data separately from Dyna. Displayed results are transferred from Dyna. Tabular values on the left side can, if desired, be copied and pasted into Microsoft Excel.

Notes on Using Dyna Version 6.1

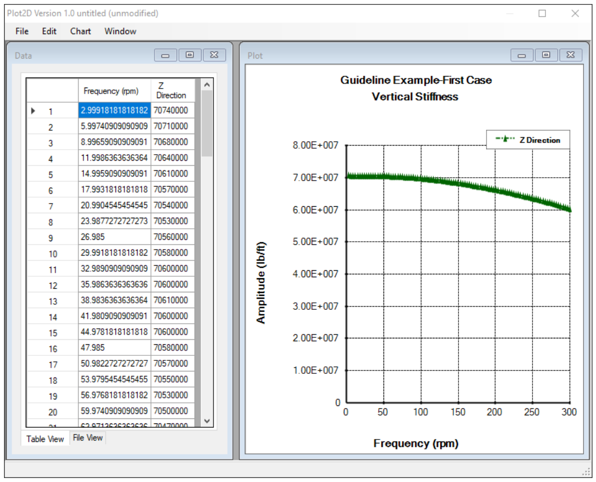

Figure 22 (Sample of Graphics Utility)

4C) Tabular Results

After a successful analysis run, tabular results can be accessed using Project Properties dialog buttons “Stiffness Table” (Figure 23A) and “Response” (Figure 23B). These tables can be copied and pasted into Microsoft Excel.

Stiffness Table

File

	Frequency (Hz)	Horizontal Stiffness X (N/m)	Horizontal Damping X (N/m/S)	Horizontal Stiffness Y (N/m)	Horizontal Damping Y (N/m/S)	Vertical Stiffness Z (N/m)	Vertical Damping Z (N/m/S)	Rocking Stiffness X (N.m/Rad)	Rocking Damping X (N.m/Rad/S)	Rocking Stiffness Y (N.m/Rad)	Rocking Damping Y (N.m/Rad/S)	Torsional Stiffness Z (N.m/Rad)	Torsional Damping Z (N.m/Rad/S)
1	1.00	3.67E+08	9.82E+06	3.67E+08	9.82E+06	3.37E+09	9.85E+07	5.12E+10	0.00E+00	5.12E+10	0.00E+00	5.38E+09	6.82E+07
2	2.00	3.65E+08	8.92E+06	3.65E+08	8.92E+06	3.31E+09	9.54E+07	4.92E+10	0.00E+00	4.92E+10	0.00E+00	5.28E+09	5.67E+07
3	3.00	3.59E+08	8.69E+06	3.59E+08	8.69E+06	3.21E+09	9.54E+07	4.65E+10	0.00E+00	4.65E+10	0.00E+00	5.14E+09	5.51E+07
4	4.00	3.49E+08	8.81E+06	3.49E+08	8.81E+06	3.05E+09	9.64E+07	4.39E+10	4.83E+07	4.39E+10	4.83E+07	4.96E+09	5.80E+07
5	5.00	3.35E+08	8.90E+06	3.35E+08	8.90E+06	2.84E+09	9.79E+07	4.15E+10	1.18E+08	4.15E+10	1.18E+08	4.76E+09	6.06E+07
6	6.00	3.18E+08	8.97E+06	3.18E+08	8.97E+06	2.56E+09	1.00E+08	3.92E+10	1.78E+08	3.92E+10	1.78E+08	4.54E+09	6.30E+07
7	7.00	2.98E+08	9.03E+06	2.98E+08	9.03E+06	2.25E+09	1.03E+08	3.73E+10	2.26E+08	3.73E+10	2.26E+08	4.32E+09	6.52E+07
8	8.00	2.74E+08	9.08E+06	2.74E+08	9.08E+06	1.90E+09	1.06E+08	3.56E+10	2.65E+08	3.56E+10	2.65E+08	4.08E+09	6.72E+07

Figure 23A (Sample of Tabular Impedance)

Notes on Using Dyna Version 6.1

Response Table

File

Center of Gravity | Resultant Points

	Frequency(H	Translational Response at CG - X (m)	Translational Response at CG - Y (m)	Translational Response at CG - Z (m)	Rotational Response at CG - X (Rad)	Rotational Response at CG - Y (Rad)	Rotational Response at CG - Z (Rad)
1	1.00	1.88E-28	2.11E-28	2.92E-07	2.39E-16	7.07E-15	5.34E-21
2	2.00	8.86E-29	2.95E-28	2.85E-07	8.25E-16	2.15E-15	8.81E-21
3	3.00	1.05E-27	4.20E-28	2.75E-07	1.63E-16	4.70E-15	3.08E-20
4	4.00	4.33E-28	9.01E-28	2.61E-07	2.04E-15	1.28E-15	9.45E-21
5	5.00	6.38E-29	2.40E-29	2.44E-07	1.06E-15	1.07E-15	1.81E-21
6	6.00	6.86E-28	3.30E-27	2.24E-07	5.99E-16	9.56E-16	2.25E-20

Figure 23B (Sample of Tabular Response)

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Sample Design: Reciprocating Machine Foundation

Purpose:

This attachment calculation illustrates the foundation design for a reciprocating compressor, soil supported, and using metric units.

Motor:

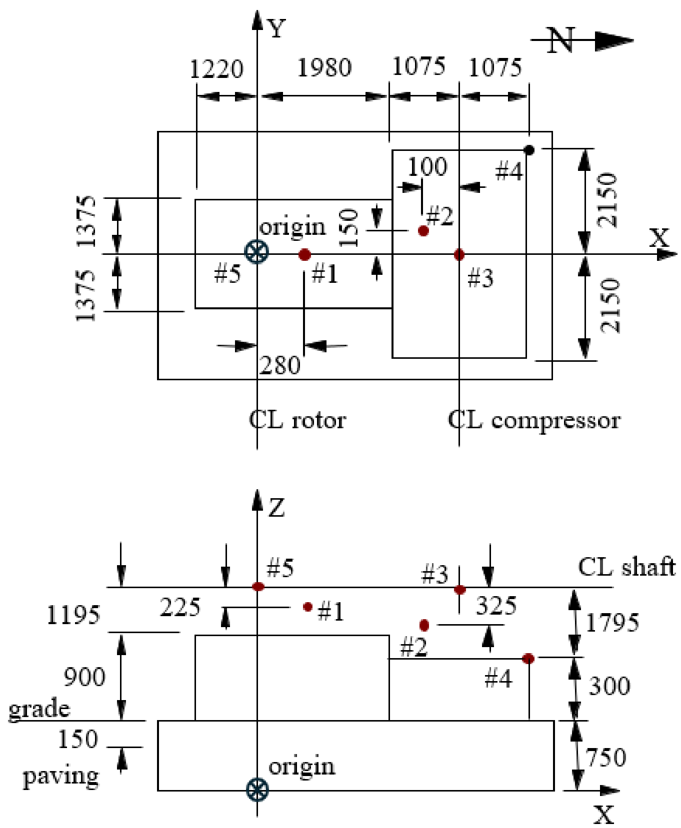
(data provided by the supplier)

Speed = 300 rpm

Weight = 66679 N (C.G. at point #1)

Dynamic Loads: (applied at point #5)

$F_Y = F_Z = 550$ N



Foundation Layout (units = mm)

Compressor:

(data provided by the supplier)

Weight = 88905 N (CG at point #2)

Primary speed = 300 rpm

Secondary speed = 600 rpm

Dynamic Loads: (applied at point #3)

	(primary)	(secondary)
F_X	0	0
F_Y	1070 N (0°)	182 N (-22°)
F_Z	782 N (31°)	0
M_X	0	0
M_Y	7595 N-m (45°)	0
M_Z	204720 N-m (35°)	2590 (45°)

Bolts: 4 – 42 mm diameter bolts per cylinder

cylinder diameter = 75 mm   maximum gas pressure = 43 MPa

rod diameter = 16 mm   minimum gas pressure = 3.5 MPa

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Sample Design: Reciprocating Machine Foundation

Existing Soil:

net bearing = 145 kPa

unit weight = 17000 N/m^3

shear modulus, G_S = 35 MPa = $3.5 * 10^7$ N/m^2

Poisson's ratio, μ_S = 0.44

material damping ratio = use 2%

Purchased Sand Backfill:

unit weight = 19000 N/m^3

shear modulus = 70 MPa = 7.0×10^7 N/m^2

Poisson's ratio = 0.35

damping ratio = use 2%

Concrete:

compressive strength f'c = 25 MPa

unit weight = 23500 N/m^3

elastic modulus, E_C = 23665 MPa

Trial Mat Size: assume 600 mm thick

L_X = pier dimension plus 600 mm
= [1.22 m) + (1.98 m) + (1.075 m) + (1.075m)] + 0.6 m
= 5.95 m, from previous trials, this is too small USE L_X = 6.4 m

L_Y = 1.5 [distance from shaft to bottom of mat]
= 1.5 [(0.6 m mat) + (0.15 m paving) + (0.9 m pier) + (1.195 m to shaft)]
= 4.268 m, from previous trials, this is too small USE L_Y = 7.9 m

verify thickness, L_Z = 600 mm

pier cantilever, d = [(7900 mm mat) - (2750 mm motor pier)] / 2 = 2575 mm

elastic modulus of soil,

$E_S = 2 (G_S)(1 + \mu_S) = 2 (35 \text{ MPa})(1 + 0.44) = 101 \text{ MPa}$

Sample Design: Reciprocating Machine Foundation

Thickness check

$[E_C / E_S]\ [L_Z / d]^3 \geq 1$ (Equation 5)

$[(23665\ \text{MPa}) / (101\ \text{MPa})]\ [(600\ \text{mm}) / (2575\ \text{mm})]^3$

$2.96 > 1$ OK, rigid USE L_z = 600 mm

Note 1: Use a thickness of 750 mm in the analysis to account for the 150 mm of concrete paving.

Note 2: Mat plan location is shown on page 17.

Resultant Locations:

Select point #4 for human tolerance, and points at extreme bearings for machine tolerance:

#4 (human)	(compr brg)	(motor brg)
Dyna #1	Dyna #2	Dyna #3
X = 4.13 m	X = 3.455 m	X = -0.3 m
Y = 2.15 m	Y = 0.0 m	Y = 0.0 m
Z = 1.05 m	Z = 2.845 m	Z = 2.845 m

Effective Embedment:

For embedment use 2/3 of embedded mat thickness, h = 2/3 (600 mm) = 400 mm

Center of Gravity Offset Check:

From 3D VIEW calculation, (page 18)

X coordinate of base relative to CG = 0.008 m

Y coordinate of base relative to CG = 0.01 m

Offset in X direction, OS_X = 0.008 m / 6.4 m = 0.00125 < 0.05, OK

Offset in Y direction OS_Y = 0.01 m / 7.9 m = 0.00127 < 0.05, OK

Soil Pressure Check:

Total weight = 132550 kg x 9.8 m/s^2 = 1,298,990 N

Moment about X axis, M_X = 1299 kN x 0.008 m = 10 kN-m

Moment about Y axis, M_Y = 1299 kN x 0.01 m = 13 kN-m

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Sample Design: Reciprocating Machine Foundation

Gross soil pressure = Total weight / (L_X x L_Y) + 6 M_X/(L_X x L_Y^2) + 6 M_Y/(L_Y x L_X^2)

= (1299) / (6.4 x 7.9) + 6(10) / (6.4 x 7.9^2) + 6(13) / (7.9 x 6.4^2) = 25.7 + 0.2 + 0.2 = 26.1 kPa

Net soil pressure = 26.1 kPa – (17.0 kN/m^3)(0.75 m) = 13.3 kPa < (145 kPa) / 2, OK

There are 3 soil cases:

COV = 0.3 (measured)

SC-1: soil properties as given (Equation 1d)

Soil below foundation,

V_S = sqrt [G_S g / U_S] = sqrt [(3.5 x 10^7 N/m^2)(9.8 m/s^2) / (17000 N/m^3)] = 142 m/s

Side soil,

V_S = sqrt [G_S g / U_S] = sqrt [(7.0 x 10^7 N/m^2)(9.8 m/s^2) / (19000 N/m^3)] = 190 m/s

SC-2: upper bound soil properties (Equations 1c and 1f)

Soil below foundation,

G_{UB} = G_S (1 + COV) = (3.5 x 10^7 N/m^2)(1 + 0.3) = 4.55 x 10^7 N/m^2

V_{UB} = sqrt [G_{UB} g / U_S] = sqrt [(4.55 x 10^7 N/m^2)(9.8 m/s^2) / (17000 N/m^3)] = 162 m/s

Side soil,

G_{UB} = G_S (1 + COV) = (7.0 x 10^7 N/m^2)(1 + 0.3) = 9.1 x 10^7 N/m^2

V_{UB} = sqrt [G_{UB} g / U_S] = sqrt [(9.1 x 10^7 N/m^2)(9.8 m/s^2) / (19000 N/m^3)] = 217 m/s

SC-3: lower bound soil properties (Equations 1b and 1e)

Soil below foundation,

G_{LB} = G_S / (1 + COV) = (3.5 x 10^7 N/m^2) / (1 + 0.3) = 2.69 x 10^7 N/m^2

V_{LB} = sqrt [G_{LB} g / U_S] = sqrt [(2.69 x 10^7 N/m^2)(9.8 m/s^2) / (17000 N/m^3)] = 125 m/s

Side soil,

G_{LB} = G_S / (1 + COV) = (7.0 x 10^7 N/m^2) / (1 + 0.3) = 5.38 x 10^7 N/m^2

V_{LB} = sqrt [G_{LB} g / U_S] = sqrt [(5.38 x 10^7 N/m^2)(9.8 m/s^2) / (19000 N/m^3)] = 167 m/s

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Sample Design: Reciprocating Machine Foundation

Frequency Analysis
Using Dyna 6 damping safety factor = 100

Based on initial analysis results, use a frequency sweep from 500 rpm to 1500 rpm at 10 rpm increments.

Nominal loading is included for the purpose of engaging each mode.

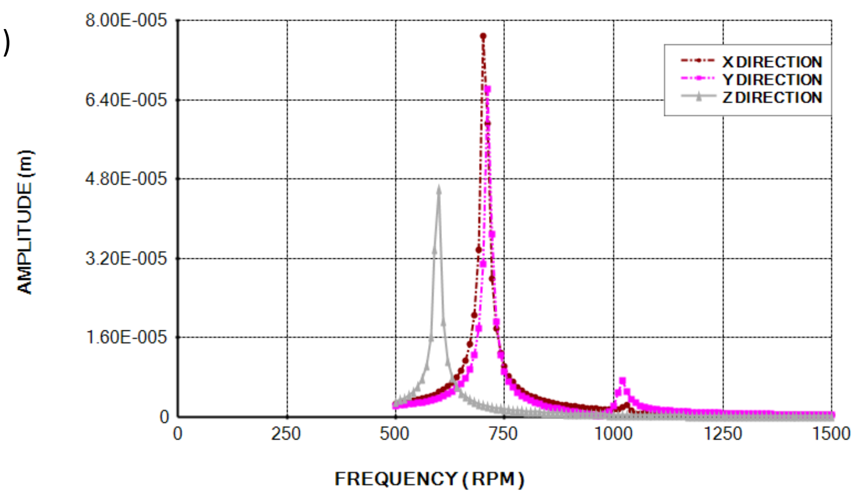
$F_X = F_Y = F_Z = 1000$ N

$M_X = M_Y = M_Z = 1000$ N-m

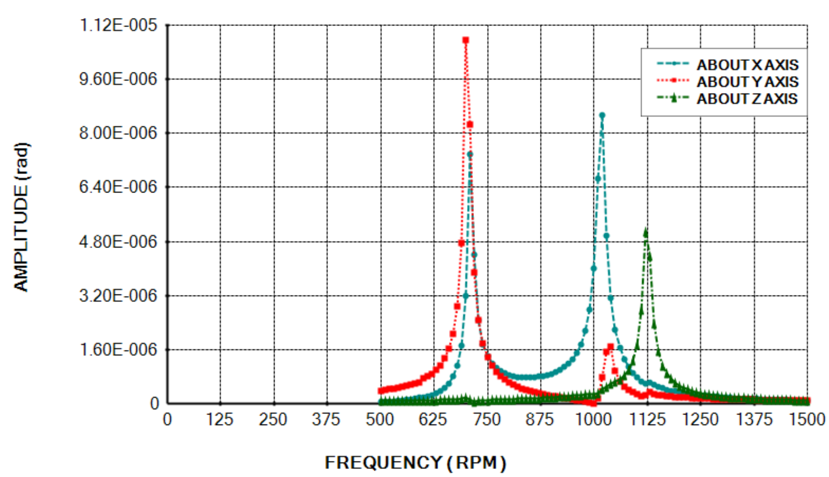
Use a resonant range from 0.8 to 1.2.

Soil Case SC-1: (soil properties as given)

Translations at CG:
X translation – 700 rpm, 1030 rpm
Y translation – 710 rpm, 1020 rpm
Z translation – 600 rpm



Rotations at CG:
X rotation – 710 rpm & 1020 rpm
Y rotation – 700 rpm & 1040 rpm
Z rotation – 1120 rpm



Sample Design: Reciprocating Machine Foundation

Summary:	Dyna 6 NF	MS(p) / NF	MS(s) / NF
Vertical	600	0.50	**1.0**
Torsional	1120	0.27	0.54
X trans / Y rot	700 & 1035	0.43, 0.29	**0.86**, 0.58
Y trans / X rot	710 & 1020	0.42, 0.29	**0.85**, 0.59

Soil Case SC-2: (upper bound soil properties)

For this guideline example, the details of this frequency sweep are not shown for brevity.

Summary:	Dyna 6 NF	MS(p) / NF	MS(s) / NF
Vertical	680	0.44	**0.88**
Torsional	1280	0.23	0.47
X trans / Y rot	800 & 1180	0.38, 0.25	0.75, 0.51
Y trans / X rot	810 & 1160	0.37, 0.26	0.75, 0.52

Soil Case SC-3: (lower bound soil properties)

For this guideline example, the details of this frequency sweep are not shown for brevity.

Summary:	Dyna 6 NF	MS(p) / NF	MS(s) / NF
Vertical	530	0.56	**1.13**
Torsional	990	0.30	0.61
X trans / Y rot	630 & 910	0.48, 0.33	**0.95**, 0.66
Y trans / X rot	620 & 900	0.48, 0.33	**0.97**, 0.67

Several secondary modes are within the resonant range. Therefore, use twice DSF for secondary force analysis.

Forced Response Analysis

Use damping safety factor = 2.0

Machine limit = 3.8 mm/s

Human limit = 0.61 m/s^2 based on 4 hours exposure, and 300/600 rpm

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RIGID FOUNDATIONS FOR VIBRATORY EQUIPMENT

Sample Design: Reciprocating Machine Foundation

Human Tolerance Calculation (using ISO 2631 Health Guidance Caution Zone)
W. Bounds (20Jan2025)

Category = health	(as opposed to comfort, perception, or motion sickness)		
Type = whole body	(as opposed to hand transmitted)		
Exposure = intermittent	(as opposed to continuous)		
<u>Input data:</u>			
Units of Measure		*metric*	Enter 1 for imperial (ft, lb), or anything else for metric (m, N).
daily exposure, DE =	14400	seconds	(range is 1 s to 24 h)
machine speed, MS	5	Hz	
<u>Permissible RMS acceleration</u>			ISO 2631-1, Section B.3.1
Amax	3	m/s2	if DE <= 10 minutes, then A rms = Amax
A rms =	0.612372436	m/s2	if DE > 10 minutes, then Arms = (Amax) sqrt [600 / exposure]
Orientation/Direction Factors			ISO 2631-1, Section 7.2.2
horizontal, kx, ky	1.4		
vertical, kz	1.0		
Frequency Weighting Factors			In lieu of ISO 2631, use SCI P354, Equations 22 & 23
horizontal, Wd	0.400		
vertical, Wb	1.000		
Permissible RMS acceleration			
X axis (horiz), Ax rms p	1.094	m/s2	= (A rms) / (kx Wd)
Y axis (horiz), Ay rms p	1.094	m/s2	= (A rms) / (ky Wd)
Z axis (vertical), Az rms p	0.612	m/s2	= (A rms) / (kz Wb)

MS	Wd	Wb
1 to 2	1	0.4
2 to 5	2 / MS	MS / 5
6 to 16	2 / MS	1
above 16	2 / MS	16 / MS

Sample Design: Reciprocating Machine Foundation

There are 3 Loading Cases

<u>LC 1: Motor Rotor</u> (300 rpm = 5 Hz = 31.4 rad/s)

For this guideline example, details are not shown for brevity

```
HARMONIC LOAD (NONQUADRATIC)
*************

MAXIMUM FREQUENCY ............   300.000  RPM
MINIMUM FREQUENCY ............   300.000  RPM
STEP FREQUENCY ...............    20.000  RPM
FORCE IN X-DIRECTION .........  0.00E+00 N
FORCE IN Y-DIRECTION .........  5.50E+02 N
FORCE IN Z-DIRECTION .........  5.50E+02 N
MOMENT ABOUT X-AXIS ..........  1.01E+03 N.m
MOMENT ABOUT Y-AXIS ..........  6.01E+02 N.m
MOMENT ABOUT Z-AXIS ..........  6.01E+02 N.m

PHASE SHIFTS OF EXTERNAL LOADS
******************************

FORCE IN X-DIRECTION .........      0.00 degrees
FORCE IN Y-DIRECTION .........      0.00 degrees
FORCE IN Z-DIRECTION .........     90.00 degrees
MOMENT ABOUT X-AXIS ..........    180.00 degrees
MOMENT ABOUT Y-AXIS ..........     90.00 degrees
MOMENT ABOUT Z-AXIS ..........    180.00 degrees
```

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Sample Design: Reciprocating Machine Foundation

LC 2: Compressor Primary (300 rpm = 5 Hz = 31.4 rad/s)

For phased loading determination, refer to page 19.

```
HARMONIC LOAD (NONQUADRATIC)
*************

MAXIMUM FREQUENCY ............   300.000  RPM
MINIMUM FREQUENCY ............   300.000  RPM
STEP FREQUENCY ...............    20.000  RPM
FORCE IN X-DIRECTION  .........  0.00E+00 N
FORCE IN Y-DIRECTION  .........  1.07E+03 N
FORCE IN Z-DIRECTION  .........  7.82E+02 N
MOMENT ABOUT X-AXIS  ..........  2.21E+03 N.m
MOMENT ABOUT Y-AXIS  ..........  6.33E+03 N.m
MOMENT ABOUT Z-AXIS  ..........  2.06E+05 N.m

PHASE SHIFTS OF EXTERNAL LOADS
******************************

FORCE IN X-DIRECTION  .........      0.00 degrees
FORCE IN Y-DIRECTION  .........      0.00 degrees
FORCE IN Z-DIRECTION  .........     31.00 degrees
MOMENT ABOUT X-AXIS  ..........    180.00 degrees
MOMENT ABOUT Y-AXIS  ..........     48.00 degrees
MOMENT ABOUT Z-AXIS  ..........     35.00 degrees
```

Sample Design: Reciprocating Machine Foundation

LC 3: Compressor Secondary (600 rpm = 10 Hz = 62.8 rad/s)

For this guideline example, details are not shown for brevity

```
HARMONIC LOAD (NONQUADRATIC)
*************

MAXIMUM FREQUENCY  ............   600.000  RPM
MINIMUM FREQUENCY  ............   600.000  RPM
STEP FREQUENCY  ...............    20.000  RPM
FORCE IN X-DIRECTION  .........  0.00E+00 N
FORCE IN Y-DIRECTION  .........  1.82E+02 N
FORCE IN Z-DIRECTION  .........  0.00E+00 N
MOMENT ABOUT X-AXIS  ..........  3.76E+02 N.m
MOMENT ABOUT Y-AXIS  ..........  0.00E+00 N.m
MOMENT ABOUT Z-AXIS  ..........  2.72E+03 N.m

PHASE SHIFTS OF EXTERNAL LOADS
******************************

FORCE IN X-DIRECTION  .........      0.00 degrees
FORCE IN Y-DIRECTION  .........    -22.00 degrees
FORCE IN Z-DIRECTION  .........      0.00 degrees
MOMENT ABOUT X-AXIS  ..........    158.00 degrees
MOMENT ABOUT Y-AXIS  ..........      0.00 degrees
MOMENT ABOUT Z-AXIS  ..........     39.00 degrees
```

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Sample Design: Reciprocating Machine Foundation

Soil Case SC-1: (soil properties as given)

Refer to Dyna 6 input & results which begin on page20.

Point #4 (Dyna #1), RMS Acceleration for human tolerance:

Peak Amplitudes: (m)	X	Y	Z	
LC 1 (Motor Rotor)	1.105E-07	9.774E-07	8.959E-07	
LC 2 (Compr Primary)	2.954E-05	3.562E-05	3.214E-06	
Sum Primary	2.965E-05	3.660E-05	4.110E-06	
LC 3 (Compr Secondary)	5.479E-07	9.979E-07	2.110E-07	
Equations 10b & 11c	A rms = (0.707)(Sum Primary)(31.4 rad/s)^2 + (0.707)(Compr Secondary)(62.8 rad/s)^2			
RMS Acceleration: (m/s2)	0.0222	0.0283	0.0035	

Utilization = (0.0293 m/s2) / (0.61 m/s^2) = 0.05 < 1.0, OK

Compressor Bearing (Dyna #2), Peak Velocity for machine tolerance:

Peak Amplitudes: (m)	X	Y	Z	
LC 1 (Motor Rotor)	2.034E-07	1.182E-06	7.315E-07	
LC 2 (Compr Primary)	2.172E-06	2.623E-05	2.232E-06	
Sum Primary	2.375E-06	2.741E-05	2.964E-06	
LC 3 (Compr Secondary)	3.705E-09	9.846E-07	1.730E-09	
Equations 10a	V peak = (Sum Primary)(31.4 rad/s) + (Compr Secondary)(62.8 rad/s)			
Peak Velocity: (mm/s)	0.0748	0.9226	0.1862	

Utilization = (0.9226 m/s) / (3.8 mm/s) = 0.24 < 1.0, OK

Sample Design: Reciprocating Machine Foundation

Motor Bearing (Dyna #3) Peak Velocity for machine tolerance:

Peak Amplitudes: (m)	X	Y	Z
LC 1 (Motor Rotor)	2.034E-07	1.041E-06	4.943E-07
LC 2 (Compr Primary)	2.172E-06	2.399E-05	8.195E-07
Sum Primary	2.375E-06	2.503E-05	1.314E-06
LC 3 (Compr Secondary)	3.705E-09	8.744E-07	9.457E-10
Equations 10a	V peak = (Sum Primary)(31.4 rad/s) + (Compr Secondary)(62.8 rad/s)		
Peak Velocity: (mm/s)	0.0748	0.8409	0.0826

Utilization = (0.8409 m/s) / (3.8 mm/s) = 0.22 < 1.0, OK

Soil Case SC-2: (soil upper bound)

For this guideline example, details are not shown for brevity.

Point #4 (Dyna #1), RMS Acceleration for human tolerance:

Peak Amplitudes: (m)	X	Y	Z
LC 1 (Motor Rotor)	8.161E-08	7.291E-07	6.590E-07
LC 2 (Compr Primary)	2.190E-05	2.641E-05	2.379E-06
Sum Primary	2.198E-05	2.714E-05	3.038E-06
LC 3 (Compr Secondary)	3.791E-07	5.885E-07	1.306E-07
Equations 10b & 11c	A rms = (0.707)(Sum Primary)(31.4 rad/s)^2 + (0.707)(Compr Secondary)(62.8 rad/s)^2		
RMS Acceleration: (m/s2)	0.0164	0.0206	0.0025

Utilization = (0.0206 m/s2) / (0.61 m/s^2) = 0.03, < 1.0, OK

Sample Design: Reciprocating Machine Foundation

Compressor Bearing (Dyna #2), Peak Velocity for machine tolerance;

Peak Amplitudes: (m)	X	Y	Z	
LC 1 (Motor Rotor)	1.480E-07	8.775E-07	5.451E-07	
LC 2 (Compr Primary)	1.578E-06	1.944E-05	1.663E-06	
Sum Primary	1.726E-06	2.032E-05	2.208E-06	
LC 3 (Compr Secondary)	1.518E-09	5.680E-07	7.729E-10	
Equations 10a	V peak = (Sum Primary)(31.4 rad/s) + (Compr Secondary)(62.8 rad/s)			
Peak Velocity: (mm/s)	0.0543	0.6736	0.1387	

Utilization = (0.6736 m/s) / (3.8 mm/s) = 0.18 < 1.0, OK

Motor Bearing (Dyna #3) Peak Velocity for machine tolerance:

Peak Amplitudes: (m)	X	Y	Z	
LC 1 (Motor Rotor)	1.480E-07	7.723E-07	3.665E-07	
LC 2 (Compr Primary)	1.578E-06	1.783E-05	5.641E-07	
Sum Primary	1.726E-06	1.860E-05	9.306E-07	
LC 3 (Compr Secondary)	1.518E-09	6.090E-07	3.553E-10	
Equations 10a	V peak = (Sum Primary)(31.4 rad/s) + (Compr Secondary)(62.8 rad/s)			
Peak Velocity: (mm/s)	0.0543	0.6224	0.0585	

Utilization = (0.6224 m/s) / (3.8 mm/s) = 0.16 < 1.0, OK

Soil Case SC-3: (soil lower bound)

For this guideline example, details are not shown for brevity.

Point #4 (Dyna #1), RMS Acceleration for human tolerance:

Peak Amplitudes: (m)	X	Y	Z	
LC 1 (Motor Rotor)	1.490E-07	1.308E-06	1.215E-06	
LC 2 (Compr Primary)	2.190E-05	2.641E-05	2.379E-06	
Sum Primary	2.205E-05	2.772E-05	3.594E-06	
LC 3 (Compr Secondary)	8.027E-07	1.731E-06	3.209E-07	
Equations 10b & 11c	A rms = (0.707)(Sum Primary)(31.4 rad/s)^2 + (0.707)(Compr Secondary)(62.8 rad/s)^2			
RMS Acceleration: (m/s2)	0.0176	0.0241	0.0034	

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Sample Design: Reciprocating Machine Foundation

Utilization = (0.0241 m/s2) / (0.61 m/s^2) = 0.04 < 1.0, OK

Compressor Bearing (Dyna #2), Peak Velocity for machine tolerance:

Peak Amplitudes: (m)	X	Y	Z
LC 1 (Motor Rotor)	2.799E-07	1.590E-06	9.770E-07
LC 2 (Compr Primary)	1.578E-06	1.944E-05	1.663E-06
Sum Primary	1.858E-06	2.103E-05	2.640E-06
LC 3 (Compr Secondary)	8.435E-09	1.702E-06	3.290E-09
Equations 10a	V peak = (Sum Primary)(31.4 rad/s) + (Compr Secondary)(62.8 rad/s)		
Peak Velocity: (mm/s)	0.0589	0.7672	0.1661

Utilization = (0.7672m/s) / (3.8 mm/s) = 0.20 < 1.0, OK

Motor Bearing (Dyna #3) Peak Velocity for machine tolerance:

Peak Amplitudes: (m)	X	Y	Z
LC 1 (Motor Rotor)	2.779E-07	1.403E-06	6.654E-07
LC 2 (Compr Primary)	1.578E-06	1.783E-05	5.641E-07
Sum Primary	1.856E-06	1.923E-05	1.230E-06
LC 3 (Compr Secondary)	8.435E-09	1.061E-06	2.570E-09
Equations 10a	V peak = (Sum Primary)(31.4 rad/s) + (Compr Secondary)(62.8 rad/s)		
Peak Velocity: (mm/s)	0.0588	0.6705	0.0775

Utilization = (0.6705 m/s) / (3.8 mm/s) = 0.18 < 1.0, OK

Anchor Bolt Check:

use maximum and minimum gas pressures for head and crank pressures.

cylinder head area,

$A_{head} = \pi\ (75\ mm)^2 / 4 = 4417.9\ mm^2$

crank area,

$A_{crank} = A_{head} - \pi\ (16\ mm)^2 / 4 = (4417.9\ mm^2) - \pi\ (16\ mm)^2 / 4 = 4216.8\ mm^2$

Sample Design: Reciprocating Machine Foundation

cylinder force:

$F_{fdn} = [(P_{head})(A_{head}) - (P_{crank})(A_{crank})]\ F_{cr} / F_{red}$ (Equation 13)

$= [(43\ MPa)(\ 4417.9\ mm^2) - (3.5\ MPa)(\ 4216.8\ mm^2)]\ 1.15 / 2.0$

$= 100{,}747\ N$

bolt force,

$F_{bolt} = (F_{fdn}) / N = (100{,}747\ N) / (4\ bolts) = 25{,}187\ N$

bolt pre-tensioning, (Equation 14)

$T_{min} = F_{bolt} / \mu - W_r = (25{,}187\ N) / 0.15 - (say\ 0\ N) = 167{,}913\ N$

bolt tensile stress area, galvanized, $A_b = 1121\ mm^2$ (PIP STE05121, Table 1M)

limit pre-tensioning to 80% of yield for A36 bolt material,

$T_{allow} = 0.8\ (F_y)(A_b) = 0.8\ (250\ MPa)(1121\ mm^2) = 224{,}200\ N\ \ > T_{min}$, OK

For this example calculation, the anchor bolt embedment check is not shown for brevity. Refer to PIP STE05121, ASCE Anchorage Design, for details.

Mat Reinforcing Design:

mat cantilever = [7.9 m mat) - (2.75 m motor pier)] / 2 = 2.575 m

By Inspection, a 2575 mm cantilever and a 600 mm thick mat does not require more than nominal reinforcing.

$A_s = (0.0018)\ (600\ mm\ depth)\ (1000\ mm\ unit\ width) = 1080\ mm^2$

USE 19 mm at 300 mm each way, top and bottom

A_s (provided) = 2 (284 mm^2) (1000 mm / 300 mm) = 1893 mm^2

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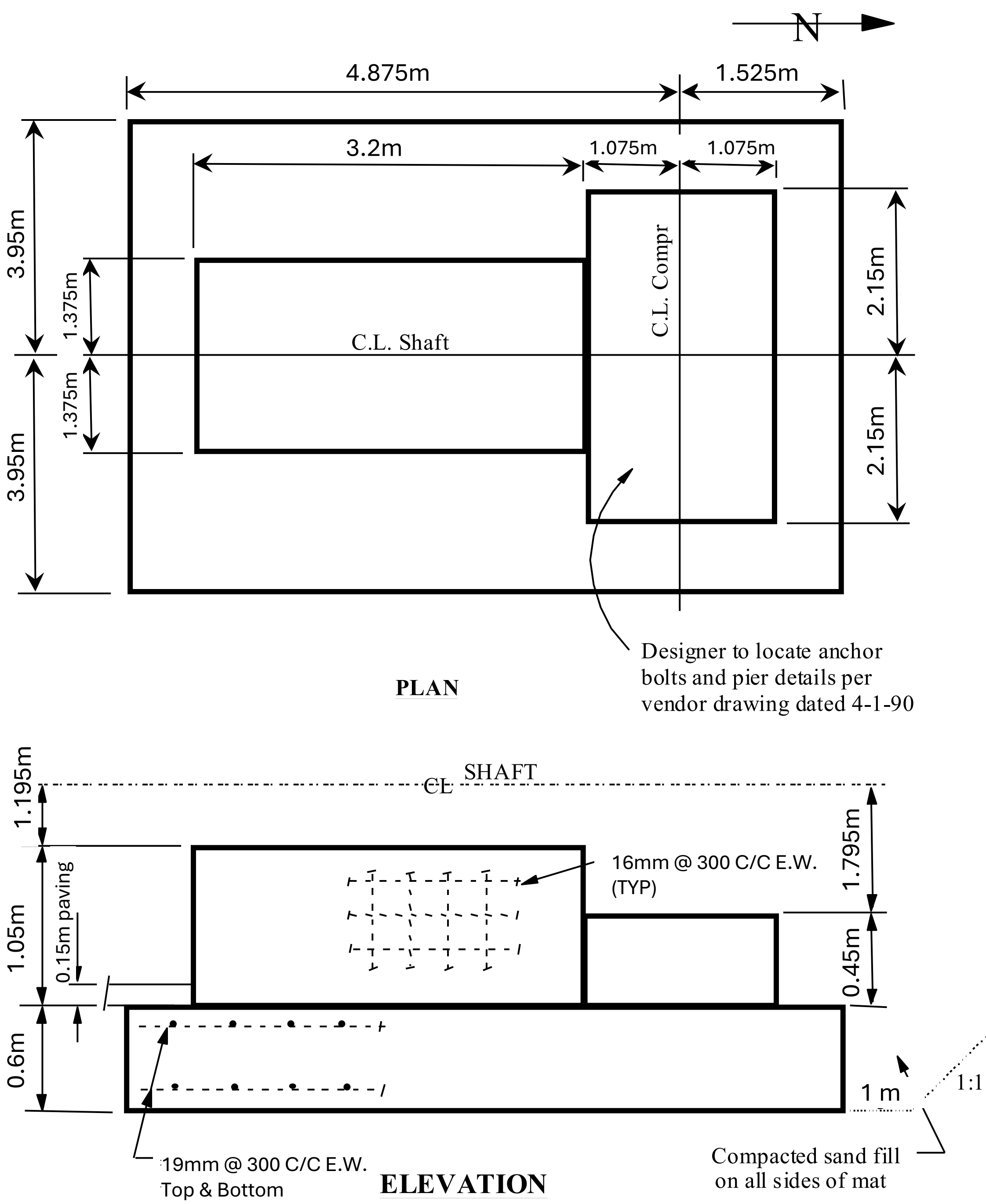
Sample Design: Reciprocating Machine Foundation



Sample Design: Reciprocating Machine Foundation

3D VIEW Sub-Program Input, (note that the vertical z axis is upward positive)

Rectangular Blocks

	Corner X	Corner Y	Corner Z	Length X	Width Y	Height Z	Density
1	-1.82	-3.95	0	6.4	7.9	0.75	2400
2	-1.22	-1.375	0.75	3.2	2.75	0.9	2400
3	1.98	-2.15	0.75	2.15	4.3	0.3	2400

Cylinderical Blocks

	Center X	Center Y	Center Z	Radius	Height	Density	Orientation

Lumped Masses

	X	Y	Z	Mass	IXX	IYY	IZZ	IXY	IXZ	IYZ
1	0.28	0	2.62	6804	0	0	0	0	0	0
2	2.955	0.15	2.52	9072	0	0	0	0	0	0

RBC

	X Length	Y Length	Base Center X	Base Center Y	Base Center Z
1	6.4	7.9	1.38	0	0

3D-VIEW Sub-Program Results, (note that the vertical z axis is upward positive)

```
Total Mass = 132548.4

Center of Mass X-Coord. = 1.37204418914148
Center of Mass Y-Coord. = 0.0102664385235884
Center of Mass Z-Coord. = 0.781724109834596

Mass Moment of Inertial IXX = 570220.021583085
Mass Moment of Inertial IYY = 472313.972511403
Mass Moment of Inertial IZZ = 893579.935789233
Mass Moment of Inertial IXY = 2154.08626741628
Mass Moment of Inertial IXZ = 4446.91452554383
Mass Moment of Inertial IYZ = 2365.44583133708

Total Force in X-Direction = 1000
Total Force in Y-Direction = 1000
Total Force in Z-Direction = 1000

Total Moment about X-Direction = 3271.45767131101
Total Moment about Y-Direction = 3090.32007930688
Total Moment about Z-Direction = 1138.22224938211

Length about X-Direction = 6.4
Length about Y-Direction = 7.9

X Coord. of BC relative to CG = 0.00795581085852382
Y Coord. of BC relative to CG = -0.0102664385235884
Z Coord. of BC relative to CG = 0.781724109834596
```

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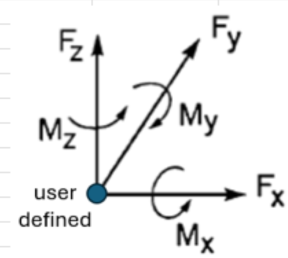
Sample Design: Reciprocating Machine Foundation

Phased Loading Spreadsheet, Loading Case LC 2 (compressor primary)

Phased Dynamic Loading Calculation
W. Bounds (20Jan2025)

Input Data: (3D VIEW coordinate system)

Title:	Loading Case LC 2 (compressor primary)			
Options:				
Metric?		*metric*	Enter 1 for imperial (ft, lb), or anything else for metric (m, N).	
Quadratic?		*non-quadratic*	Enter 1 for quadratic loads (mass x eccentricity)	
Center of Gravity:				
X coordinate, CGx	1.372	m		
Y coordinate, CGy	0.01	m		
Z coordinate, CGz	0.782	m		
Point of Load Application:				
X coordinate, Lx	3.055	m		
Y coordinate, Ly	0	m		
Z coordinate, Lz	2.845	m		
Forces:	Magnitude		Phase Angle	
X force, Fx	0	N	0	deg
Y force, Fy	1070	N	0	deg
Z force, Fz	782	N	31	deg
Moments:	Magnitude		Phase Angle	
X moment, Mx	0	N-m	0	deg
Y moment, My	7595	N-m	45	deg
Z moment, Mz	204720	N-m	35	deg



Results: (for entry into DYNA 6)

	magnitude		phase	
X force	0.0	N	0	deg
Y force	1,070.0	N	0	deg
Z force	782.0	N	31	deg
X moment	2,214.1	N-m	-180	deg
Y moment	6,326.0	N-m	48	deg
Z moment	206,197.7	N-m	35	deg

Sample Design: Reciprocating Machine Foundation

DYNA 6 Output File (note that vertical Z axis is downward positive)

Soil Case SC-1 (Soil Properties As Given)

Loading Case LC 1 (Motor Rotor)

Sample Design: Reciprocating Machine Foundation

```
*************************************************
*                                               *
*        D Y N A 6   S I M U L A T I O N        *
*                                               *
*          Copyright (c) Western - 2011         *
*             RUN DATE -  2025/ 1/24            *
*             TIME     -    13:21:38            *
*             REVISION -  Jan.21/201            *
*                                               *
*************************************************

Guideline 1234 Att 5-Compressor Primary

                     DATA ECHO
                     *********

GRAVITATIONAL CONSTANT SET TO   9.81 m/s**2
*****************************

FREQUENCY UNITS :  RPM
***************

LENGTH UNITS : m
************

FORCE UNITS : N
***********

MASS UNITS : kg
**********

STIFFNESS AND DAMPING CONSTANTS TO BE PRINTED
*********************************************


DAMPING SAFETY FACTOR -    2.00
*********************


FOUNDATION TYPE - HALF-SPACE
***************

RECTANGULAR FOUNDATION  -     6.400 m   BY   7.900 m
**********************
```

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Sample Design: Reciprocating Machine Foundation

```
MASS CONSTANTS
**************

TOTAL MASS OF FOOTING  ......... 1.3255E+05  kg
MASS MOMENT ABOUT X-AXIS  ...... 5.7022E+05  kg.m**2
MASS MOMENT ABOUT Y-AXIS  ...... 4.7231E+05  kg.m**2
MASS MOMENT ABOUT Z-AXIS  ...... 8.9358E+05  kg.m**2
CROSS PRODUCT X-Y  ............. 2.1541E+03  kg.m**2
CROSS PRODUCT X-Z  ............. 4.4469E+03  kg.m**2
CROSS PRODUCT Y-Z  ............. 2.3654E+03  kg.m**2

COORDINATES OF BASE CENTRE
**************************

X-COORD. OF BASE CENTRE REL. TO C.G.  ..     0.008  m
Y-COORD. OF BASE CENTRE REL. TO C.G.  ..    -0.010  m
Z-COORD. OF BASE CENTRE REL. TO C.G.  ..     0.782  m

SOIL
****

SOIL CONSTANTS  NUMBER OF SIDE LAYERS -   1
**************

LAYER  LAYER  SHEAR WAVE   UNIT   POISSON'S  MATERIAL
 NO.   DEPTH   VELOCITY   WEIGHT    RATIO     DAMPING
         m       m/s      N/m**3

  1    0.400    190.00  19000.00    0.350     0.040

HALF-SPACE SOIL
***************

SHEAR WAVE    UNIT    POISSON'S  MATERIAL
 VELOCITY    WEIGHT     RATIO     DAMPING
   m/s       N/m**3

 1.42E+02   17000.00   0.440     0.040
```

Sample Design: Reciprocating Machine Foundation

```
HARMONIC LOAD (NONQUADRATIC)
*************

MAXIMUM FREQUENCY  ............   300.000  RPM
MINIMUM FREQUENCY  ............   300.000  RPM
STEP FREQUENCY  ...............    20.000  RPM
FORCE IN X-DIRECTION  .........  0.00E+00 N
FORCE IN Y-DIRECTION  .........  1.07E+03 N
FORCE IN Z-DIRECTION  .........  7.82E+02 N
MOMENT ABOUT X-AXIS  ..........  2.21E+03 N.m
MOMENT ABOUT Y-AXIS  ..........  6.33E+03 N.m
MOMENT ABOUT Z-AXIS  ..........  2.06E+05 N.m

PHASE SHIFTS OF EXTERNAL LOADS
******************************

FORCE IN X-DIRECTION  .........      0.00 degrees
FORCE IN Y-DIRECTION  .........      0.00 degrees
FORCE IN Z-DIRECTION  .........     31.00 degrees
MOMENT ABOUT X-AXIS  ..........    180.00 degrees
MOMENT ABOUT Y-AXIS  ..........     48.00 degrees
MOMENT ABOUT Z-AXIS  ..........     35.00 degrees


RESPONSE PHASE SHIFT TO BE PRINTED
**********************************
RESULTANT TRANSLATIONS REQUIRED AT  3 POINTS
*******************************************

COORDINATES OF POINTS :

POINT     X         Y         Z
 NO.      m         m         m

  1      2.758     2.140    -0.268
  2      2.083    -0.010    -2.063
  3     -1.672    -0.010    -2.063
```

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Sample Design: Reciprocating Machine Foundation

```
*************************************************
*                                               *
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*                                               *
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*             TIME     -    13:21:38            *
*             REVISION -  Jan.21/201            *
*                                               *
*************************************************

Guideline 1234 Att 5-Compressor Primary

                                RESULTS
                                *******
STIFFNESS CONSTANTS (K)
***********************

HORIZONTAL TRANSLATION  (X) ...  8.202E+08  N/m
HORIZONTAL TRANSLATION  (Y) ...  8.202E+08  N/m
VERTICAL TRANSLATION  (Z) .....  9.242E+08  N/m
ROTATION ABOUT  (X) ...........  1.246E+10  N.m/rad
ROTATION ABOUT  (Y) ...........  9.562E+09  N.m/rad
TORSION ABOUT  (Z) ............  1.626E+10  N.m/rad
CROSS-STIFFNESS (YZ PLANE) ....  6.194E+08  N/rad
CROSS-STIFFNESS (XZ PLANE) .... -6.194E+08  N/rad

DAMPING CONSTANTS (C)
*********************

HORIZONTAL TRANSLATION  (X) ...  9.688E+06  N/m/s
HORIZONTAL TRANSLATION  (Y) ...  9.688E+06  N/m/s
VERTICAL TRANSLATION  (Z) .....  1.480E+07  N/m/s
ROTATION ABOUT  (X) ...........  4.917E+07  N.m/rad/s
ROTATION ABOUT  (Y) ...........  3.333E+07  N.m/rad/s
TORSION ABOUT  (Z) ............  4.718E+07  N.m/rad/s
CROSS-DAMPING (YZ PLANE) ......  6.946E+06  N/rad/s
CROSS-DAMPING (XZ PLANE) ...... -6.946E+06  N/rad/s
```

Sample Design: Reciprocating Machine Foundation

```
*************************************************
*                                               *
*         D Y N A 6   S I M U L A T I O N       *
*                                               *
*           Copyright (c) Western - 2011        *
*              RUN DATE -  2025/ 1/24           *
*              TIME     -    13:21:38           *
*              REVISION -  Jan.21/201           *
*                                               *
*************************************************

Guideline 1234 Att 5-Compressor Primary

                                RESULTS
                                *******
```

FLUOR®

Sample Design: Reciprocating Machine Foundation

```
                     FOOTING RESPONSE AMPLITUDES
                     ---------------------------

 FREQ.    TRANS. X  TRANS. Y  VERTICAL  ROT ABOUT ROT ABOUT TORSIONAL
  RPM     DIRECTION DIRECTION DIRECTION  X AXIS    Y AXIS    Z AXIS
             (m)       (m)       (m)     (rad)     (rad)     (rad)
-------   --------- --------- --------- --------- --------- ---------

 300.00    7.96E-07  1.75E-06  8.48E-07  2.81E-07  7.34E-07  1.33E-05


                    PHASE SHIFT OF RESPONSE (RAD)
                    -----------------------------

 FREQ.    TRANS. X  TRANS. Y  VERTICAL  ROT ABOUT ROT ABOUT TORSIONAL
  RPM     DIRECTION DIRECTION DIRECTION  X AXIS    Y AXIS    Z AXIS
-------   --------- --------- --------- --------- --------- ---------

 300.00     0.617    -0.333     0.005     3.009     0.734     0.515


MAXIMA OF FOOTING RESPONSE AMPLITUDES
*************************************

                                                     FREQ.
                                                      RPM
                                                    -------

MAX. TRANS. IN X-DIRECTION  -     7.960E-07  m       300.00
MAX. TRANS. IN Y-DIRECTION  -     1.745E-06  m       300.00
MAX. TRANS. IN Z-DIRECTION  -     8.484E-07  m       300.00
MAX. ROT. ABOUT X AXIS      -     2.812E-07  rad     300.00
MAX. ROT. ABOUT Y AXIS      -     7.339E-07  rad     300.00
MAX. ROT. ABOUT Z AXIS      -     1.335E-05  rad     300.00
```

Sample Design: Reciprocating Machine Foundation

```
                           RESULTANT TRANSLATIONS (m)
                           **********************

                 POINT NO.  1                POINT NO.  2
                 ------------                ------------

 FREQ.    HORIZ. X  HORIZ. Y  VERTICAL  HORIZ. X  HORIZ. Y  VERTICAL
  RPM    DIRECTION DIRECTION DIRECTION DIRECTION DIRECTION DIRECTION
-------   --------- --------- --------- --------- --------- ---------

 300.00    2.95E-05  3.56E-05  3.21E-06  2.17E-06  2.62E-05  2.23E-06



MAXIMUM RESULTANT TRANSLATIONS AT POINT NO.  1
**********************************************
                                                      FREQ.
                                                       RPM
                                                     -------

MAX. TRANS. IN X-DIRECTION  -     2.954E-05  m       300.00
MAX. TRANS. IN Y-DIRECTION  -     3.562E-05  m       300.00
MAX. TRANS. IN Z-DIRECTION  -     3.214E-06  m       300.00
MAX. TRANS. IN HORIZ. PLANE -     4.627E-05  m       300.00



MAXIMUM RESULTANT TRANSLATIONS AT POINT NO.  2
**********************************************
                                                      FREQ.
                                                       RPM
                                                     -------

MAX. TRANS. IN X-DIRECTION  -     2.172E-06  m       300.00
MAX. TRANS. IN Y-DIRECTION  -     2.623E-05  m       300.00
MAX. TRANS. IN Z-DIRECTION  -     2.232E-06  m       300.00
MAX. TRANS. IN HORIZ. PLANE -     2.632E-05  m       300.00
```

Sample Design: Reciprocating Machine Foundation

```
             RESULTANT TRANSLATIONS (m)
             **********************

                  POINT NO.   3
                  ------------

 FREQ.    HORIZ. X  HORIZ. Y  VERTICAL
  RPM    DIRECTION DIRECTION DIRECTION
-------   --------- --------- ---------

 300.00    2.17E-06  2.40E-05  8.19E-07



MAXIMUM RESULTANT TRANSLATIONS AT POINT NO.  3
**********************************************
                                                     FREQ.
                                                      RPM
                                                    -------

MAX. TRANS. IN X-DIRECTION  -     2.172E-06  m       300.00
MAX. TRANS. IN Y-DIRECTION  -     2.399E-05  m       300.00
MAX. TRANS. IN Z-DIRECTION  -     8.195E-07  m       300.00
MAX. TRANS. IN HORIZ. PLANE -     2.409E-05  m       300.00
```

FLUOR

Sample Design: Centrifugal Machine Foundation

Purpose:

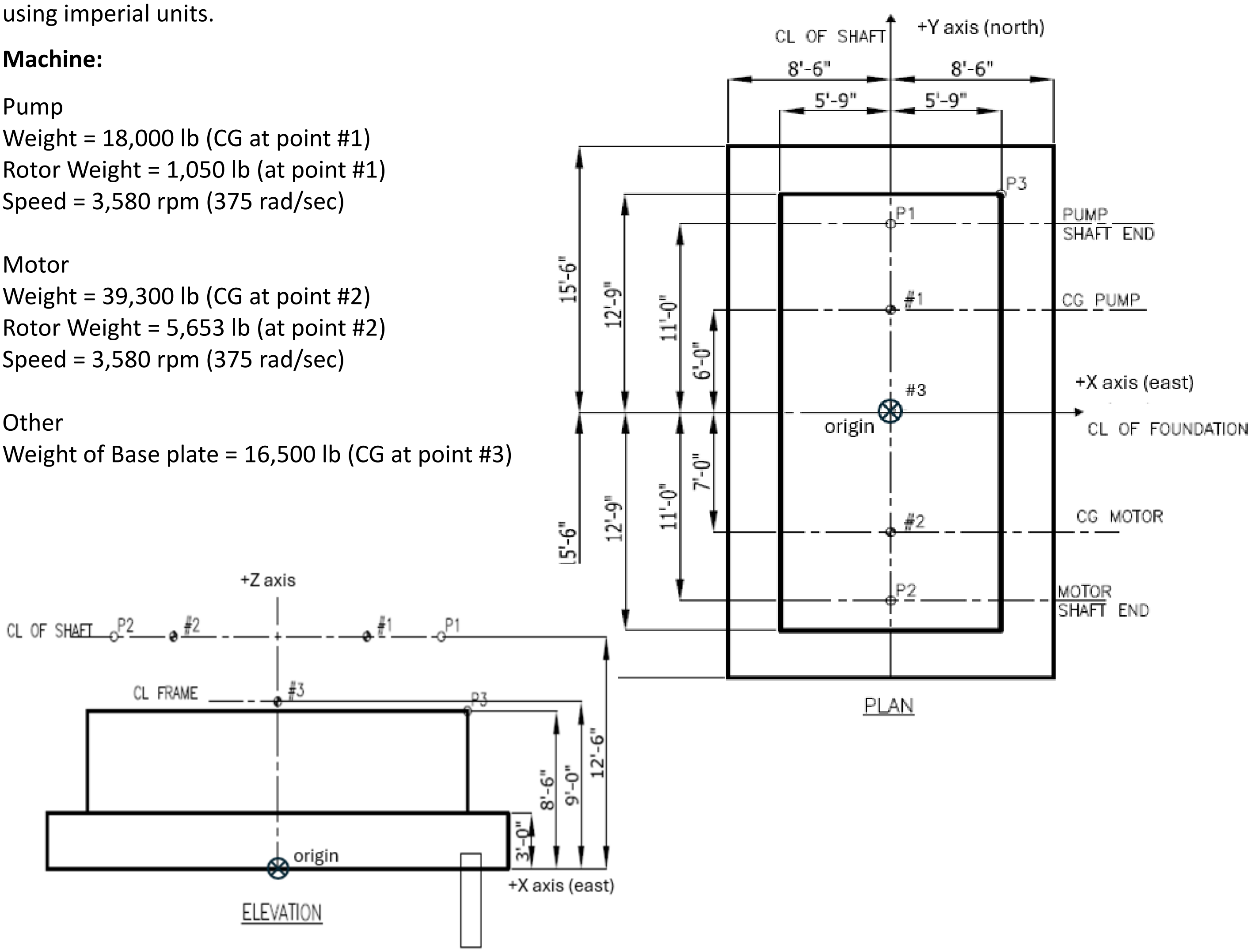
This attachment calculation illustrates the pile-supported foundation design for a centrifugal pump, using imperial units.

Machine:

Pump
Weight = 18,000 lb (CG at point #1)
Rotor Weight = 1,050 lb (at point #1)
Speed = 3,580 rpm (375 rad/sec)

Motor
Weight = 39,300 lb (CG at point #2)
Rotor Weight = 5,653 lb (at point #2)
Speed = 3,580 rpm (375 rad/sec)

Other
Weight of Base plate = 16,500 lb (CG at point #3)



FLUOR

RIGID FOUNDATIONS FOR VIBRATORY EQUIPMENT

Sample Design: Centrifugal Machine Foundation

Pile Properties:

Use Steel Pipe Piles

Wall Thickness, t = 0.5 inch = 0.042 ft

Outer diameter, D_O = 16 inch = 1.33 ft

Inner diameter, D_I = 15 inch = 1.25 ft

Cross section area = $\pi (D_O^2 - D_I^2)/4 = 0.169$ ft^2

Moment of inertia = $\pi (D_O^4 - D_I^4)/64 = 0.035$ ft^4

Torsional constant = 2 x 0.035 ft^4 = 0.071 ft^4

Length = 30 feet

Static capacity = 150 kips

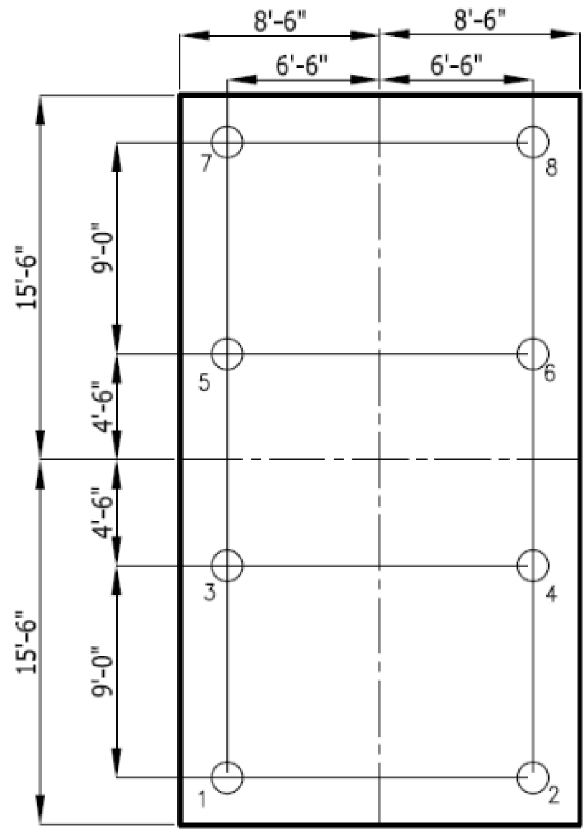
Rigidity factor = 2 (thin tube)

Modulus of elasticity, E_P = 2,9000 ksi = 4.176x10^9 lb/ft^2

Unit weight, U_P = 490 pcf

Nos of piles = 8

Pile material damping ratio = 0.025



PILE LAYOUT PLAN

Sample Design: Centrifugal Machine Foundation

Soil Properties:

There are 3 soil cases, defined using a coefficient of variation, COV = 0.3.

SC-1: shear wave velocity as given

Layer	Layer Depth (ft)	Shear Wave Velocity (ft/s)	Unit Weight (pcf)	Poisson's Ratio	Damping Ratio
Backfill	3.00	384	95	0.45	0.02
1	3.00	384	95	0.45	0.02
2	3.00	335	95	0.45	0.02
3	3.00	374	95	0.45	0.02
4	3.00	345	108	0.45	0.02
5	3.00	413	108	0.45	0.02
6	3.00	374	108	0.45	0.02
7	3.00	387	108	0.45	0.02
8	3.00	410	120	0.45	0.02
9	3.00	492	120	0.45	0.02
10	3.00	525	120	0.45	0.02
Below Tip	-	525	120	0.45	0.02

Sample Design: Centrifugal Machine Foundation

SC-2: upper bound & SC-2: lower bound (Equations 1a thru 1f)

Except for the shear modulus and shear wave velocity, properties are the same as SC-1.

$G_S = (V_S)^2 (U_S) / g$ $G_{UB} = G_S (1 + COV)$ $G_{LB} = G_S / (1 + COV)$

$V_{UB} = \text{sqrt} [G_{UB}\, g / U_S]$ $V_{LB} = \text{sqrt} [G_{LB}\, g / U_S]$

Layer	SC-1			SC-2		SC-3	
	V_S (ft/s)	U_S (pcf)	G_S (psf)	G_{UB} (psf)	V_{UB} (ft/s)	G_{LB} (psf)	V_{LB} (ft/s)
backfill	384	95	435,041	565,553	438	334,647	337
1	384	95	435,041	565,553	438	334,647	337
2	335	95	331,099	430,428	382	254,691	294
3	374	95	412,678	536,481	426	317,444	328
4	345	108	399,214	518,979	393	307,088	303
5	413	108	572,095	743,723	471	440,073	362
6	374	108	469,149	609,894	426	360,884	328
7	387	108	502,331	653,030	441	386,408	339
8	410	120	626,460	814,398	467	481,892	360
9	492	120	902,102	1,172,732	561	693,925	432
10	525	120	1,027,174	1,335,326	599	790,134	460
below tip	525	120	1,027,174	1,335,326	599	790,134	460

Concrete:

Compressive strength f'c = 3000 psi

Unit weight = 150 lb/ft^3

Density = 150 /32.2 = 4.6584 slug/ft^3

Elastic modulus, $E_C = 57{,}000 \sqrt{f'_C} = 57{,}000 \sqrt{3{,}000}$ psi = 3,122,000 psi

Pile Cap Thickness:

Considered thickness, Lz = 3' 0"

Pier cantilever, d = [(31 ft pile cap) – (25.5 ft pier)] / 2 = 2.75 ft

Try using top layer properties.

Upper bound elastic modulus of soil, $E_s = 2 (G_{UB})(1 + \mu_s)$ = 2 (565,553 psf /144)(1 + 0.45) = 11,390 psi

$[E_c / E_s][L_z / d]^3 \geq 1$ (Equation 5)

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Sample Design: Centrifugal Machine Foundation

[(3,122,000 psi) / (11,390 psi)] [(3.0 ft) / (2.75 ft)]^3 ≥ 1

356 > 1 OK, rigid USE 3' 0" thickness

Comparison is not close, so assumption of top layer is justified.

Resultant Locations:

Select point P3 for human tolerance and points at extreme bearings for machine tolerance:

P3 (human)	P1 (pump brg)	P2 (motor brg)
Dyna #1	Dyna #1	Dyna #3
X = 5.75 ft	X = 0 ft	X = 0 ft
Y = 12.75 ft	Y = 11 ft	Y = -11 ft
Z = 8.5 ft	Z = 12.5 ft	Z = 12.5 ft

Effective Embedment:

Use 2/3 of pile cap thickness, h = 2/3 (3.0 ft) = 2 ft

Center of Gravity Offset Check:

From 3D VIEW calculation (page 18)

X coordinate of base relative to CG = 0

Y coordinate of base relative to CG = 0.30 ft

Offset in X direction, OSX = 0 < 5%, OK

Offset in Y direction, OSY = 0.30 ft / 31 ft = 0.97% < 5%, OK

Pile Capacity Check:

Total dead weight = (17,170 slug) (32.2 ft/s2) = 552,874 lb

Half pile capacity = 0.5 (150 kips x 1000) = 75,000 lb

Pile load = (552,874 lb) / (8 piles) = 69,109 lb < 75,000 lb, OK

Frequency Analysis:

Using Dyna 6 damping safety factor = 100

Based on initial analysis results, use a frequency sweep from 300 rpm to 1500 rpm at 10 rpm increments.

Nominal loading is included for the purpose of engaging each mode.

$F_X = F_Y = F_Z = 1000$ N

$M_X = M_Y = M_Z = 1000$ N-m

Sample Design: Centrifugal Machine Foundation

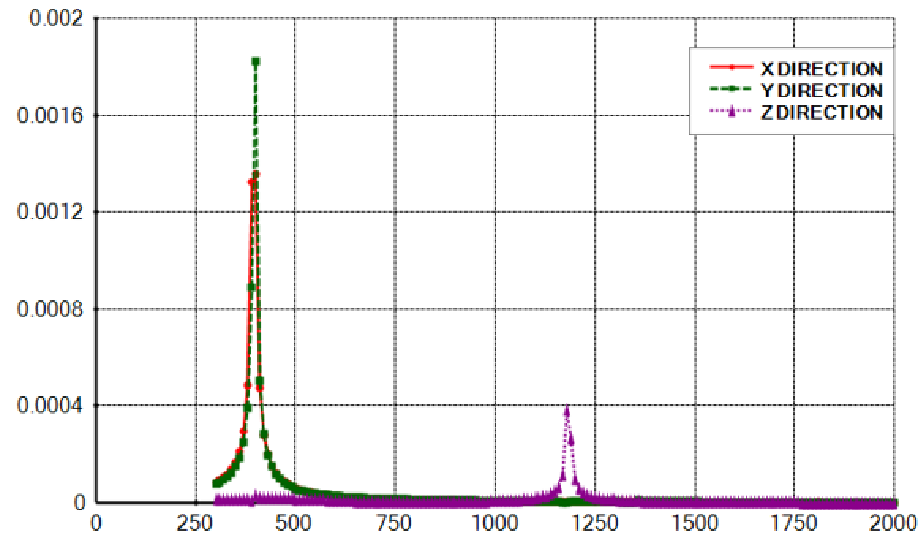
```
HARMONIC LOAD (NONQUADRATIC)
*************

MAXIMUM FREQUENCY ............  2000.000  RPM
MINIMUM FREQUENCY ............   300.000  RPM
STEP FREQUENCY ...............    10.000  RPM
FORCE IN X-DIRECTION  .........  1.00E+03 lb
FORCE IN Y-DIRECTION  .........  1.00E+03 lb
FORCE IN Z-DIRECTION  .........  1.00E+03 lb
MOMENT ABOUT X-AXIS  ..........  1.00E+03 lb.ft
MOMENT ABOUT Y-AXIS  ..........  1.00E+03 lb.ft
MOMENT ABOUT Z-AXIS  ..........  1.00E+03 lb.ft

PHASE SHIFTS OF EXTERNAL LOADS
******************************

FORCE IN X-DIRECTION  .........      0.00 degrees
FORCE IN Y-DIRECTION  .........      0.00 degrees
FORCE IN Z-DIRECTION  .........      0.00 degrees
MOMENT ABOUT X-AXIS  ..........      0.00 degrees
MOMENT ABOUT Y-AXIS  ..........      0.00 degrees
MOMENT ABOUT Z-AXIS  ..........      0.00 degrees
```

Use a resonant range from 0.8 to 1.2.

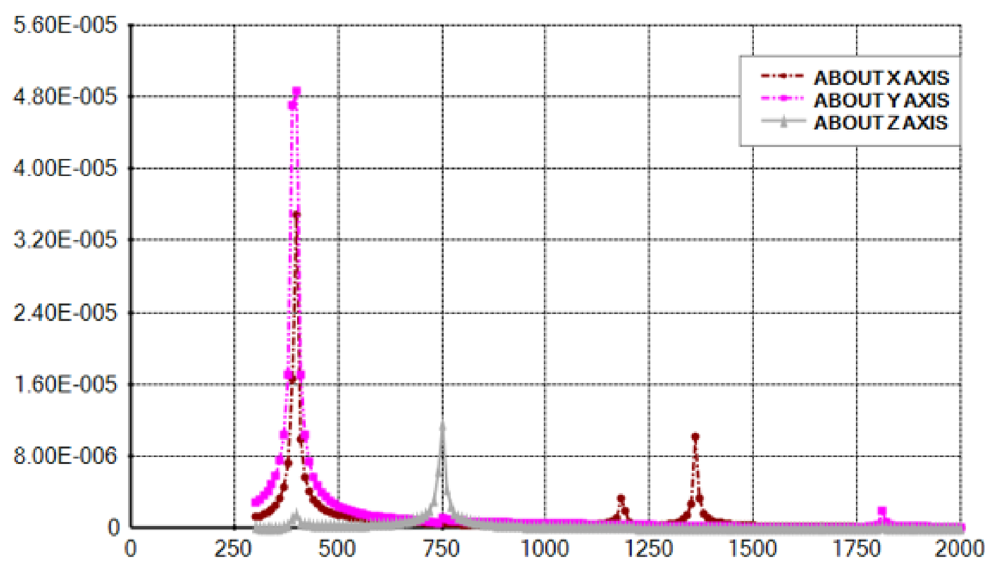
Sample Design: Centrifugal Machine Foundation

Soil Case SC-1 (soil properties as given)

Translations at CG:
X translation - 400 rpm
Y translation - 400 rpm
Z translation - 1180 rpm



Rotations at CG
X rotation = 400 rpm, 1360 rpm
Y rotation = 400 rpm, 1810 rpm
Z rotation = 750 rpm



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Sample Design: Centrifugal Machine Foundation

Summary: (rpm) MS = 3580 rpm

	Dyna 6 NF	MS / NF
Vertical	1180	3.0
Torsional	750	4.8
X trans / Y rot	400 & 1810	9.0 & 2.0
Y trans / X rot	400 & 1360	9.0 & 2.6

Frequencies are outside of resonant range

Soil Case SC-2: (upper bound soil properties)

For this guideline example, details of this frequency sweep are omitted for brevity.

Summary: (rpm) MS = 3580 rpm

	Dyna 6 NF	MS / NF
Vertical	1290	2.8
Torsional	820	4.4
X trans / Y rot	440, 1980	8.1, 1.8
Y trans / X rot	440, 1490	8.1, 2.4

Frequencies are outside of resonant range

Soil Case SC-3: (lower bound soil properties)

For this guideline example, details of this frequency sweep are omitted for brevity.

Summary: (rpm) MS = 3580 rpm

	Dyna 6 NF	MS / NF
Vertical	1080	3.3
Torsional	690	5.2
X trans / Y rot	350, 1640	10.2, 2.2
Y trans / X rot	350, 1240	10.2, 2.9

Frequencies are outside of resonant range

FLUOR

Sample Design: Centrifugal Machine Foundation

Forced Response Analysis:

Use damping safety factor = 2.0

Machine limit = 0.15 in/s

Human limit = 7.5 ft/s2 based on 4 hour exposure and 3580 rpm

Human Tolerance Calculation (using ISO 2631 Health Guidance Caution Zone)
W. Bounds (20Jan2025)

Category = health	(as opposed to comfort, perception, or motion sickness)		
Type = whole body	(as opposed to hand transmitted)		
Exposure = intermittent	(as opposed to continuous)		
Input data:			
Units of Measure	1	imperial	Enter 1 for imperial (ft, lb), or anything else for metric (m, N).
daily exposure, DE =	14400	seconds	(range is 1 s to 24 h)
machine speed, MS	59.66666667	Hz	
Permissible RMS acceleration			ISO 2631-1, Section B.3.1
Amax	9.84	ft/s2	if DE <= 10 minutes, then A rms = Amax
A rms =	2.008581589	ft/s2	if DE > 10 minutes, then Arms = (Amax) sqrt [600 / exposure]
Orientation/Direction Factors			ISO 2631-1, Section 7.2.2
horizontal, kx, ky	1.4		
vertical, kz	1.0		
Frequency Weighting Factors			In lieu of ISO 2631, use SCI P354, Equations 22 & 23
horizontal, Wd	0.034		
vertical, Wb	0.268		
Permissible RMS acceleration			
X axis (horiz), Ax rms p	42.802	ft/s2	= (A rms) / (kx Wd)
Y axis (horiz), Ay rms p	42.802	ft/s2	= (A rms) / (ky Wd)
Z axis (vertical), Az rms p	7.490	ft/s2	= (A rms) / (kz Wb)

MS	Wd	Wb
1 to 2	1	0.4
2 to 5	2 / MS	MS / 5
6 to 16	2 / MS	1
above 16	2 / MS	16 / MS

The foundation undergoes analysis for dynamic loads of both pump and motor. Two separate analyses are performed for each force: one for the pump load at 3580 rpm and another for the motor load at 3580 rpm. The resultant amplitude is determined by algebraically adding the amplitudes from both analyses at the specified measurement locations.

Note that Dyna 6 material damping input expects twice the damping ratio.

Use SF = 1 and Qg increased by one level, Qg = 0.62 in/s

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Sample Design: Centrifugal Machine Foundation

Force = (RW / g)(Qg)(MS)(S_F)/12 (Equation A1-3)

LC1: Motor Rotor (3580 rpm = 375 rad/s)

Rotor mass = 5653 lb / 32.2 ft/s2 = 175.6 slug

Unbalance force = (175.6 slug)(0.615 in/sec)(375 rad/sec)(1.0) / 12 = 3374 lb

3D VIEW input:

	X	Y	Z	Fx	Fy	Fz	Mx	My	Mz
1	0	-7	12.5	3374	0	3374	0	0	0
2									

DYNA 6 input at CG: (added applicable phase angles)

```
HARMONIC LOAD (NONQUADRATIC)
*************

MAXIMUM FREQUENCY  ............  3580.000  RPM
MINIMUM FREQUENCY  ............  3580.000  RPM
STEP FREQUENCY ...............     1.000  RPM
FORCE IN X-DIRECTION  .........  3.37E+03 lb
FORCE IN Y-DIRECTION  .........  0.00E+00 lb
FORCE IN Z-DIRECTION  .........  3.37E+03 lb
MOMENT ABOUT X-AXIS  .......... -2.26E+04 lb.ft
MOMENT ABOUT Y-AXIS  ..........  2.62E+04 lb.ft
MOMENT ABOUT Z-AXIS  ..........  2.26E+04 lb.ft

PHASE SHIFTS OF EXTERNAL LOADS
******************************

FORCE IN X-DIRECTION  .........      0.00 degrees
FORCE IN Y-DIRECTION  .........      0.00 degrees
FORCE IN Z-DIRECTION  .........     90.00 degrees
MOMENT ABOUT X-AXIS  ..........     90.00 degrees
MOMENT ABOUT Y-AXIS  ..........      0.00 degrees
MOMENT ABOUT Z-AXIS  ..........      0.00 degrees
```

Sample Design: Centrifugal Machine Foundation

<u>LC2: Pump Rotor</u> (3580 rpm = 375 rad/s)

Rotor mass = 1050 lb / 32.2 ft/s2 = 32.6 slug

Unbalance force at Pump = (32.6 slug)(0.615 in/s)(375 rad/s)(1.0) / 12 = 627 lb

3D VIEW input:

	X	Y	Z	Fx	Fy	Fz	Mx	My	Mz
1	0	6	12.5	627	0	627	0	0	0
2									

DYNA 6 input: (added applicable phase angles)

```
HARMONIC LOAD (NONQUADRATIC)
*************

MAXIMUM FREQUENCY  ............  3580.000  RPM
MINIMUM FREQUENCY  ............  3580.000  RPM
STEP FREQUENCY  ...............     1.000  RPM
FORCE IN X-DIRECTION  .........  6.27E+02 lb
FORCE IN Y-DIRECTION  .........  0.00E+00 lb
FORCE IN Z-DIRECTION  .........  6.27E+02 lb
MOMENT ABOUT X-AXIS  ..........  3.95E+03 lb.ft
MOMENT ABOUT Y-AXIS  ..........  4.88E+03 lb.ft
MOMENT ABOUT Z-AXIS  .......... -3.95E+03 lb.ft

PHASE SHIFTS OF EXTERNAL LOADS
******************************

FORCE IN X-DIRECTION  .........      0.00 degrees
FORCE IN Y-DIRECTION  .........      0.00 degrees
FORCE IN Z-DIRECTION  .........     90.00 degrees
MOMENT ABOUT X-AXIS  ..........     90.00 degrees
MOMENT ABOUT Y-AXIS  ..........      0.00 degrees
MOMENT ABOUT Z-AXIS  ..........      0.00 degrees
```

FLUOR®

Sample Design: Centrifugal Machine Foundation

Soil Case SC-1: (soil properties as given)

Refer to Dyna 6 output file which begins on page 19.

Point P3 (Dyna #1), RMS Acceleration for Human Tolerance

Peak Amplitude: (ft)	X	Y	Z
LC1 (motor rotor)	6.177E-06	1.096E-06	3.170E-06
LC2 (pump rotor)	6.648E-07	1.900E-07	8.579E-09
Sum	6.842E-06	1.286E-06	3.179E-06

Arms = (0.707)(sum)(375 rad/s)^2

RMS Acceleration (ft/s2)	0.68	0.13	0.32

Utilization = (0.68 ft/s2) / (7.5 ft/s2) = 0.09 < 1, OK

Point P1 (Dyna #2), Velocity for Pump Tolerance

Peak Amplitude: (ft)	X	Y	Z
LC1 (motor rotor)	3.705E-06	1.096E-06	5.035E-08
LC2 (pump rotor)	1.216E-06	1.900E-07	5.521E-07
Sum	4.921E-06	1.286E-06	6.020E-07

V = (sum)(375 rad/s) 12

Peak Velocity (in/s)	0.022	0.0058	0.0027

Utilization = (0.022 in/s) / (0.15 in/s) = 0.15 < 1, OK

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Sample Design: Centrifugal Machine Foundation

Point P2 (Dyna #3), Velocity for Motor Tolerance

Peak Amplitude: (ft)	X	Y	Z
LC1 (motor rotor)	4.550E-06	8.318E-07	4.402E-06
LC2 (pump rotor)	2.626E-07	1.668E-07	5.245E-07
Sum	4.813E-06	9.990E-07	4.927E-06

V = (sum)(*375 rad/s) 12

Peak Velocity (in/s)	0.022	0.0045	0.022

Utilization = (0.022 in/2) / (0.15 in/s) = 0.15 < 1, OK

Soil Case SC-2: (soil upper bound)

For this guideline example, details are omitted for brevity.

Point P3 (Dyna #1), RMS Acceleration for Human Tolerance

Peak Amplitude: (ft)	X	Y	Z
LC1 (motor rotor)	6.105E-06	1.132E-06	3.226E-06
LC2 (pump rotor)	6.443E-07	1.962E-07	1.705E-08
Sum	6.749E-06	1.328E-06	3.243E-06

Arms = (0.707)(sum)(375 rad/s)^2

RMS Acceleration (ft/s2)	0.67	0.13	0.32

Utilization = (0.67 ft/s2) / (7.5 ft/s2) = 0.09 < 1, OK

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Sample Design: Centrifugal Machine Foundation

Point P1 (Dyna #2), Velocity for Pump Tolerance

Peak Amplitude: (ft)	X	Y	Z
LC1 (motor rotor)	3.592E-06	1.132E-06	1.020E-07
LC2 (pump rotor)	1.201E-06	1.962E-07	5.612E-07
Sum	4.793E-06	1.328E-06	6.632E-07

V = (sum)(375 rad/s) 12

Peak Velocity (in/s)	0.022	0.0060	0.0030

Utilization = (0.022 in/s) / (0.15 in/s) = 0.15 < 1, OK

Point P2 (Dyna #3), Velocity for Motor Tolerance

Peak Amplitude: (ft)	X	Y	Z
LC1 (motor rotor)	4.530E-06	8.455E-07	4.460E-06
LC2 (pump rotor)	2.518E-07	1.687E-07	5.165E-07
Sum	4.782E-06	1.014E-06	4.977E-06

V = (sum)(*375 rad/s) 12

Peak Velocity (in/s)	0.022	0.0046	0.022

Utilization = (0.022 in/2) / (0.15 in/s) = 0.15 < 1, OK

Soil Case SC-3: (soil lower bound)

For this guideline example, details are omitted for brevity.

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Sample Design: Centrifugal Machine Foundation

Point P3 (Dyna #1), RMS Acceleration for Human Tolerance

Peak Amplitude: (ft)	X	Y	Z
LC1 (motor rotor)	6.165E-06	1.105E-06	3.221E-06
LC2 (pump rotor)	6.652E-07	1.910E-07	1.708E-08
Sum	6.830E-06	1.296E-06	3.238E-06

Arms = (0.707)(sum)(375 rad/s)^2

RMS Acceleration (ft/s2)	0.68	0.13	0.32

Utilization = (0.68 ft/s2) / (7.5 ft/s2) = 0.09 < 1, OK

Point P1 (Dyna #2), Velocity for Pump Tolerance

Peak Amplitude: (ft)	X	Y	Z
LC1 (motor rotor)	3.707E-06	1.105E-06	8.679E-08
LC2 (pump rotor)	1.214E-06	1.910E-07	5.600E-07
Sum	4.921E-06	1.296E-06	6.468E-07

V = (sum)(375 rad/s) 12

Peak Velocity (in/s)	0.022	0.0058	0.0029

Utilization = (0.022 in/s) / (0.15 in/s) = 0.15 < 1, OK

Point P2 (Dyna #3), Velocity for Motor Tolerance

Peak Amplitude: (ft)	X	Y	Z
LC1 (motor rotor)	4.541E-06	8.413E-07	4.452E-06
LC2 (pump rotor)	2.638E-07	1.686E-07	5.296E-07
Sum	4.805E-06	1.010E-06	4.982E-06

V = (sum)(375 rad/s) 12

Peak Velocity (in/s)	0.022	0.0045	0.022

Utilization = (0.022 in/2) / (0.15 in/s) = 0.15 < 1, OK

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Sample Design: Centrifugal Machine Foundation

Anchor Bolt Check:

Note: For the brevity of this guideline example, the anchor bolt embedment check is not shown. Refer to PIP STE05121, ASCE Anchorage Design.

Pier Reinforcing Design:

USE #5 at 12 inches each way

Mat Reinforcing Design:

Mat cantilever = 2.75 ft

By inspection, 2.75 ft cantilever and a 3 ft thick mat does not require more than nominal reinforcing.

As = (0.0018)(36 in depth)(12 in unit width) = 0.78 in^2

USE #6 @ 12 inches each way, top and bottom.

Sample Design: Centrifugal Machine Foundation

3D VIEW INPUT

Coordinates & dimensions are in feet.
Density is in slugs/ft^3
Mass is in slugs

Rectangular Blocks

	Corner_X	Corner_Y	Corner_Z	Length_X	Width_Y	Height_Z	Density
1	-8.5	-15.5	0	17	31	3	4.6584
2	-5.75	-12.75	3	11.5	25.5	5.5	4.6584

Lumped Masses

	X	Y	Z	Mass	IXX	IYY	IZZ	IXY	IXZ	IYZ
1	0	6	12.5	559.006	0	0	0	0	0	0
2	0	-7	12.5	1220.497	0	0	0	0	0	0
3	0	0	9	512.422	0	0	0	0	0	0

Base Center

	X_Length	Y_Length	Base_Center_X	Base_Center_Y	Base_Center_Z
1	17	31	0	0	0

Pile Location Input Relative to User Origin

	X-Coord	Y-Coord
1	-6.5	-13.5
2	6.5	-13.5
3	-6.5	-4.5
4	6.5	-4.5
5	-6.5	4.5
6	6.5	4.5
7	-6.5	13.5
8	6.5	13.5

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Sample Design: Centrifugal Machine Foundation

Resultant Locations Relative to User Origin

	X-Coord	Y-Coord	Z-Coord
1	0	11	12.5
2	0	-11	12.5
3	5.75	12.75	8.5

3D VIEW Mass Calculation (loading is not indicated)

Total Mass = 17169.7473

Center of Mass X-Coord. = 0
Center of Mass Y-Coord. = -0.302042884463419
Center of Mass Z-Coord. = 4.72334515808512

Mass Moment of Inertial IXX = 1301142.53761635
Mass Moment of Inertial IYY = 486039.666765179
Mass Moment of Inertial IZZ = 1335455.24928867
Mass Moment of Inertial IXY = 0
Mass Moment of Inertial IXZ = 0
Mass Moment of Inertial IYZ = -40329.7320101706

Total Force in X-Direction = 0
Total Force in Y-Direction = 0
Total Force in Z-Direction = 0

Total Moment about X-Direction = 0
Total Moment about Y-Direction = 0
Total Moment about Z-Direction = 0

Length about X-Direction = 17
Length about Y-Direction = 31

X Coord. of BC relative to CG = 0
Y Coord. of BC relative to CG = 0.302042884463419
Z Coord. of BC relative to CG = 4.72334515808512

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Sample Design: Centrifugal Machine Foundation

DYNA 6 Output File (Soil Properties As Given) – Dynamic Load at Pump

```
*************************************************
*                                               *
*       D Y N A 6   S I M U L A T I O N         *
*                                               *
*         Copyright (c) Western - 2011          *
*            RUN DATE -  2024/12/20             *
*            TIME     -    13:32: 2             *
*            REVISION -  Jan.21/201             *
*                                               *
*************************************************

Guideline 1234 Att 6-Motor Rotor

                     DATA ECHO
                     *********

GRAVITATIONAL CONSTANT SET TO  32.20 ft/s**2
*****************************

FREQUENCY UNITS :  RPM
***************

LENGTH UNITS : ft
************

FORCE UNITS : lb
***********

MASS UNITS : slug
**********

STIFFNESS AND DAMPING CONSTANTS TO BE PRINTED
*********************************************


DAMPING SAFETY FACTOR -    2.00
*********************


FOUNDATION TYPE - PILE
***************
```

FLUOR®

Sample Design: Centrifugal Machine Foundation

```
RIGID CAP
*********

RECTANGULAR FOUNDATION  -    17.000 ft  BY  31.000 ft
**********************


MASS CONSTANTS
**************

TOTAL MASS OF FOOTING  ......... 1.7170E+04  slug
MASS MOMENT ABOUT X-AXIS  ...... 1.3011E+06  slug.ft**2
MASS MOMENT ABOUT Y-AXIS  ...... 4.8604E+05  slug.ft**2
MASS MOMENT ABOUT Z-AXIS  ...... 1.3355E+06  slug.ft**2
CROSS PRODUCT X-Y  ............. 0.0000E+00  slug.ft**2
CROSS PRODUCT X-Z  ............. 0.0000E+00  slug.ft**2
CROSS PRODUCT Y-Z  .............-4.0330E+04  slug.ft**2

NUMBER OF LAYERS OR ELEMENTS -  10
****************************

PILES ARE FIXED AT BUTT
***********************

LOCATION OF PILES
*****************

PILE        X           Y
 NO.     LOCATION    LOCATION
            ft          ft

  1       -6.500     -13.198
  2        6.500     -13.198
  3       -6.500      -4.198
  4        6.500      -4.198
  5       -6.500       4.802
  6        6.500       4.802
  7       -6.500      13.802
  8        6.500      13.802
```

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```
COORDINATES OF BASE CENTRE
**************************

X-COORD. OF BASE CENTRE REL. TO C.G.  ..     0.000  ft
Y-COORD. OF BASE CENTRE REL. TO C.G.  ..     0.302  ft
Z-COORD. OF BASE CENTRE REL. TO C.G.  ..     4.723  ft

PILE PROPERTIES
***************

TOTAL LENGTH  .................    30.000  ft
UNIT WEIGHT  ...................   490.000  lb/ft**3
STATIC LOAD ON PILE  ........... 7.000E+04  lb
POISSON'S RATIO  ...............     0.300
MATERIAL DAMPING  ..............     0.050
COEFF. OF RIGIDITY  ...........     2.000
YOUNG'S MODULUS  ............... 4.176E+09  lb/ft**2

ELEMENT
*******
```

LAYER NO.	LAYER DEPTH ft	PILE RADIUS (X) ft	PILE RADIUS (Y) ft	CROSS SECTIONAL AREA ft**2	MOMENT OF INERTIA ABOUT Y ft**4	MOMENT OF INERTIA ABOUT X ft**4	TORSIONAL CONSTANT ft**4
1	3.000	0.667	0.667	0.169	3.50E-02	3.50E-02	7.10E-02
2	3.000	0.667	0.667	0.169	3.50E-02	3.50E-02	7.10E-02
3	3.000	0.667	0.667	0.169	3.50E-02	3.50E-02	7.10E-02
4	3.000	0.667	0.667	0.169	3.50E-02	3.50E-02	7.10E-02
5	3.000	0.667	0.667	0.169	3.50E-02	3.50E-02	7.10E-02
6	3.000	0.667	0.667	0.169	3.50E-02	3.50E-02	7.10E-02
7	3.000	0.667	0.667	0.169	3.50E-02	3.50E-02	7.10E-02
8	3.000	0.667	0.667	0.169	3.50E-02	3.50E-02	7.10E-02
9	3.000	0.667	0.667	0.169	3.50E-02	3.50E-02	7.10E-02
10	3.000	0.667	0.667	0.169	3.50E-02	3.50E-02	7.10E-02

```
PILE TIP CONDITION : FLOATING (DEFAULT)
******************
```

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```
PILE-SOIL-PILE INTERACTION IS CONSIDERED
****************************************


SOIL
****

SOIL CONSTANTS
**************

LAYER   SHEAR WAVE     UNIT     POISSON'S  MATERIAL
 NO.     VELOCITY     WEIGHT     RATIO      DAMPING
           ft/s      lb/ft**3

  1       384.000       95.00    0.450      0.040
  2       335.000       95.00    0.450      0.040
  3       374.000       95.00    0.450      0.040
  4       345.000      108.00    0.450      0.040
  5       413.000      108.00    0.450      0.040
  6       374.000      108.00    0.450      0.040
  7       387.000      108.00    0.450      0.040
  8       410.000      120.00    0.450      0.040
  9       492.000      120.00    0.450      0.040
 10       525.000      120.00    0.450      0.040

SOIL BELOW PILE
***************

SHEAR WAVE    UNIT    POISSON'S  MATERIAL
 VELOCITY    WEIGHT     RATIO     DAMPING
   ft/s     lb/ft**3

 5.25E+02     120.00   0.450     0.040

EMBEDDED OPTION
**************

NUMBER OF SIDE LAYERS  -    1

LAYER  THICKNESS  SHEAR WAVE   UNIT   POISSON'S  MATERIAL
 NO.    OF LAYER   VELOCITY   WEIGHT    RATIO     DAMPING
          ft         ft/s    lb/ft**3

  1      2.000      384.00      95.00   0.450     0.040
```

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Sample Design: Centrifugal Machine Foundation

```
HARMONIC LOAD (NONQUADRATIC)
*************

MAXIMUM FREQUENCY ...........  3580.000  RPM
MINIMUM FREQUENCY ...........  3580.000  RPM
STEP FREQUENCY  ..............    1.000  RPM
FORCE IN X-DIRECTION  ........  3.37E+03 lb
FORCE IN Y-DIRECTION  ........  0.00E+00 lb
FORCE IN Z-DIRECTION  ........  3.37E+03 lb
MOMENT ABOUT X-AXIS  ......... -2.26E+04 lb.ft
MOMENT ABOUT Y-AXIS  .........  2.62E+04 lb.ft
MOMENT ABOUT Z-AXIS  .........  2.26E+04 lb.ft

PHASE SHIFTS OF EXTERNAL LOADS
******************************

FORCE IN X-DIRECTION  ........      0.00 degrees
FORCE IN Y-DIRECTION  ........      0.00 degrees
FORCE IN Z-DIRECTION  ........     90.00 degrees
MOMENT ABOUT X-AXIS  .........     90.00 degrees
MOMENT ABOUT Y-AXIS  .........      0.00 degrees
MOMENT ABOUT Z-AXIS  .........      0.00 degrees


RESPONSE PHASE SHIFT TO BE PRINTED
**********************************

RESULTANT TRANSLATIONS REQUIRED AT  3 POINTS
**********************************************

COORDINATES OF POINTS :

POINT     X         Y         Z
 NO.      ft        ft        ft

  1     0.000    11.302    -7.776
  2     0.000   -10.698    -7.776
  3     5.750    13.052    -3.776
```

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Sample Design: Centrifugal Machine Foundation

```
****************************************************
*                                                  *
*         D Y N A 6   S I M U L A T I O N          *
*                                                  *
*           Copyright (c) Western - 2011           *
*              RUN DATE -  2024/12/20              *
*              TIME     -    13:32: 2              *
*              REVISION -  Jan.21/201              *
*                                                  *
****************************************************

 Guideline 1234 Att 6-Motor Rotor

                                  RESULTS
                                  *******

FREQUENCY  -  3580.000   RPM
*********


STIFFNESS CONSTANTS (K)
***********************

HORIZONTAL TRANSLATION  (X) ...  6.731E+07  lb/ft
HORIZONTAL TRANSLATION  (Y) ...  6.459E+07  lb/ft
VERTICAL TRANSLATION  (Z) .....  2.228E+08  lb/ft
ROTATION ABOUT  (X) ...........  2.656E+10  lb.ft/rad
ROTATION ABOUT  (Y) ...........  1.457E+10  lb.ft/rad
TORSION ABOUT  (Z) ............  1.025E+10  lb.ft/rad
CROSS-STIFFNESS (YZ PLANE) ....  4.505E+08  lb/rad
CROSS-STIFFNESS (XZ PLANE) .... -4.654E+08  lb/rad

DAMPING CONSTANTS (C)
*********************

HORIZONTAL TRANSLATION  (X) ...  3.247E+05  lb/ft/s
HORIZONTAL TRANSLATION  (Y) ...  2.925E+05  lb/ft/s
VERTICAL TRANSLATION  (Z) .....  4.640E+05  lb/ft/s
ROTATION ABOUT  (X) ...........  4.575E+07  lb.ft/rad/s
ROTATION ABOUT  (Y) ...........  2.507E+07  lb.ft/rad/s
TORSION ABOUT  (Z) ............  3.293E+07  lb.ft/rad/s
CROSS-DAMPING (YZ PLANE) ......  1.300E+06  lb/rad/s
CROSS-DAMPING (XZ PLANE) ...... -1.455E+06  lb/rad/s
```

Sample Design: Centrifugal Machine Foundation

```
**************************************************
*                                                *
*        D Y N A 6   S I M U L A T I O N         *
*                                                *
*          Copyright (c) Western - 2011          *
*             RUN DATE -  2024/12/20             *
*             TIME     -     13:32: 2            *
*             REVISION -  Jan.21/201             *
*                                                *
**************************************************

Guideline 1234 Att 6-Motor Rotor

                              RESULTS
                              *******
                    FOOTING RESPONSE AMPLITUDES
                    ---------------------------

 FREQ.     TRANS. X  TRANS. Y  VERTICAL  ROT ABOUT ROT ABOUT TORSIONAL
  RPM     DIRECTION DIRECTION DIRECTION   X AXIS    Y AXIS    Z AXIS
             (ft)      (ft)      (ft)      (rad)     (rad)     (rad)
-------   --------- --------- --------- --------- --------- ---------

3580.00    1.37E-06  4.07E-08  1.54E-06  1.44E-07  4.57E-07  1.13E-07

                   PHASE SHIFT OF RESPONSE (RAD)
                   -----------------------------

 FREQ.     TRANS. X  TRANS. Y  VERTICAL  ROT ABOUT ROT ABOUT TORSIONAL
  RPM     DIRECTION DIRECTION DIRECTION   X AXIS    Y AXIS    Z AXIS
-------   --------- --------- --------- --------- --------- ---------

3580.00     3.108     2.556    -1.489     1.685    -3.000    -3.092

MAXIMA OF FOOTING RESPONSE AMPLITUDES
*************************************

                                                     FREQ.
                                                      RPM
                                                    -------

MAX. TRANS. IN X-DIRECTION  -     1.368E-06  ft     3580.00
MAX. TRANS. IN Y-DIRECTION  -     4.073E-08  ft     3580.00
MAX. TRANS. IN Z-DIRECTION  -     1.540E-06  ft     3580.00
MAX. ROT. ABOUT X AXIS      -     1.443E-07  rad    3580.00
MAX. ROT. ABOUT Y AXIS      -     4.571E-07  rad    3580.00
MAX. ROT. ABOUT Z AXIS      -     1.125E-07  rad    3580.00
```

Sample Design: Centrifugal Machine Foundation

```
                         RESULTANT TRANSLATIONS (ft)
                         **********************

              POINT NO.  1                 POINT NO.  2
              ------------                 ------------

 FREQ.    HORIZ. X  HORIZ. Y  VERTICAL  HORIZ. X  HORIZ. Y  VERTICAL
  RPM    DIRECTION DIRECTION DIRECTION DIRECTION DIRECTION DIRECTION
-------   --------- --------- --------- --------- --------- ---------

3580.00    6.18E-06  1.10E-06  3.17E-06  3.70E-06  1.10E-06  5.04E-08



MAXIMUM RESULTANT TRANSLATIONS AT POINT NO.  1
**********************************************
                                                     FREQ.
                                                      RPM
                                                    -------

MAX. TRANS. IN X-DIRECTION  -     6.177E-06  ft     3580.00
MAX. TRANS. IN Y-DIRECTION  -     1.096E-06  ft     3580.00
MAX. TRANS. IN Z-DIRECTION  -     3.170E-06  ft     3580.00
MAX. TRANS. IN HORIZ. PLANE -     6.274E-06  ft     3580.00


MAXIMUM RESULTANT TRANSLATIONS AT POINT NO.  2
**********************************************
                                                     FREQ.
                                                      RPM
                                                    -------

MAX. TRANS. IN X-DIRECTION  -     3.705E-06  ft     3580.00
MAX. TRANS. IN Y-DIRECTION  -     1.096E-06  ft     3580.00
MAX. TRANS. IN Z-DIRECTION  -     5.035E-08  ft     3580.00
MAX. TRANS. IN HORIZ. PLANE -     3.863E-06  ft     3580.00
```

Sample Design: Centrifugal Machine Foundation

```
              RESULTANT TRANSLATIONS (ft)
              **********************

                   POINT NO.  3
                   ------------

  FREQ.    HORIZ. X  HORIZ. Y  VERTICAL
   RPM    DIRECTION DIRECTION DIRECTION
 -------   --------- --------- ---------

 3580.00    4.55E-06  8.32E-07  4.40E-06



 MAXIMUM RESULTANT TRANSLATIONS AT POINT NO.  3
 **********************************************
                                                     FREQ.
                                                      RPM
                                                    -------

 MAX. TRANS. IN X-DIRECTION  -     4.550E-06  ft     3580.00
 MAX. TRANS. IN Y-DIRECTION  -     8.318E-07  ft     3580.00
 MAX. TRANS. IN Z-DIRECTION  -     4.402E-06  ft     3580.00
 MAX. TRANS. IN HORIZ. PLANE -     4.626E-06  ft     3580.00
```